

$$\frac{dE}{dt} = \Phi + \mathcal{A}$$



$$\frac{dP}{dt} = F + G$$



$$\frac{dL}{dt} = M + \mathcal{T}$$



$$\frac{dS}{dt} \geq \Psi$$



$$\frac{dN}{dt} = J + \mathcal{R}$$



$$\frac{dQ}{dt} = I$$



$$\frac{d\mathcal{B}}{dt} = -\mathcal{E}$$

The Seven Wonders of the World

Exam tips, exercises & solutions

updated 21 March 2026

11:54:41

“Finally, I said that I couldn’t see how anyone could be educated by this self-propagating system in which people pass exams, and teach others to pass exams, but nobody knows anything.”

R. P. Feynman (*Surely You’re Joking, Mr. Feynman!*, Bantam Books 1985)

- **Know the most important notions:**

You must be able to explain them in intuitive terms, at least by giving examples.

Exercise by trying to explain them to a friend, illustrating with simple examples.

- Control volumes and control surfaces
- Volume contents and fluxes
- Symmetry of fluxes
- Extensivity
- Difference between universal laws and constitutive relations
- Meaning of a balance law (it's a budget, involving also future values)

- **Know the universal balance laws we studied:**

*The solution of **every** physical problem is based on one or more universal balance laws.*

- Have an idea of which balance law is most relevant in a given problem
- Be able to find an unknown term in a balance law, given the others
- Approximate form for short times

- **Review the constitutive relations we discussed in the course:**

- Understand when each constitutive equation can be applied, and when it cannot
- Be able to find an unknown term in a constitutive equation, given the others

- **Know how a basic script for physical simulation is structured:**

The best exercise: try to set up a simulation of a simple physical phenomenon you're curious about.

- What drives the simulation forward in time (balance laws!)
- The role of constitutive equations (connect the quantities of the different balance laws)

- **Be able to sum/subtract vectors, and to find unknown vectors from others**

The most important concepts that you must know are signalled by green boxes with a bookmark in the lecture notes Example:

■ **Constitutive relation**

A **constitutive relation**, also called *constitutive equation*, or *closure equation*, or *constitutive property*, is a physical relationship or that is true only under specific conditions: for example: only for specific physical

Also see the *Pensumliste* on Canvas for the sections that can be skipped

Example questions and solutions

*Note: some answers are more verbose than they need to be at the exam.
This is because these solutions are also an occasion to review the study material*

Examples of conceptual or theoretic questions



Tell which of the following quantities are *extensive* and which aren't. For those that are not extensive, give a simple example showing that if we treat that quantity as extensive we would get nonsense results.

- Energy content
- Temperature
- Velocity
- Momentum content
- Entropy content
- Pressure
- Heat flux
- Time lapse
- Force



Tell which of the following quantities are *extensive* and which aren't. For those that are not extensive, give a simple example showing that if we treat that quantity as extensive we would get nonsense results.

- Energy content
- Temperature
- Velocity
- Momentum content
- Entropy content
- Pressure
- Heat flux
- Time lapse
- Force

(The question does **not** say to give examples for the extensive quantities, but some examples are added in this example answer anyway.)

Energy content: **extensive** (over volume). For example, if we divide a volume into two parts and one part has energy content 5J, and the other part has energy content 7J, then the full volume has energy content $(5+7)\text{J} = 12\text{J}$.

Temperature: **not extensive**. Example: suppose we divide a room (or the air in it) in two parts. The air in one part has temperature 25°C , the air in the other part also has temperature 25°C . If temperature were extensive, we should say that the whole room has temperature $(25+25)^\circ\text{C} = 50^\circ\text{C}$. Nonsense!

Velocity: **not extensive**. Example: let's say we're in a car moving at 80 km/h. Obviously the front part of the car is moving at 80 km/h, and also the rear part of the car is moving at 80 km/h. If velocity were extensive, we would have to say that the whole car is moving at $(80+80)\text{km/h} = 160\text{km/h}$. Nonsense!

Momentum content: **extensive** (over volume). For example, let's say a car with weight 1000 kg is moving at 25 m/s. Let's say that the front part has weight 500 kg, and is moving at 25 m/s; its momentum content is $(500 \cdot 25)\text{Ns} = 12500\text{Ns}$. The rear part has the same weight and velocity; its momentum content is also 12500 Ns. Then the momentum content of the whole car is $(12500+12500)\text{Ns} = 25000\text{Ns}$. This indeed agrees with calculating it as (whole mass)·(velocity).

Entropy content: **extensive** (over volume).

Pressure: **not extensive**. Example: Pressure is force/area. Let's say that air is exerting a force of 100000 N on a table with area 2m^2 . The pressure over the table is then $(100000/2)\text{N/m}^2 = 50000\text{N/m}^2$. Now instead let's divide the table surface into two halves of 1m^2 each. The force on one half is 50000 N (because force is extensive), so the pressure there is $(50000/1)\text{N/m}^2$. The pressure on the other half is also 50000N/m^2 . If pressure were extensive, then the total pressure on the table would be $(50000+50000)\text{N/m}^2 = 100000\text{N/m}^2$. This contradicts our initial statement that the pressure was 50000N/m^2 .

Heat flux: **extensive** (over area). Example: let's say that through half of a window there's a heat flux of 700J/s , and through the other half also 700J/s . Then through the whole window there's a heat flux of 1400J/s .

Time lapse: **not extensive**. In fact it is not even clear what "extensive" would mean in this case! Let's say a given control volume exists for a time lapse of 3 s. Then one half exists for 3 s, and the other half also exists for 3 s. We can't say that the whole volume exists for $(3+3)\text{s} = 6\text{s}$!

Force: **extensive**. Force is momentum flux, and momentum flux is extensive over surfaces.

All seven basic physical quantities (matter, momentum, energy, ang. momentum, entropy, charge, magnetic flux) are extensive. Their volume contents and their supplies are extensive over volumes; their fluxes are extensive over areas.

Consider two separate control volumes called A and B, with corresponding closed control surfaces. Call " P_A " the momentum content, " F_A " the influx of momentum, and " G_A " the supply of momentum for control volume A; and analogously for control volume B.

- (1) Write the balance equations of momentum for control volume A and control volume B, in their differential expression.
- (2) Write the balance equation of momentum, in its differential expression, for the two control volumes combined. Call " P " the momentum content, " F " the momentum influx, and " G " the momentum supply of the two control volumes combined. Relate these quantities to those of the individual volumes; explain the reason or physical principle behind this relation.
- (3) If the two volumes were not separate, that is, if they have an overlap, could we then connect P , P_A , P_B with a simple formula?

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$$\frac{dP_A(t)}{dt} = F_A(t) + G_A(t) \qquad \frac{dP_B(t)}{dt} = F_B(t) + G_B(t)$$

(2)

$$\frac{dP(t)}{dt} = F(t) + G(t)$$

and, because of the principle of extensivity (or additivity):

$$P(t) = P_A(t) + P_B(t) \qquad F(t) = F_A(t) + F_B(t) \qquad G(t) = G_A(t) + G_B(t)$$

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- (3) No. The extensivity formulae for momentum

$$P(t) = P_A(t) + P_B(t) \qquad F(t) = F_A(t) + F_B(t) \qquad G(t) = G_A(t) + G_B(t)$$

are only valid for two **disjoint**, that is, non-overlapping (although they may touch, that is, have part of their control surfaces in common).

The same is true for all other extensive quantities.

- 
- i. Explain what is the difference between a universal law and a constitutive relation. Give at least two examples of constitutive relations.

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☞ A **universal** law (like the name says) applies to **all possible** physical phenomena, **without exceptions**

We know seven universal laws. They apply in all our physical theories, even approximate theories

☞ A **constitutive** relation (like the name says) applies **only to particular situations**:

- particular physical phenomena
- particular physical theories

but there are phenomena or theories in which it cannot be applied

☞ Universal laws are disconnected from one another

☞ Constitutive relations **connect** the quantities in the universal laws

☞ Universal laws usually allow us to make predictions about future values of quantities

☞ Constitutive relations usually do not allow us to make predictions about future values of quantities
(but there are exceptions)

Why do we consider Newton's formula $\mathbf{P} = m\mathbf{v}$ (relating momentum, mass, velocity) to be a constitutive relation, rather than a universal law?

Why do we consider Newton's formula $\mathbf{P} = m\mathbf{v}$ (relating momentum, mass, velocity) to be a constitutive relation, rather than a universal law?

☞ Because it is **not universally** valid. It applies **only**:

- to matter
- at speeds much smaller than the speed of light
- in not too strong gravitational fields

What is the difference between a *conservation* law and a *balance* law? Which quantities do you know that are conserved, and which that are balanced but not conserved?

What is the difference between a *conservation* law and a *balance* law? Which quantities do you know that are conserved, and which that are balanced but not conserved?

☞ Conservation law is a particular (but important) case of balance law

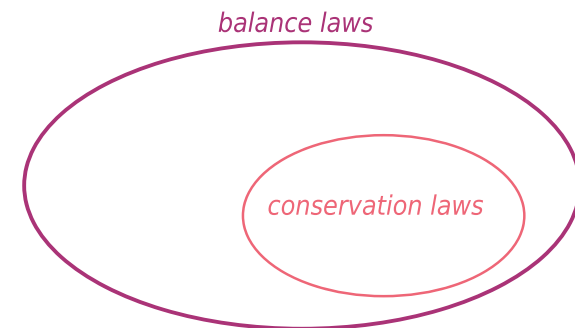
☞ Conservation law is a balance law where the supply is always zero
→ conserved quantity in a volume can only change by coming in-out of surface

☞ *Conserved:*

- matter (all matter together)
- electric charge
- magnetic flux

☞ *Balanced but not conserved:*

- matter (individual kinds)
- momentum
- angular momentum
- energy
- entropy



Take the balance of energy (without supply): $E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) \, dt$

Write it in an approximate form that can be used in a simulation, and briefly explain how you get this approximate form.

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Write it in an approximate form that can be used in a simulation, and briefly explain how you get this approximate form.

☞ We take a time t_1 that is just a short duration after t_0 :

$$t_1 = t_0 + \Delta t$$

The flux in the integral can then be considered approximately constant:

$$\int_{t_0}^{t_1} \Phi(t) dt \approx \Phi(t_0) (t_1 - t_0) = \Phi(t_0) \Delta t$$

So

$$E(t_0 + \Delta t) \approx E(t_0) + \Phi(t_0) \Delta t$$



What does it mean to say that a quantity is **extensive**? (Explain it at least with an example.)
Give two examples of extensive quantities



What does it mean to say that a quantity is **extensive**? (Explain it at least with an example.)
Give two examples of extensive quantities

A quantity – more precisely a **volume content** or a **flux** – is extensive if:

☞ It makes sense to say that a volume *contains* a given amount of that quantity

☞ If we divide the volume into separate parts,
the total amount in the volume is the *sum* of the amounts in the parts.

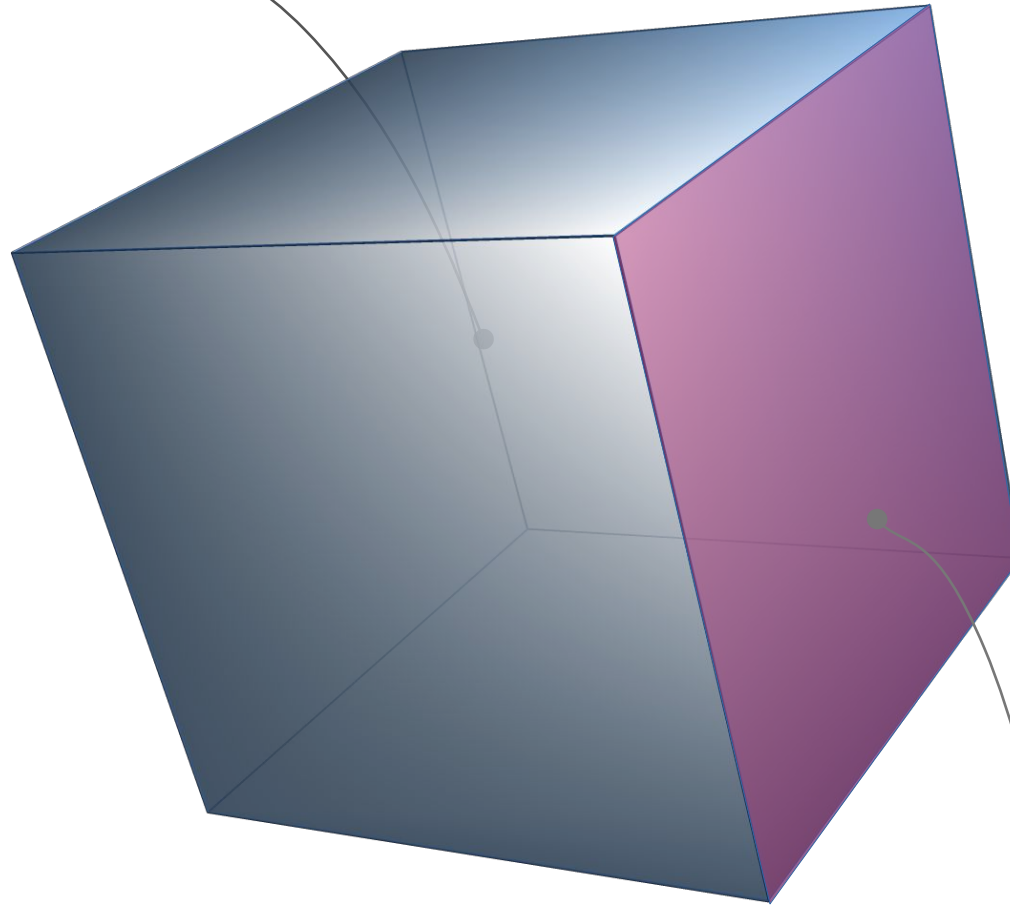
☞ It makes sense to say that through a surface there's a *flux* of that quantity

☞ If we divide the surface into separate parts,
the total flux through the surface is the *sum* of the fluxes through the parts.

[see illustrations on the next pages]

☞ The seven fundamental quantities are all extensive!
(matter, momentum, energy, angular momentum, entropy, electric charge, magnetic flux)

how much {
matter
energy-mass
momentum
angular momentum
entropy
} is in this volume?

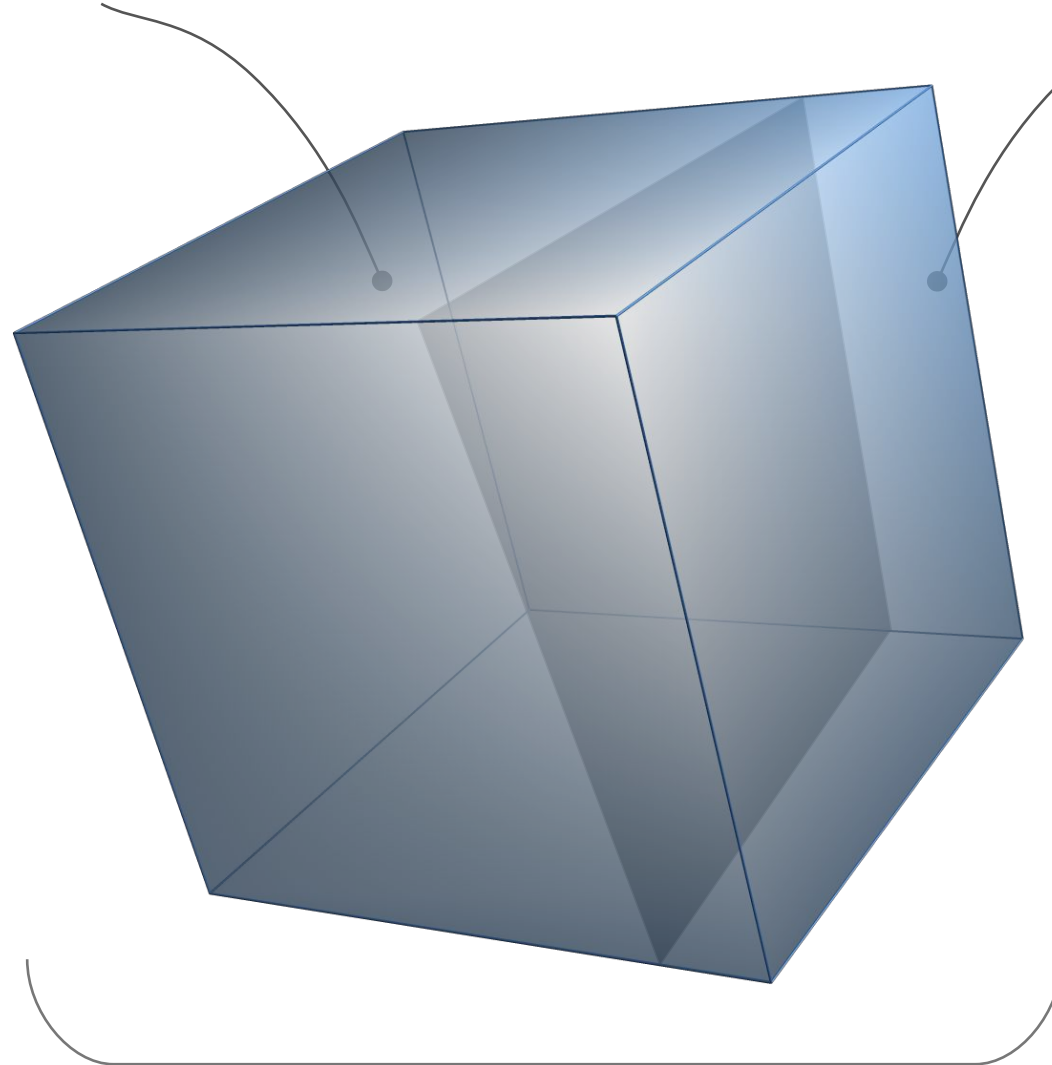


how much {
matter
energy-mass
momentum
angular momentum
entropy
} is flowing through this surface per unit time?

Example of property 3.

5 J of energy in this subvolume

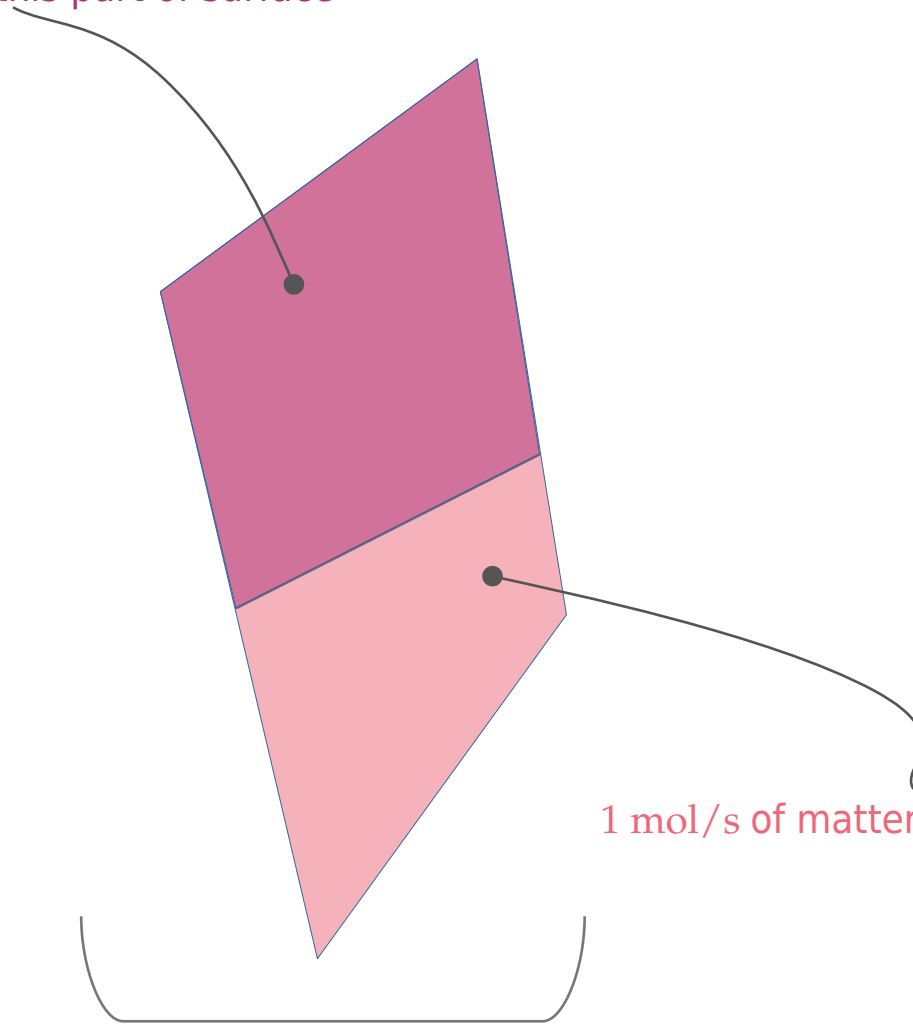
-2 J of energy in this subvolume



$5\text{ J} - 2\text{ J} = 3\text{ J}$ of energy in the whole volume

Example of property 3.

3 mol/s of matter through this part of surface



1 mol/s of matter through this part of surface

$3 \text{ mol/s} + 1 \text{ mol/s} = 4 \text{ mol/s}$ of matter through whole surface



Explain the principle of the *symmetry of flux* in the case of the flux of a vector quantity. You can explain it in abstract and general terms, or by means of one or more examples.

Explain the principle of the *symmetry of flux* in the case of the flux of a vector quantity. You can explain it in abstract and general terms, or by means of one or more examples.

Reminder: The notion of flux is related to surfaces. The flux through a particular surface is the *net* amount of a quantity (like energy or momentum or substance etc) that crosses that surface in the unit of time. In order to make sense of this “crossing”, we must specify the crossing direction of the surface, otherwise it is unclear on which side of the quantity is increasing or decreasing.

The principle of **symmetry of flux** is connected to this crossing direction. It says that **the flux in one crossing direction is always minus the flux in the opposite crossing direction**. This makes intuitive sense if we think of a couple of non-rigorous illustrative examples: If we say that **7** people **exit** through a door in one minute, it also means that **-7** people **enter** that door in one minute.

One example **vector** quantity for which the notion of flux makes sense is **momentum** (another one is angular momentum). The flux of momentum is also called “force”. The principle of symmetry of flux is exactly the same in this case: if the flux in one crossing direction through a surface is

$[F_x, F_y, F_z]$

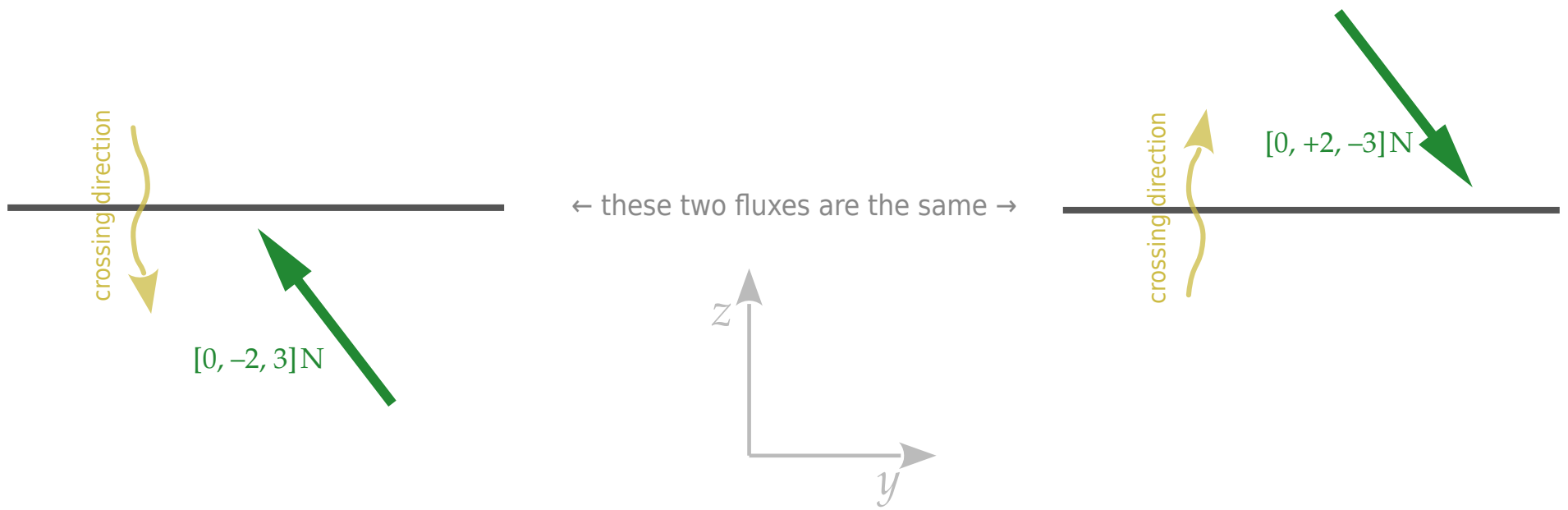
then the flux in the opposite crossing direction is

$-[F_x, F_y, F_z]$

or equivalently

$[-F_x, -F_y, -F_z]$

For example, if we have a horizontal surface and the flux of momentum from top to bottom is $[0, -2, 3]\text{N}$, then the flux of momentum from bottom to top is $[0, +2, -3]\text{N}$





Is **temperature** an extensive quantity?

Explain why or why not, for instance with an example



Is **temperature** an extensive quantity?
Explain why or why not, for instance with an example

No, it isn't!

Imagine that half of a room is at a temperature of $20\text{ }^{\circ}\text{C}$, and the other half also at $20\text{ }^{\circ}\text{C}$.
If temperature were extensive, we should say that the whole room is at a temperature of $40\text{ }^{\circ}\text{C}$!
(Or even more: imagine to divide the temperature in four parts...)

?

Consider a Hookean spring attached to two small bodies a and b , as in the side figure. Let's assume there's no gravity. The force (momentum flux) exerted by the spring on body a is equal and opposite to the force exerted by the spring on body b : $\mathbf{F}_a = -\mathbf{F}_b$

Is this an example of the *principle of symmetry of flux*? or an example of *balance of momentum*? Explain why or why not. (Note: a Hookean spring has no mass.)





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Is this an example of the *principle of symmetry of flux*? or an example of *balance of momentum*? Explain why or why not. (Note: a Hookean spring has no mass.)

No, it isn't an example of the principle of symmetry of flux. This principle only applies to the flux through **one given surface**, not to the fluxes through two or more different surfaces.

In the case of this spring, the force F_a is the flux of momentum through the small imaginary surface separating the spring and body a , while the force F_b is the flux of momentum through the small imaginary surface separating the spring and body b . These are two different surfaces, it doesn't make sense to invoke the symmetry of flux for their two fluxes!

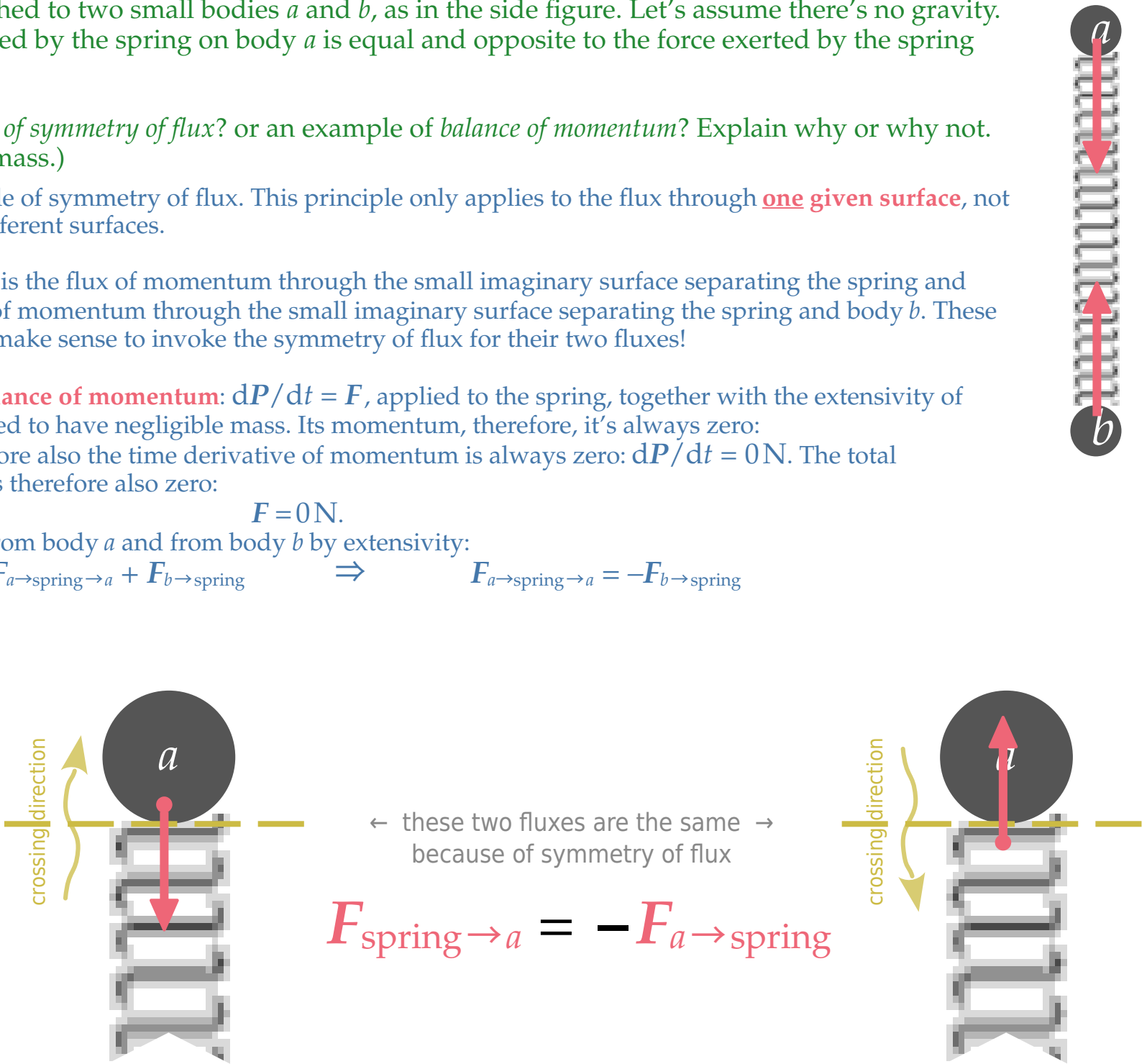
Instead, this is a consequence of the **balance of momentum**: $dP/dt = F$, applied to the spring, together with the extensivity of fluxes. A Hookean spring is considered to have negligible mass. Its momentum, therefore, it's always zero: $P = m_{\text{spring}} v_{\text{spring}} \approx 0 \text{ N}\cdot\text{s}$, and therefore also the time derivative of momentum is always zero: $dP/dt = 0 \text{ N}$. The total momentum influx F into the spring is therefore also zero:

$$F = 0 \text{ N}.$$

This total momentum influx comes from body a and from body b by extensivity:

$$0 = F = F_{a \rightarrow \text{spring} \rightarrow a} + F_{b \rightarrow \text{spring}} \Rightarrow F_{a \rightarrow \text{spring} \rightarrow a} = -F_{b \rightarrow \text{spring}}$$

Additional note: The principle of symmetry of flux applies instead if we focus on the **imaginary surface** between the spring and only one of the bodies, say a , as in the side figure. In this case the principle says that the flux of momentum **from spring to body a** through this surface (that is, the force exerted from the spring to the body a) is *equal and opposite* to the momentum flux **from body a to spring** (that is, the force exerted from body a to the spring). Clearly the body b and any fluxes to/from it are not involved.



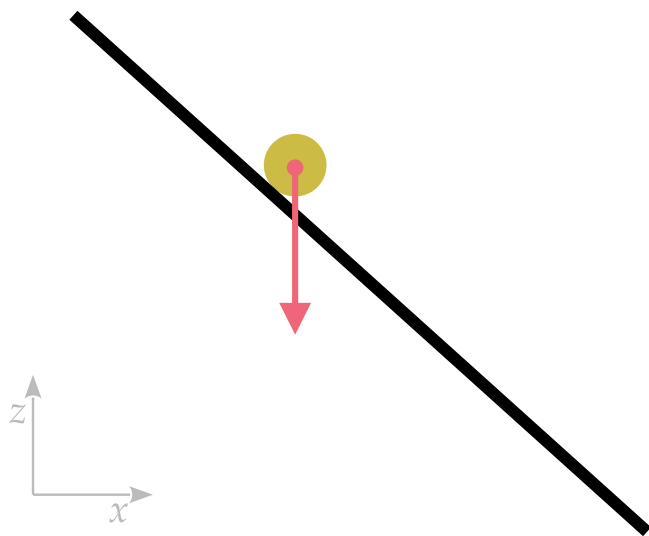
The following prompt was given to ChatGPT-5.1:

We have the following physical situation at a given time instant t .

A small solid object of constant mass 1 kg is in contact with an inclined plane at this time instant. The plane forms an angle of $\pi/4$ rad with the vertical, inclined from bottom-right to top-left. There cannot be friction between the object and the inclined plane. We use a coordinate system (x, z) fixed with the ground, with z upwards. The gravitational acceleration is 9.1 N/kg .

We know that the object has zero velocity (both components) at this time instant. Is it possible to calculate its acceleration (both components) at this time instant? Answer first with a simple "yes" or "no", and explain afterwards.

ChatGPT's output is shown on the side. Is ChatGPT correct?



Yes.

We assume the only forces acting are:

- Gravity: $\vec{W} = (0, -mg)$ with $m = 1 \text{ kg}$ and $g = 9.1 \text{ m/s}^2$
- The normal reaction from the plane, perpendicular to the plane
- No friction (so no tangential contact force)

The plane makes an angle $\pi/4$ with the **vertical**, so it also makes an angle $\pi/4$ with the **horizontal** (since $\pi/2 - \pi/4 = \pi/4$).

Take:

- x horizontal, positive to the right
- z vertical, positive upwards

The plane is inclined from bottom-right to top-left, so the **down-the-plane** direction (the direction in which it will accelerate) is from top-left to bottom-right. That direction has unit vector

$$\hat{t} = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right).$$

For a frictionless plane, the acceleration is simply the component of gravity **along** the plane:

$$a_{\parallel} = g \sin\left(\frac{\pi}{4}\right) = 9.1 \cdot \frac{\sqrt{2}}{2} = \frac{9.1}{\sqrt{2}} \approx 6.4 \text{ m/s}^2.$$

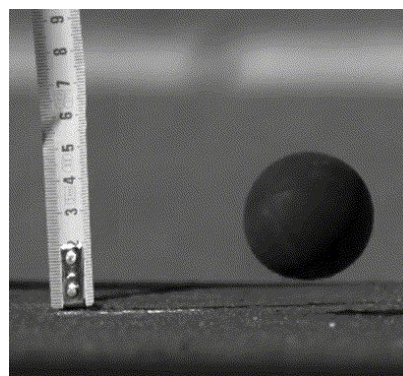
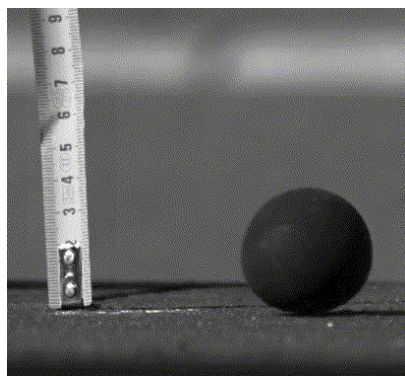
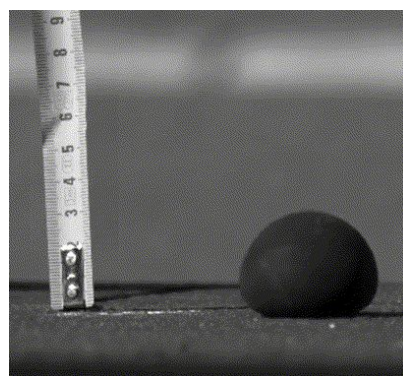
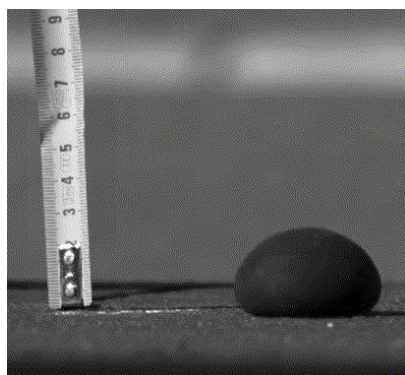
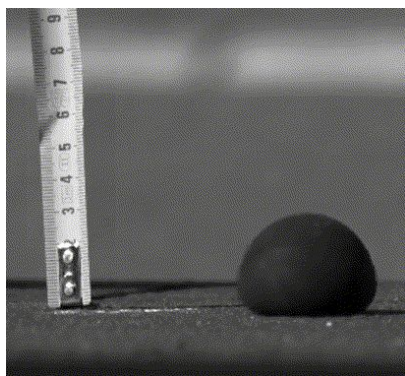
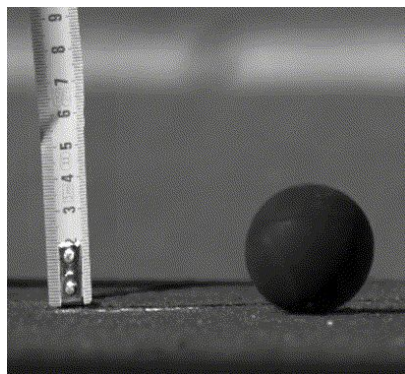
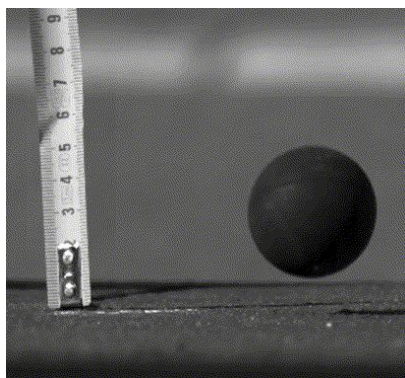
So the acceleration vector is

$$\vec{a} = a_{\parallel} \hat{t} = \frac{9.1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right) = \left(\frac{9.1}{2}, -\frac{9.1}{2} \right) \text{ m/s}^2.$$

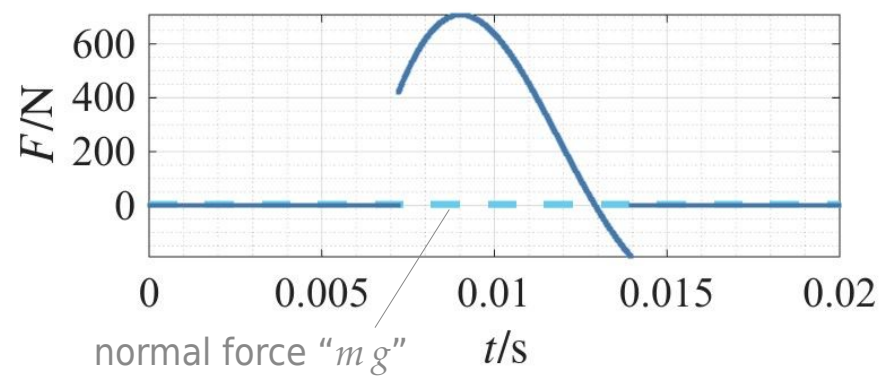
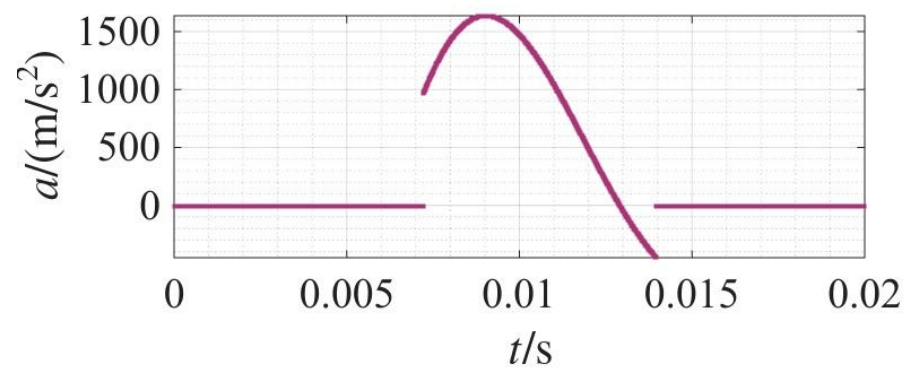
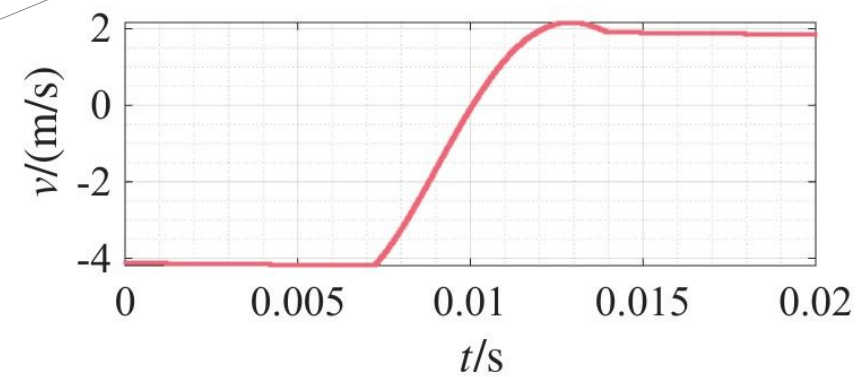
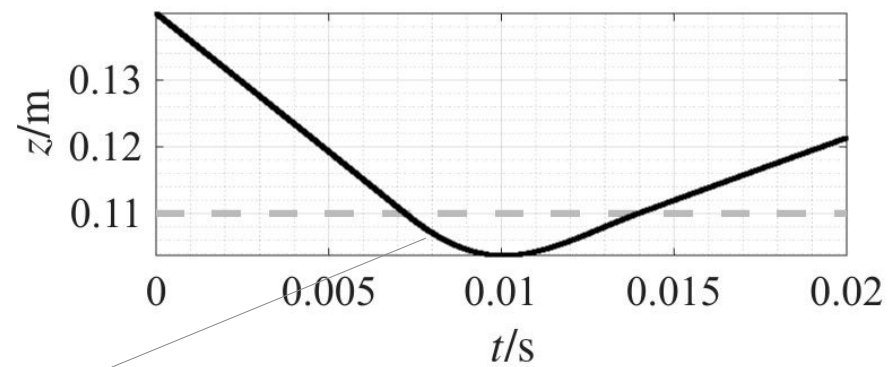
Numerically,

- $a_x = +4.55 \text{ m/s}^2$ (to the right)
- $a_z = -4.55 \text{ m/s}^2$ (downward)

The fact that the velocity is zero at that instant does **not** prevent the object from having a nonzero acceleration; the acceleration is determined by the forces, which we can compute from the given information.



compression



Exclusive: Meta planning sweeping layoffs as AI costs mount

By Katie Paul, Jeff Horwitz and Deepa Seetharaman

March 14, 2026 1:17 AM GMT+1 · Updated 20 hours ago



People pose for a photo in front of a sign of Meta, the new name for the company formerly known as Facebook, at its headquarters in Menlo Park, California, U.S. October 28, 2021. REUTERS/Nathan Frandino/File Photo [Purchase Licensing Rights](#)

Summary Companies

- Meta's workforce could shrink by 20% amid AI focus
- Meta's AI spending includes \$600 billion for data centers by 2028
- AI expected to create efficiencies

Is ChatGPT's output (on the side) correct?

Prompt:

An engineer has measured the forces applied on a block of a very light material at the points (small surfaces)

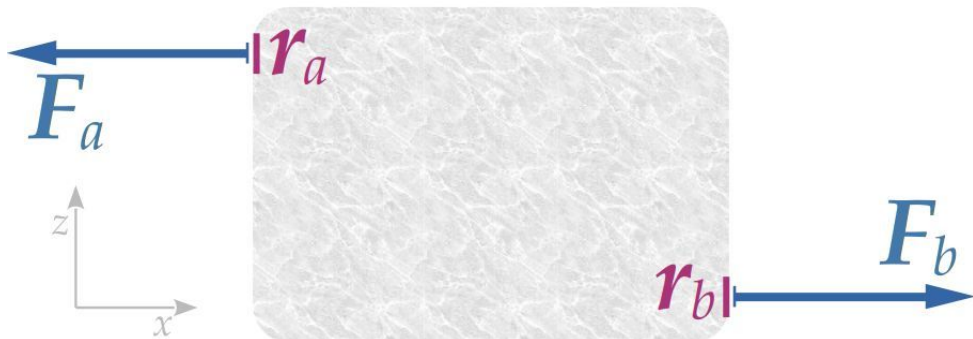
$$\mathbf{r}_a = [-0.2, 0, 0.1] \text{ m} \quad \text{and} \quad \mathbf{r}_b = [0.2, 0, -0.1] \text{ m}$$

and found that these forces are equal to

$$\mathbf{F}_a = [-3, 0, 0] \text{ N} \quad \text{and} \quad \mathbf{F}_b = [3, 0, 0] \text{ N}.$$

Since the material is very light, the engineer decides to model its physical simulation as if it had zero mass, and use the usual constitutive relations for matter.

Is the engineer's decision reasonable?



No, the decision is not reasonable.

Compute the resultant force and moment of the two applied forces.

Given

- Positions (in m):

$$\mathbf{r}_a = [-0.2, 0, 0.1],$$

$$\mathbf{r}_b = [0.2, 0, -0.1]$$

- Forces (in N):

$$\mathbf{F}_a = [-3, 0, 0],$$

$$\mathbf{F}_b = [3, 0, 0].$$

Net force

$$\mathbf{F}_{\text{res}} = \mathbf{F}_a + \mathbf{F}_b = [-3, 0, 0] + [3, 0, 0] = [0, 0, 0].$$

So there is no resultant force, which would be compatible with a body of negligible mass not translating.

Net moment about the origin

$$\mathbf{M}_a = \mathbf{r}_a \times \mathbf{F}_a, \quad \mathbf{M}_b = \mathbf{r}_b \times \mathbf{F}_b.$$

Compute:

$$\mathbf{M}_a = [-0.2, 0, 0.1] \times [-3, 0, 0] = [0, -0.3, 0], \quad \mathbf{M}_b = [0.2, 0, -0.1] \times [3, 0, 0] = [0, -0.3, 0]$$

Thus

$$\mathbf{M}_{\text{res}} = \mathbf{M}_a + \mathbf{M}_b = [0, -0.6, 0] \neq 0.$$

So the two forces form a non-zero couple.

For a material point with mass m , a couple produces angular acceleration governed by the rotational equations of motion (moments of inertia, etc.). But if the body is modeled as having **zero mass** (and thus zero mass moment of inertia), the angular momentum balance implies that the net moment must be zero; otherwise finite moments would cause infinite angular accelerations. This contradicts the usual continuum mechanics relations, which assume finite inertia.

Therefore a massless body cannot support this non-zero couple while obeying the standard constitutive and balance laws. The engineer's zero-mass model is not physically consistent for this loading.



An engineer has measured the forces applied on a block of a very light material at the points (small surfaces)

$$\mathbf{r}_a = [-0.2, 0, 0.1] \text{ N} \quad \text{and} \quad \mathbf{r}_b = [0.2, 0, -0.1] \text{ m}$$

and found that these forces are equal to

$$\mathbf{F}_a = [-3, 0, 0] \text{ N} \quad \text{and} \quad \mathbf{F}_b = [3, 0, 0] \text{ N} .$$

Since the material is very light, the engineer decides to model its physical simulation as if it had zero mass, and use the usual constitutive relations for matter.

Is the engineer's decision reasonable?





The block of material in the figure is assumed to have zero mass and obey the usual constitutive relations for matter.

Two surface forces

$$\mathbf{F}_a = [-3, 0, 0] \text{ N} \quad \text{and} \quad \mathbf{F}_b = [3, 0, 0] \text{ N}$$

are applied at the points (small surfaces)

$$\mathbf{r}_a = [-0.2, 0, 0.1] \text{ m} \quad \text{and} \quad \mathbf{r}_b = [0.2, 0, -0.1] \text{ m} .$$

Is this configuration of forces physically possible, under the stated assumptions?





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- ☞ The choice is made in the most convenient way to solve a physical problem

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- ☞ Because we are completely free to choose their position, size, shape, and movement since they are imaginary
- ☞ The choice is made in the most convenient way to solve a physical problem
- ☞ Because they allow us to keep control of:
 - the amount of a quantity contained in a volume
 - the amount of a quantity flowing through a surface

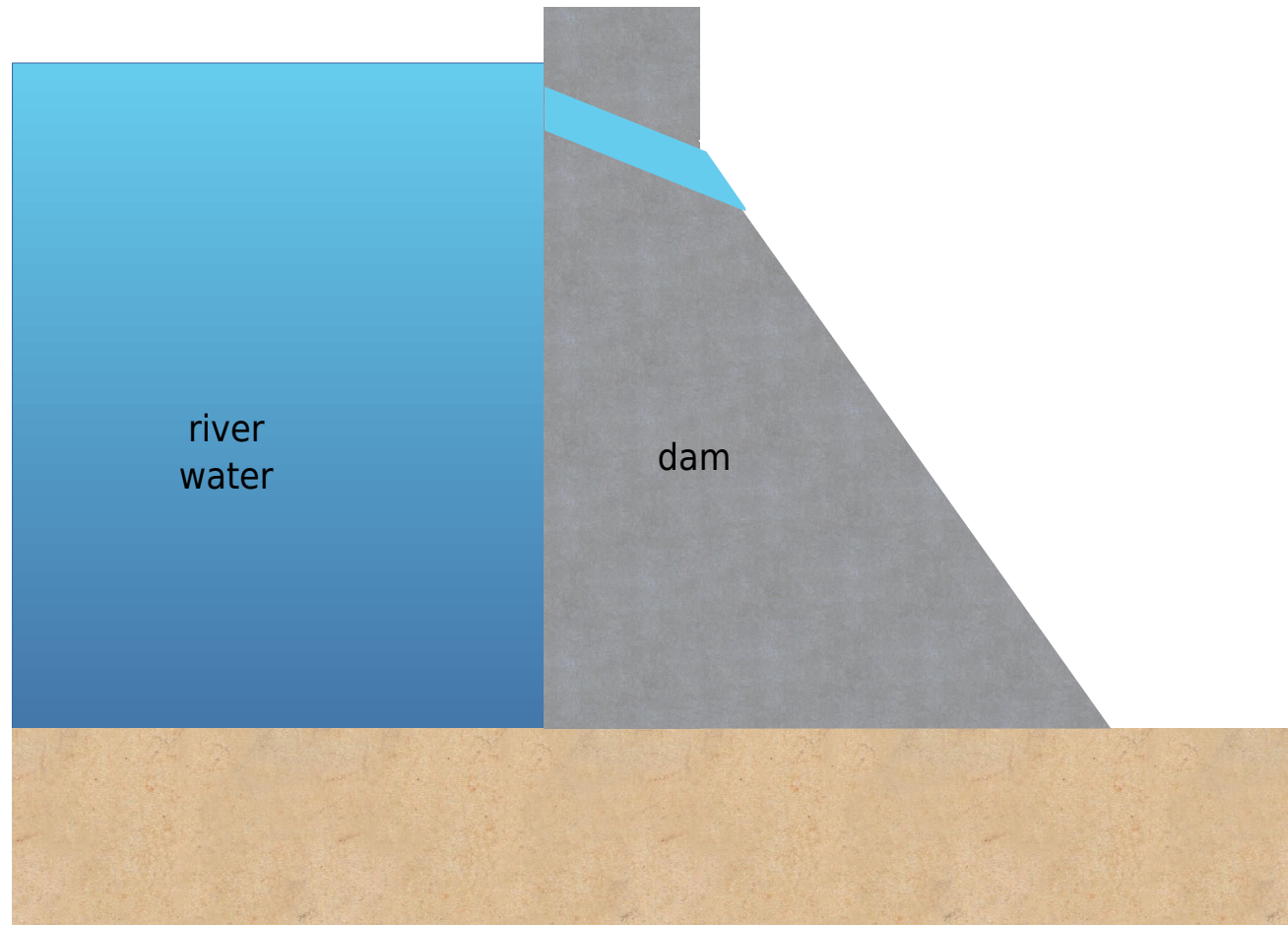


This is the cross-section of a dam. An engineer is interested in:

- The force (momentum flux) that the water exerts on the dam.
- How much water flows out of the river through the dam.

What kind of control surfaces or volumes would you use to solve this problem?
Explain your choices.

(You cannot solve this problem, there are not enough data. Just set up the control surfaces that in your opinion it would make sense to use)





The balance of entropy is peculiar because it has an inequality ' $\leq$ ' sign, rather than an equality '=' like the other balances.

Discuss some consequences or reasons of this peculiarity

The balance of entropy is peculiar because it has an inequality ' $\leq$ ' sign, rather than an equality '=' like the other balances.

Discuss some consequences or reasons of this peculiarity

- ☞ One consequence is that it cannot be used to predict future values

- ☞ Therefore it cannot be used to “drive forward” a simulation

- ☞ It is a physical ‘meta-law’,
that is, a physical law that says which constitutive relations are or are not allowed

For example, it is because of this law that the coefficient of heat transfer, the coefficient of viscosity, or the coefficient of kinetic friction *must* be positive



A rectangular box of lateral area A and horizontal length h contains an amount of gas.

- The gas is in complete equilibrium, at rest.
- Total amount of gas is N , total energy content E , internal pressure p ; all constant in time.
- There is no heat flux through any control surface within or around the box.

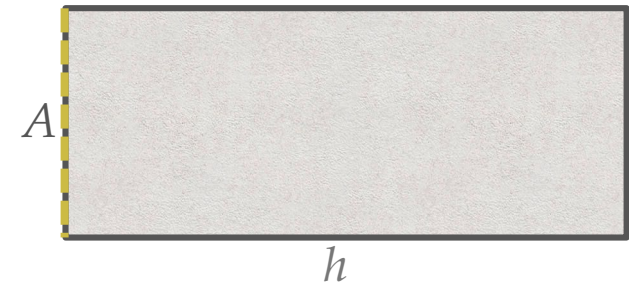
Consider an imaginary vertical control surface at the left side of the box (in yellow in the side picture).

I asked ChatGPT 4 how much is the flux of horizontal momentum F through this imaginary control surface, in the left-to-right crossing direction.

ChatGPT answered:

“Since the gas is at rest and not moving, there is no transfer of momentum across the control surface.”

ChatGPT’s answer is wrong. Explain why. (You don’t need to find the correct numerical answer.)



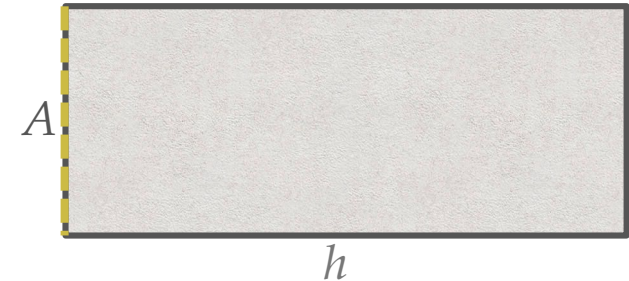


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ChatGPT’s answer is wrong. Explain why. (You don’t need to find the correct numerical answer.)

We know that the box wall exerts a pressure force on the gas, that is, there is a flux of horizontal momentum from the wall to the gas. This momentum has left-to-right orientation, and its magnitude is the pressure times the area, $F = pA$.

Vice versa, there is also a flux of horizontal momentum from the gas to the wall (the gas exerts a pressure on its container). This momentum has right-to-left orientation. This is consistent with the symmetry of fluxes.

Examples of numerics-based questions

A strategy to solve some kinds of physics problems

This is not a universal strategy, but it can be useful in some cases

NB: These steps are only roughly chronological. It may happen that you need to look at a later step, or go back to a previous one.

- 1 Choose one or more control volumes & surfaces that seem relevant to the problem.
- 2 Identify which balance laws should be applied to the chosen control volumes.
→ At the exam: explain shortly why these balance laws should be used.
- 3 Identify which constitutive relations should be applied to the chosen control volumes.
→ At the exam: explain shortly why these constitutive relations should be used.
Remember in which conditions the constitutive relations can be used or not used
- 4 If needed, apply the principles of *extensivity* and *symmetry of flux* to connect sub-volumes.
- 5 List additional info given in the problem, expressing it mathematically.
- 6 Put all equations from the steps above together in a system.

– At this point, the “physics part” is done –

- 7 Identify the unknowns requested by the problem, and solve the equations for them.

This is often just a linear or second-degree system of equations.

This is essentially a maths problem, even if physical intuition can guide to its solution.

Exam tips

- ☞ Use as few control volumes as possible.
- ☞ Use the largest control volume for which all forces explicitly given in the problem are *external*.

❓ How many balance laws do I need?

- ☞ Typically you need *as many as the unknowns* in the problem.

3D: 7 possible balances

$$P_x(t_1) = P_x(t_0) + \int_{t_0}^{t_1} F_x(t) dt + \int_{t_0}^{t_1} G_x(t) dt$$

$$P_y(t_1) = P_y(t_0) + \int_{t_0}^{t_1} F_y(t) dt + \int_{t_0}^{t_1} G_y(t) dt$$

$$P_z(t_1) = P_z(t_0) + \int_{t_0}^{t_1} F_z(t) dt + \int_{t_0}^{t_1} G_z(t) dt$$

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt$$

$$L_x(t_1) = L_x(t_0) + \int_{t_0}^{t_1} M_x(t) dt + \int_{t_0}^{t_1} \mathcal{T}_x(t) dt$$

$$L_y(t_1) = L_y(t_0) + \int_{t_0}^{t_1} M_y(t) dt + \int_{t_0}^{t_1} \mathcal{T}_y(t) dt$$

$$L_z(t_1) = L_z(t_0) + \int_{t_0}^{t_1} M_z(t) dt + \int_{t_0}^{t_1} \mathcal{T}_z(t) dt$$

2D (x, z): 4 possible balances

$$P_x(t_1) = P_x(t_0) + \int_{t_0}^{t_1} F_x(t) dt + \int_{t_0}^{t_1} G_x(t) dt$$

$$P_z(t_1) = P_z(t_0) + \int_{t_0}^{t_1} F_z(t) dt + \int_{t_0}^{t_1} G_z(t) dt$$

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt$$

$$L_y(t_1) = L_y(t_0) + \int_{t_0}^{t_1} M_y(t) dt + \int_{t_0}^{t_1} \mathcal{T}_y(t) dt$$

❓ Which balances do I need to use?

- ☞ Look for hints:

- motion or static problem → *momentum balance*
- mention of “energy”, “heat”, “power”, “work” → *energy balance*
- mention of rotation, rigid body, torques → *angular-momentum balance*
- the *application points* of the forces are explicitly given → *angular-momentum balance*

Known:

- Masses m_1, m_2 0.2 kg
- F from air is zero.

• Constit. relation for gravitational force:

$$\mathbf{G} = -m \mathbf{g} [0, 0, 1]$$

with $\mathbf{g} = 9.8 \text{ N/kg}$

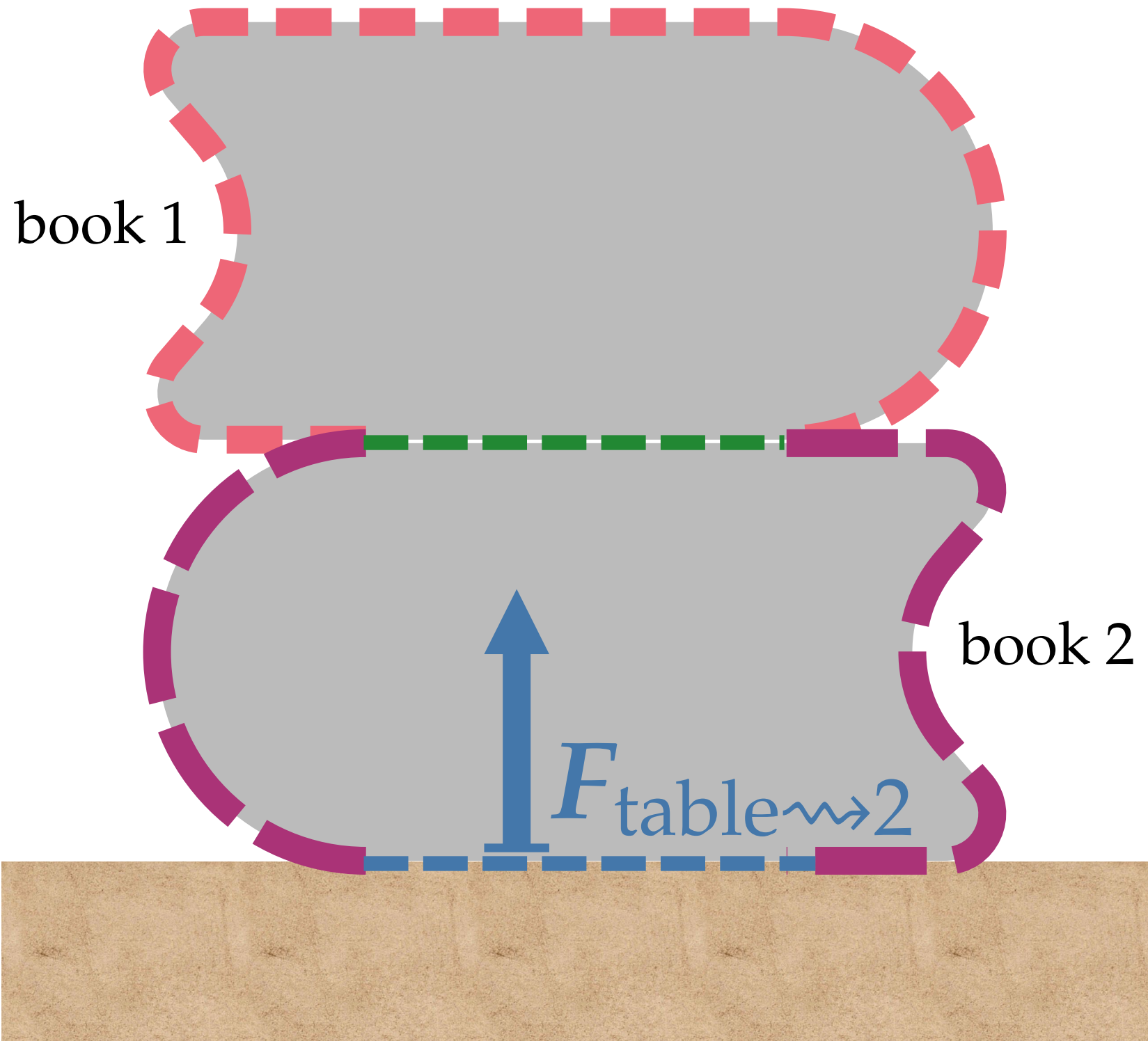
book 1

book 2

$F_{\text{table} \rightarrow 2}$



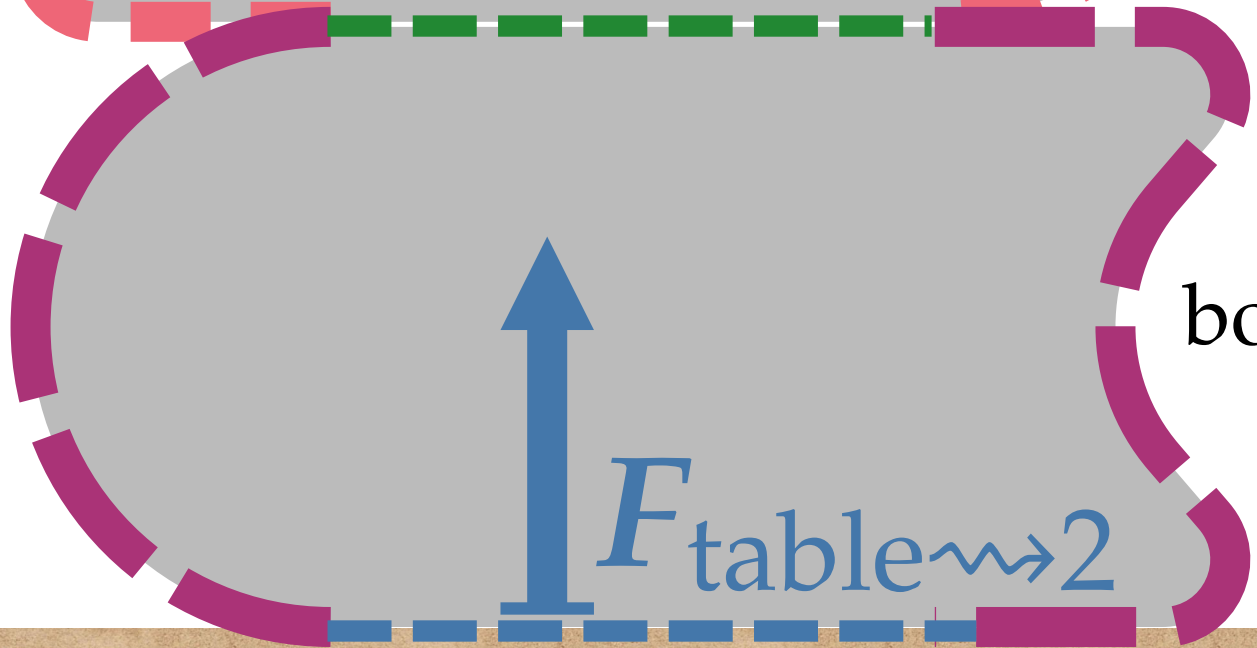
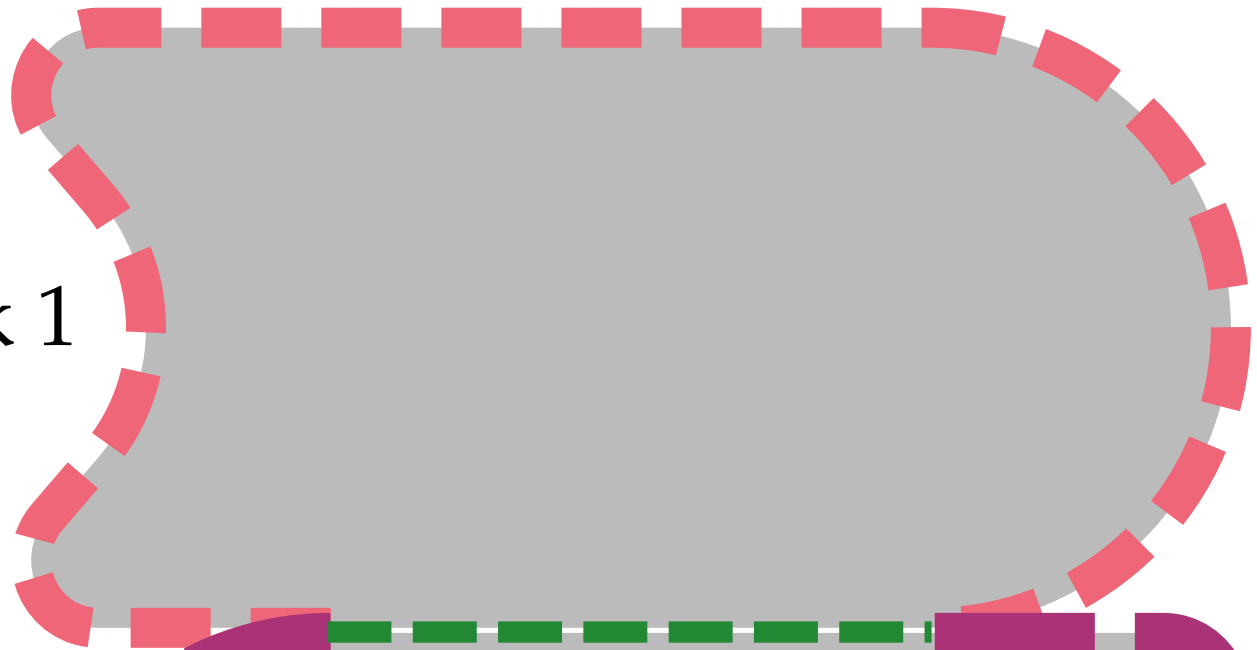
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2 Identify which balance laws should be applied to the chosen control volumes.

$$\frac{d\mathbf{P}_1}{dt} = \mathbf{F}_1 + \mathbf{G}_1$$
$$\frac{d\mathbf{P}_2}{dt} = \mathbf{F}_2 + \mathbf{G}_2$$

book 1



book 2

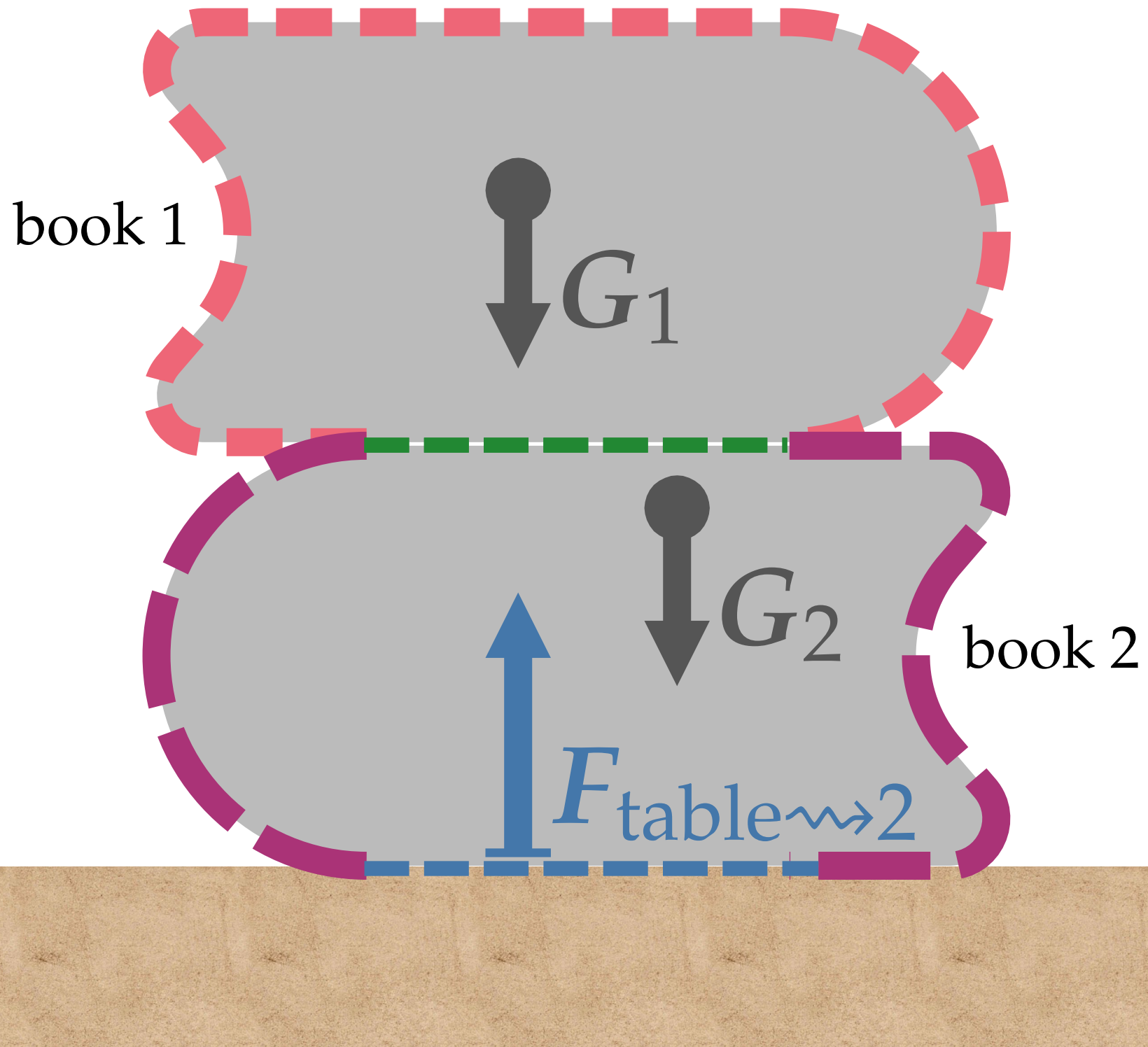
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$$\frac{d\mathbf{P}_1}{dt} = \mathbf{F}_1 + \mathbf{G}_1$$
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$$\mathbf{G}_1 = m_1 \mathbf{g} \approx -[0, 0, 2.0] \text{ N}$$

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4 If needed, apply the principles of *extensivity* and *symmetry of flux* to connect sub-volumes.

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book 1

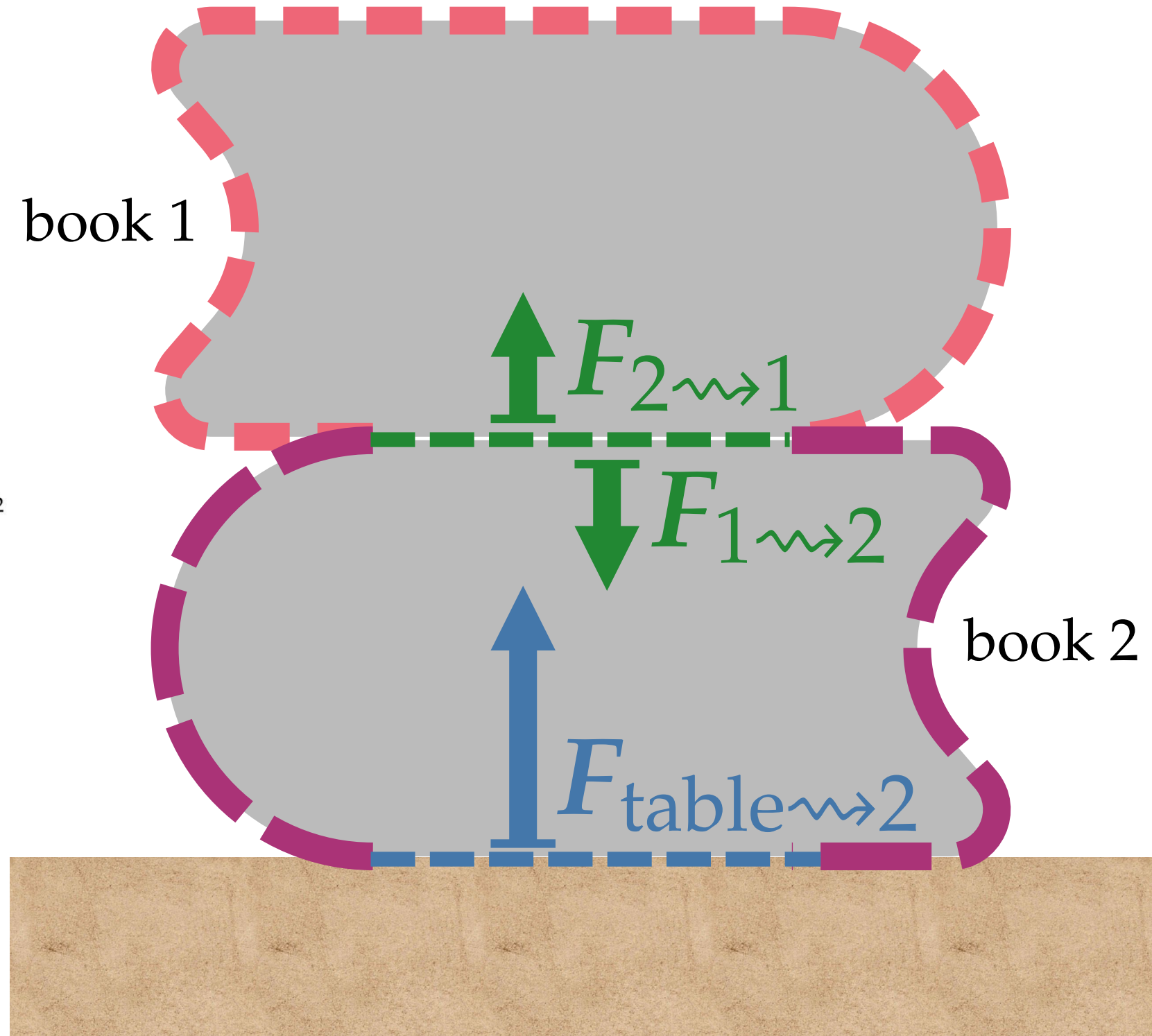
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$$\mathbf{F}_{2 \rightsquigarrow 1} = -\mathbf{F}_{1 \rightsquigarrow 2}$$



5 List additional info given in the problem, expressing it mathematically.

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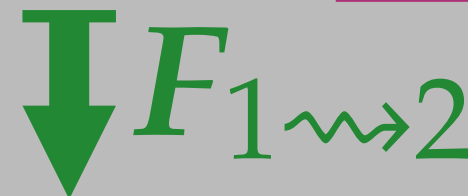
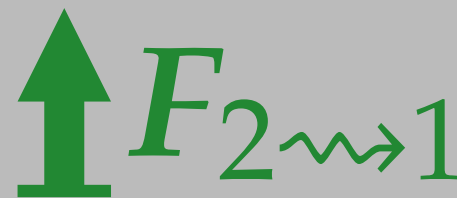
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$$\mathbf{F}_{\text{air} \rightsquigarrow 1} \approx \mathbf{0}\text{ N}$$

$$\mathbf{F}_{\text{air} \rightsquigarrow 2} \approx \mathbf{0}\text{ N}$$

$$\mathbf{F}_{\text{table} \rightsquigarrow 2} \approx [0, 0, 4.0]\text{ N}$$

book 1



book 2



6 Put all equations from the steps above together in a system.

$$\frac{d\mathbf{P}_1}{dt} = \mathbf{F}_1 + \mathbf{G}_1$$

$$\frac{d\mathbf{P}_2}{dt} = \mathbf{F}_2 + \mathbf{G}_2$$

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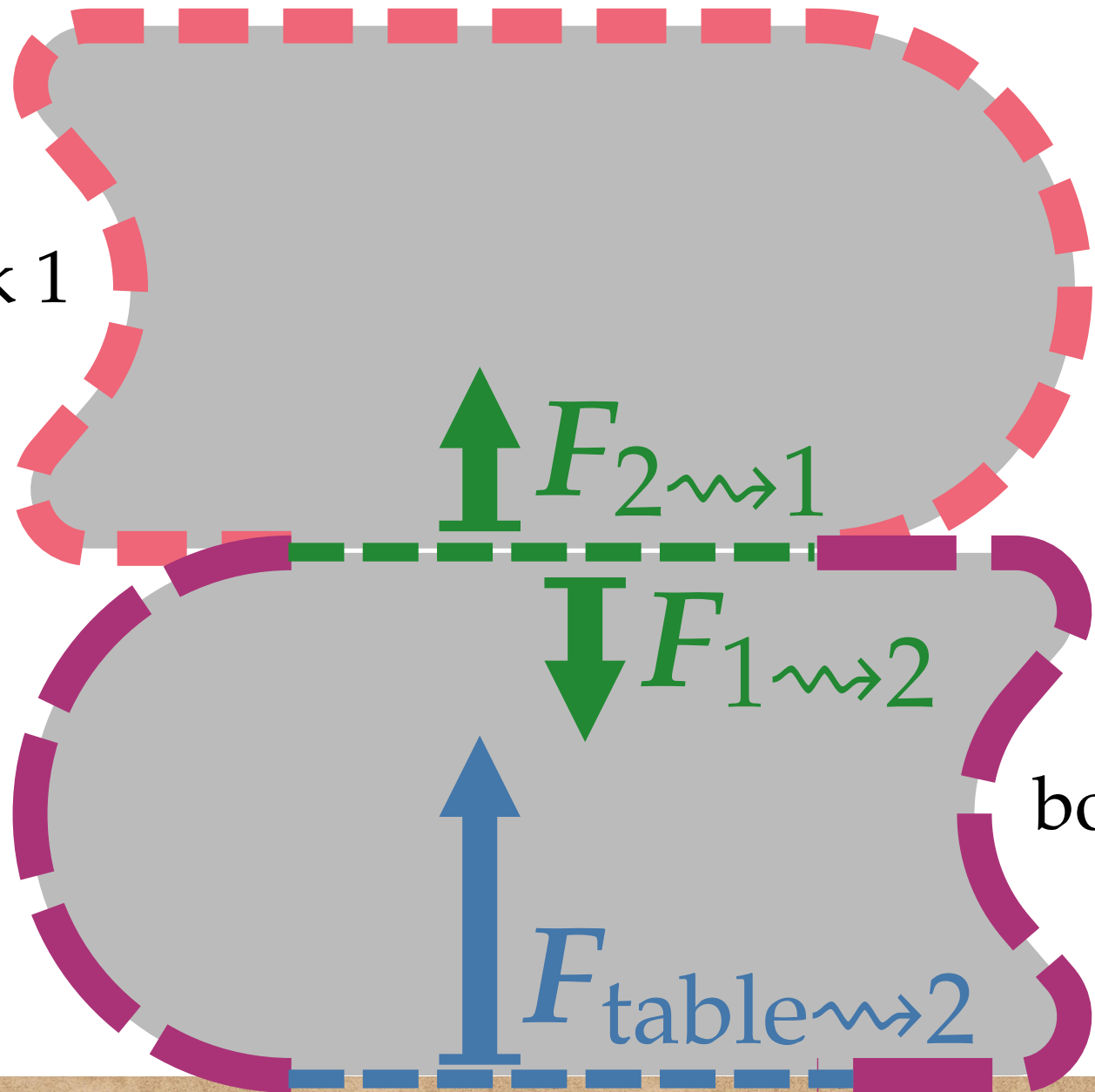
$$\mathbf{F}_2 = \mathbf{F}_{1 \rightsquigarrow 2} + \mathbf{F}_{\text{table} \rightsquigarrow 2} + \mathbf{F}_{\text{air} \rightsquigarrow 2}$$

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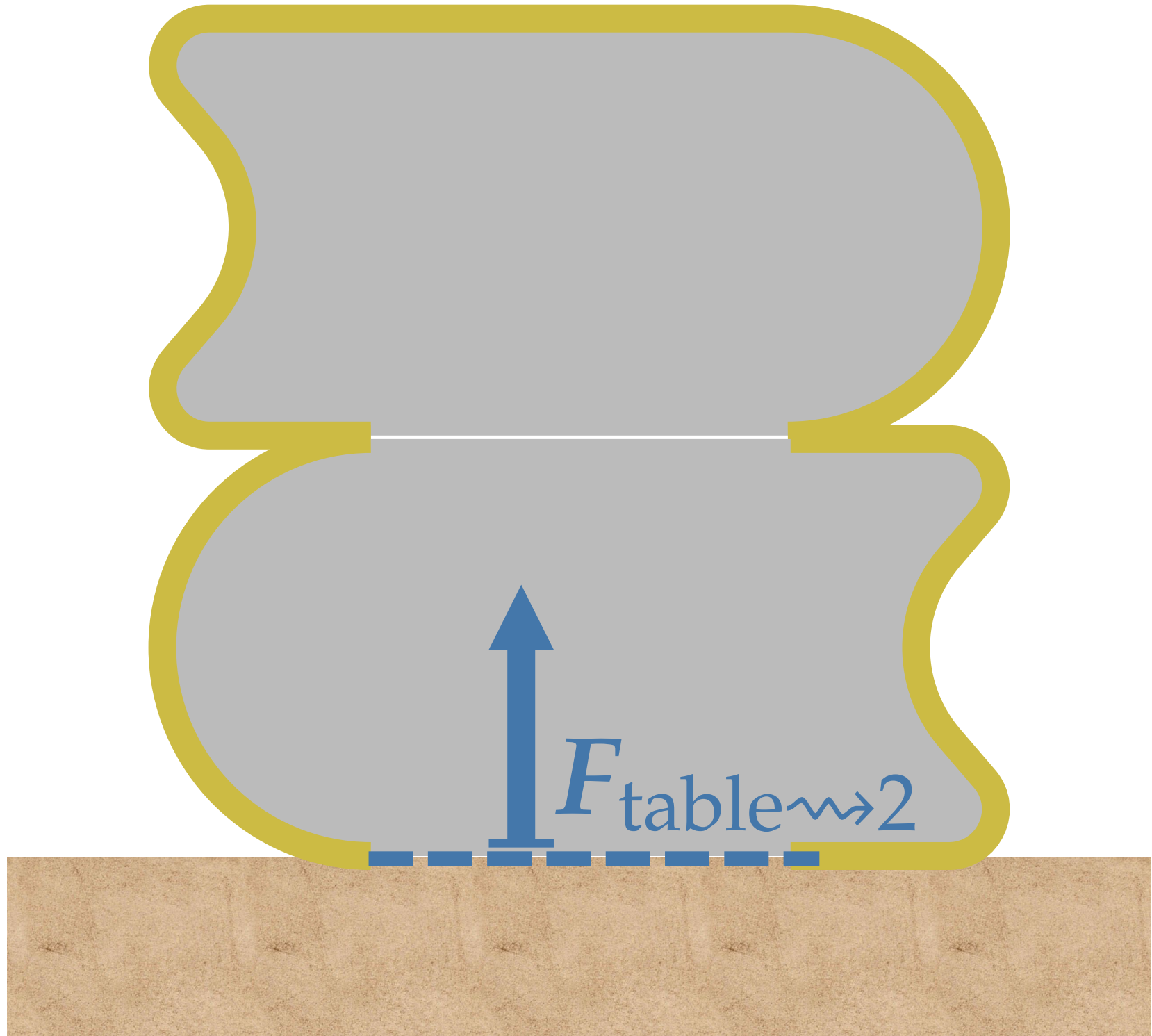
$$\mathbf{F}_{\text{air} \rightsquigarrow 2} \approx \mathbf{0}\text{N}$$

book 1



book 2

1 Choose one or more control volumes & surfaces that seem relevant to the problem.



2 Identify which balance laws should be applied to the chosen control volumes.

$$\frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G}$$

book 1

book 2

$F_{\text{table} \rightarrow 2}$

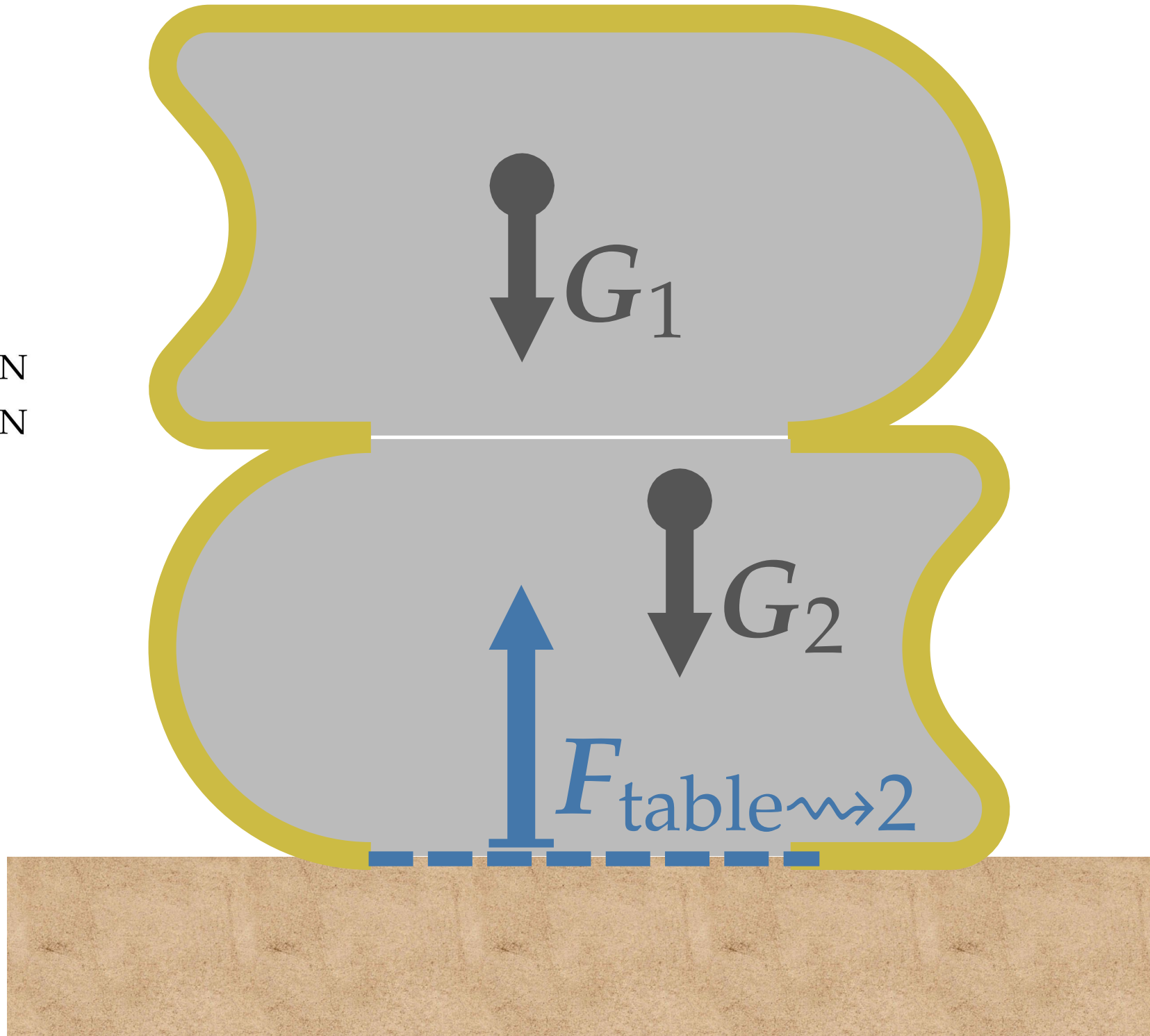


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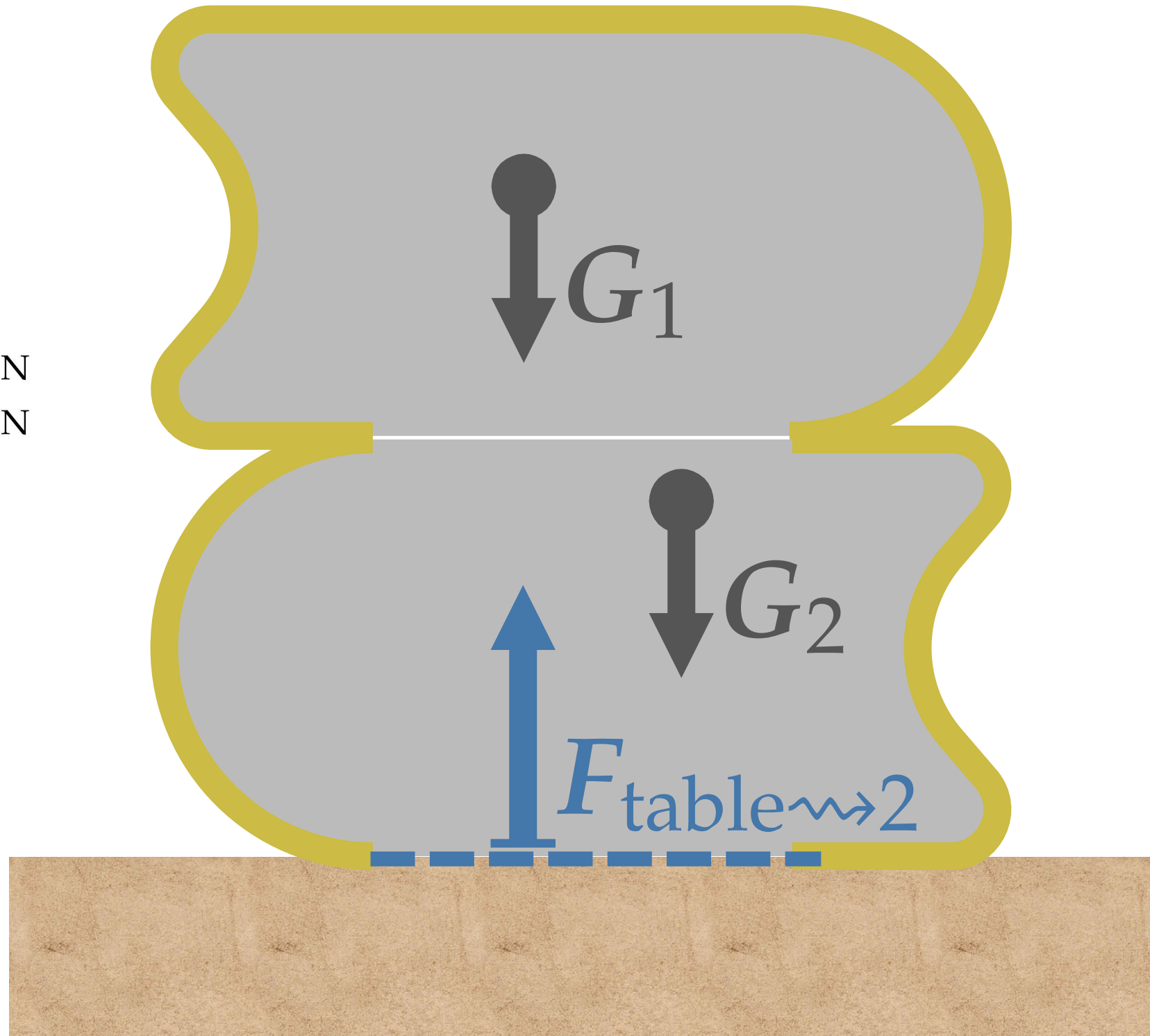
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$$\frac{d\mathbf{P}}{dt} = \mathbf{F} + \mathbf{G}$$

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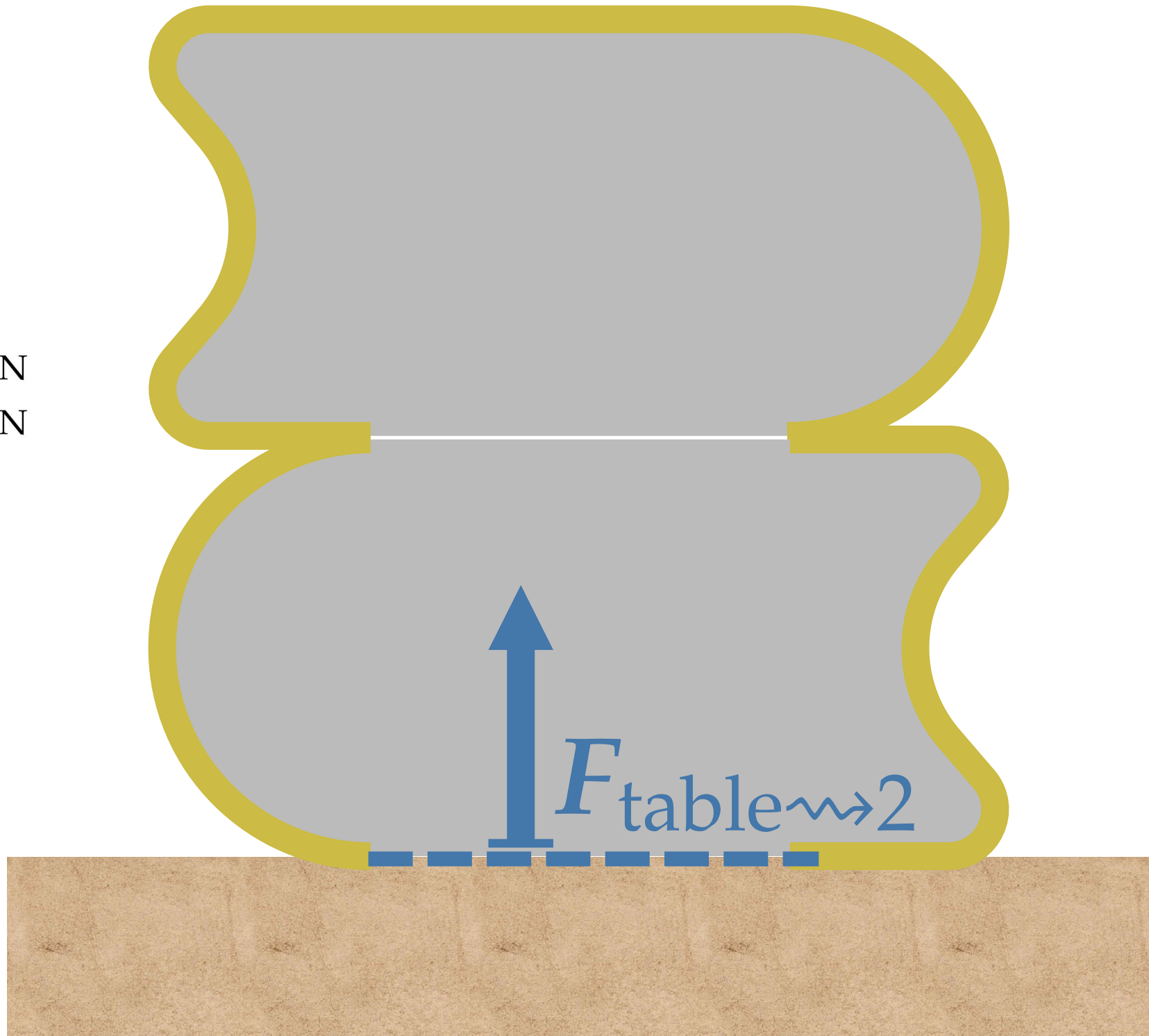
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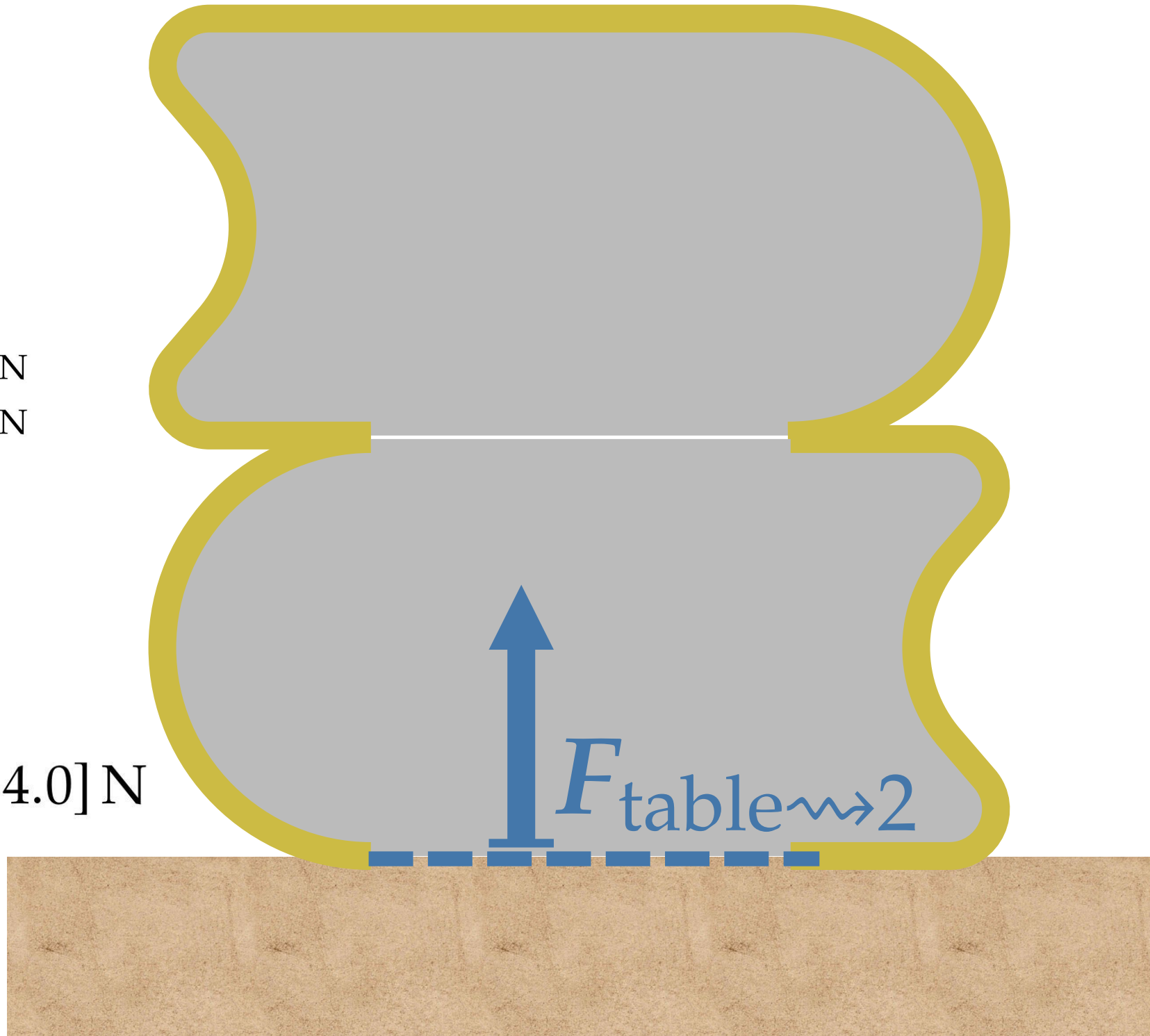
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$$\mathbf{F}_{\text{table} \rightsquigarrow 2} \approx [0, 0, 4.0] \text{ N}$$



Two walls of granite 'Left' and 'Right' are leaning as illustrated in the figure, with a beam of steel between their tops. Wall L has mass 27 000 kg, and wall R has mass 9000 kg. The mass of the steel beam is practically zero.

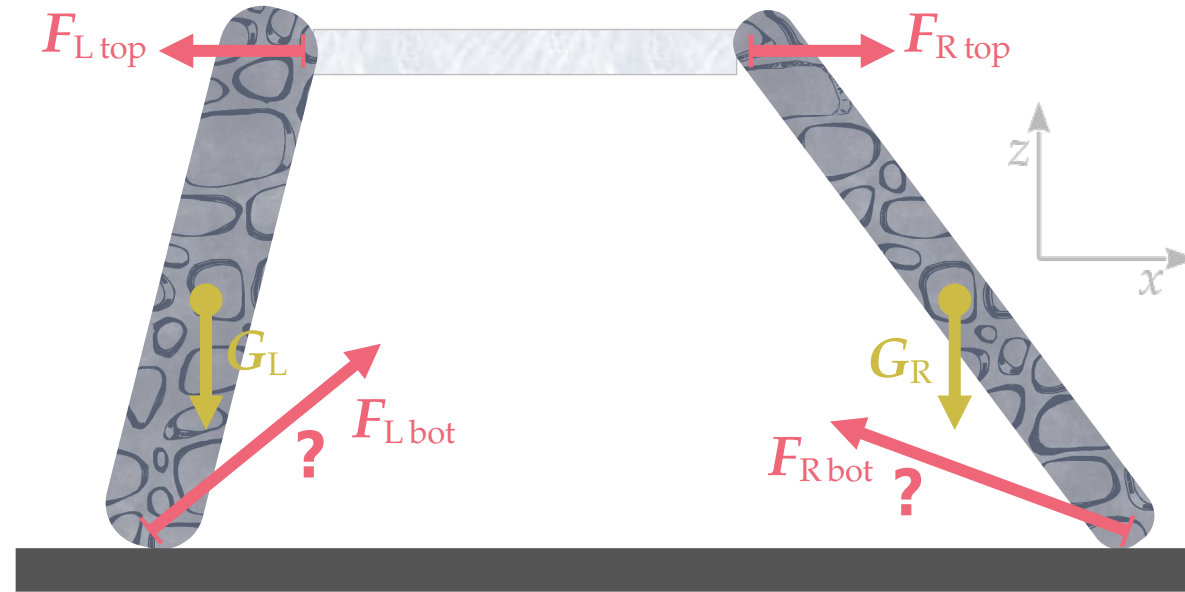
The volume forces and the surface forces (momentum fluxes) on the walls are shown in the figure. The actual directions of $F_{L\text{bot}}$ and $F_{R\text{bot}}$ are unknown. The surface forces on the steel beam are not shown.

You know this:

- The z-components of $F_{R\text{top}}$ and $F_{L\text{top}}$ are zero.
- The steel beam is compressed by an amount 0.006 62 m with respect to its natural length.

② How much is $F_{R\text{bot}}$ (both components)?

② Can you manage to solve this problem using only one control volume?



Useful constitutive relations:

- Gravitational force (momentum supply)

$$\mathbf{G} = -mg[0, 0, 1], \quad \text{with } g = 9.8 \text{ N/kg}$$

where m is the mass in the control volume.

- Hooke's law: the outward force (efflux) F exerted by an elastic material (like steel) is

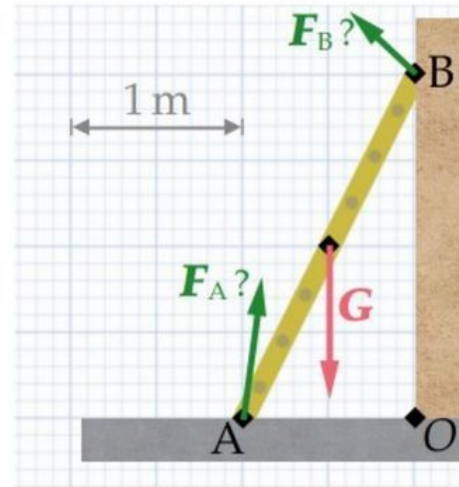
$$F = -k \Delta l$$

where Δl is the shortening of the material and k is the elastic constant. The elastic constant for steel beam is $5 \times 10^6 \text{ N/m}$.

Oppgave 5 (18.75 poeng)

Se figuren til høyre. En **stiv** stige (tegnet i **gult**) med masse $m = 10$ kg står stille på gulvet ved punkt A og lener seg mot en vegg ved punkt B. Den er helt **statisk**. Bruk et koordinatsystem (x, y, z) hvor x -aksen peker horisontalt til høyre, y -aksen peker inn i papiret, og z -aksen peker vertikalt oppover. Med hensyn til origo O får du oppgitt at:

- Punkt A har koordinater $[-1, 0, 0]$ m,
- Punkt B har koordinater $[0, 0, 2]$ m,
- Massesentrum til stigen er: $[-0.5, 0, 1]$ m.



Tre krefter (momentuminnflukser og kilde) virker på stigen:

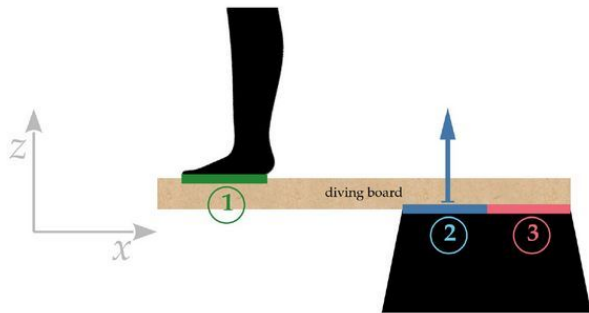
- kraft $\mathbf{F}_A$ ved punkt A; denne kraften har z -komponent $F_{A,z} = +90$ N;
- kraft $\mathbf{F}_B$ ved punkt B;
- tyngdekraften $\mathbf{G} = -m g [0, 0, 1]$ med $g = 9.8$ N/kg.

(NB: $\mathbf{F}_A$ og $\mathbf{F}_B$ i figuren er hypotetiske, retning og størrelse er ikke tegnet presist!)

Bruk momentumbalansen for å finne så mange komponenter som mulig til $\mathbf{F}_A$ og $\mathbf{F}_B$.

See solution in "Past exams" Module on Canvas

A diver is standing on a diving board, which is bolted on a base, as schematized in this picture:

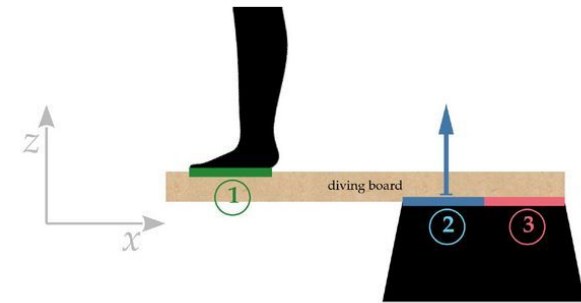


Use coordinates (x, z) as in the picture. Consider the diving board as a control volume. There are momentum influxes (forces) $\mathbf{F}_1$, $\mathbf{F}_2$, $\mathbf{F}_3$ through the control surfaces denoted ①, ②, ③. You have the following information:

- The diver has mass 60 kg, the gravitational acceleration is $g = 9.8 \text{ N/kg}$
- The diving board is completely static
- The mass of the diving board is negligible
- Through surface ② there is a momentum influx $\mathbf{F}_2 = \begin{bmatrix} 0 \\ +2142 \end{bmatrix} \text{ N}$ (shown in the picture)
- Atmospheric pressure is negligible

- Find the momentum *influxes* $\mathbf{F}_1$ and $\mathbf{F}_3$ through control surfaces ① and ③. Express them as vectors; be careful with the signs.
- Explain (unless you already did) which universal laws, constitutive relations, and physical properties you needed to use, or would need to use, to solve this problem.

En stuper står på et stupebrett som er boltet til en base, som skjematisk vist her:



Bruk koordinater (x, z) som i bildet. Tenk på stupebrettet som et kontrollvolum. Det er momentuminnflukser (krefter) $\mathbf{F}_1$, $\mathbf{F}_2$, $\mathbf{F}_3$ gjennom kontrollflatene betegnet med ①, ②, ③. Du har følgende informasjon:

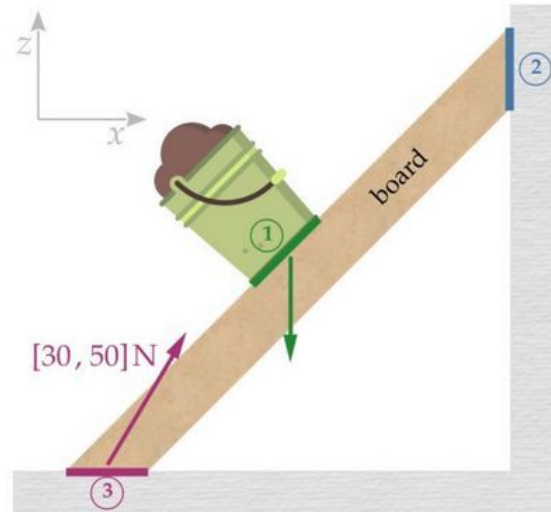
- Stuperen har en masse på 60 kg, tyngdeakselerasjonen er $g = 9.8 \text{ N/kg}$
- Stupebrettet er helt statisk
- Massen til stupebrettet er ubetydelig
- Gjennom overflate ② er det en momentuminnfluks $\mathbf{F}_2 = \begin{bmatrix} 0 \\ +2142 \end{bmatrix} \text{ N}$ (vist i bildet)
- Atmosfærisk trykk er ubetydelig

- Finn momentuminnflukser $\mathbf{F}_1$ og $\mathbf{F}_3$ gjennom kontrollflatene ① og ③. Uttrykk dem som vektorer; vær forsiktig med tegnene.
- Forklar (med mindre du allerede har gjort det) hvilke universelle lover, konstitutive relasjoner og fysiske egenskaper du trengte å bruke, eller ville trenge å bruke, for å løse dette problemet.

See solution in "Past exams" Module on Canvas

Oppgave 4

A filled bucket is placed on an inclined board, as schematized in the side figure. The whole arrangement is completely static. Use coordinates (x, z) as in the figure, and consider the board as a control volume. There are momentum influxes (forces) $\mathbf{F}_1$, $\mathbf{F}_2$, $\mathbf{F}_3$ through the control surfaces denoted ① (under bucket), ② (right), ③ (bottom). You have the following information:



- Atmospheric pressure is negligible.
 - The mass of the board is negligible.
 - The filled bucket has mass 4.08 kg , the gravitational acceleration is $g = 9.8 \text{ N/kg}$.
 - Through surface ③ there is a momentum influx $\mathbf{F}_3 = [+30, +50] \text{ N}$ (shown in the figure).
- (i) Find the momentum *influxes* $\mathbf{F}_1$ and $\mathbf{F}_2$ through control surfaces ① and ②. Express them as vectors; be careful with the signs.
- (ii) Explain (unless you already did) which universal laws, constitutive relations, and physical properties you needed to use, or would need to use, to solve this problem.

See solution in “Past exams” Module on Canvas

A tennis ball of mass $m = 0.05 \text{ kg}$ is at time t_0 at a particular height above the ground and is flying downward and to the right. It bounces off the ground, and at a later time t_1 is at a new height above ground and is flying upward, still to the right (see picture; note that the velocity vectors are only illustrative). Use a coordinate system (x, z) as in the picture.

You're given the following information:

- At t_0 , the height of the ball is $z(t_0) = 3 \text{ m}$ and its velocity $\mathbf{v}(t_0) = [1, -1] \text{ m/s}$
- At t_1 , the height of the ball is $z(t_1) = 2 \text{ m}$
- During the bounce off the ground there is a time-integrated flux of x -momentum equal to

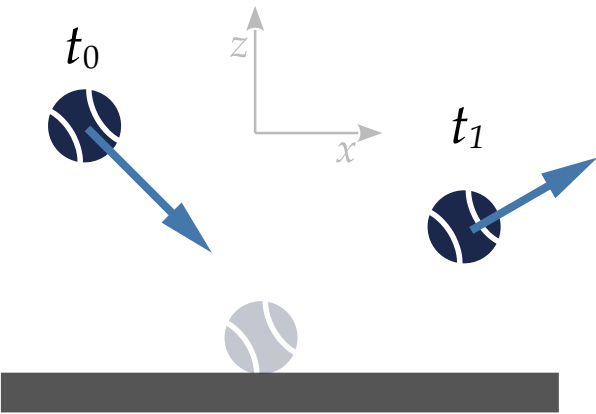
$$\int F_x(t) dt = -0.01 \text{ N s}$$

(Note that we don't know how much is the flux of z -momentum)

- During the bounce, there is a time-integrated heat flux from ball to ground equal to

$$\int Q(t) dt = -0.4 \text{ J}$$

- There is no flux of mechanical power
- The internal energy of the ball at t_1 is 0.05 J larger than at t_0
- Air effects are negligible



Find the velocity of the ball at t_1 , $\mathbf{v}(t_1) = [?, ?]$, from the information given.

Explain which physical laws and principles you use in your solution.

(Hint: use the balance of x -momentum together with the balance of energy.)

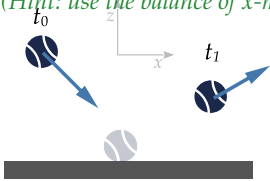
You're given the following information:

- $m = 0.05 \text{ kg}$
- At t_0 , the height of the ball is $z(t_0) = 3 \text{ m}$ and its velocity $v(t_0) = [1, -1] \text{ m/s}$
- At t_1 , the height of the ball is $z(t_1) = 2 \text{ m}$
- During the bounce off the ground there is a time-integrated flux of x -momentum equal to $\int F_x(t) dt = -0.01 \text{ N s}$

(Note that we don't know how much is the flux of z -momentum)

- During the bounce, there is a time-integrated heat flux from ball to ground equal to $\int Q(t) dt = -0.4 \text{ J}$
- The internal energy of the ball at t_1 is 0.05 J larger than at t_0
- There is no flux of mechanical power
- Air effects are negligible

Find the velocity of the ball at t_1 , $v(t_1) = [?, ?]$, from the information given.
(Hint: use the balance of x -momentum together with the balance of energy.)



We follow the hint and write down the **balances of x -momentum and of energy** for the tennis ball, clearly in integral form:

$$P_x(t_1) = P_x(t_0) + \int_{t_0}^{t_1} F_x(t) dt + \int_{t_0}^{t_1} G_x(t) dt \quad E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt$$

Let's see which of the terms above we can get from the info in the problem and constitutive relations:

- **Newton's relation for momentum** gives us $P_x(t_0) = m v_x(t_0) = 0.05 \text{ N s}$
- We have $\int F_x(t) dt$
- From the **constitutive relation for gravitational momentum supply** we know that $G_x = 0 \text{ N} \Rightarrow \int G_x(t) dt = 0 \text{ N s}$

We can therefore find $P_x(t_1) = (0.05 - 0.01 + 0) \text{ N s} = 0.04 \text{ N s}$ from the x -momentum balance, and using again Newton's relation for momentum we find

$$v_x(t_1) = P_x(t_1) / m = 0.8 \text{ m/s}$$

You're given the following information:

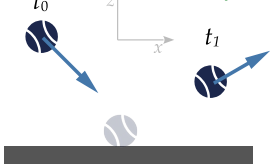
- $m = 0.05 \text{ kg}$
- At t_0 , the height of the ball is $z(t_0) = 3 \text{ m}$ and its velocity $v(t_0) = [1, -1] \text{ m/s}$
- At t_1 , the height of the ball is $z(t_1) = 2 \text{ m}$
- During the bounce off the ground there is a time-integrated flux of x -momentum equal to $\int F_x(t) dt = -0.01 \text{ N s}$

(Note that we don't know how much is the flux of z -momentum)

- During the bounce, there is a time-integrated heat flux from ball to ground equal to $\int Q(t) dt = -0.4 \text{ J}$
- The internal energy of the ball at t_1 is 0.05 J larger than at t_0
- There is no flux of mechanical power
- Air effects are negligible

Find the velocity of the ball at t_1 , $v(t_1) = [?, ?]$, from the information given.

(Hint: use the balance of x -momentum together with the balance of energy.)



We follow the hint and write down the **balances of x -momentum and of energy** for the tennis ball, clearly in integral form:

$$P_x(t_1) = P_x(t_0) + \int_{t_0}^{t_1} F_x(t) dt + \int_{t_0}^{t_1} G_x(t) dt \quad E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt$$

Let's see which of the terms above we can get from the info in the problem and constitutive relations:

- **Newton's relation for momentum** gives us $P_x(t_0) = m v_x(t_0) = 0.05 \text{ N s}$
- We have $\int F_x(t) dt$
- From the **constitutive relation for gravitational momentum supply** we know that $G_x = 0 \text{ N} \Rightarrow \int G_x(t) dt = 0 \text{ N s}$

We can therefore find $P_x(t_1) = (0.05 - 0.01 + 0) \text{ N s} = 0.04 \text{ N s}$ from the x -momentum balance, and using again Newton's relation for momentum we find

$$v_x(t_1) = P_x(t_1) / m = 0.8 \text{ m/s}$$

Now let's see which terms in the energy balance we can get from the info in the problem and constitutive relations:

- The **constitutive relation for energy in matter** gives us

$$E(t_0) = U(t_0) + \frac{1}{2} m [v_x(t_0)^2 + v_z(t_0)^2] + m g z(t_0) = U(t_0) + [0.5 \cdot 0.05 \cdot (1 + 1)] \text{ J} + 0.05 \cdot 9.8 \cdot 3 \text{ J} = U(t_0) + 1.52 \text{ J}$$

$$E(t_1) = U(t_1) + \frac{1}{2} m [v_x(t_1)^2 + v_z(t_1)^2] + m g z(t_1)$$

in this last equation we know:

- $U(t_1) = U(t_0) + 0.05 \text{ J}$ from the problem
- $v_x(t_1) = 0.8 \text{ m/s}$ from our previous calculation $\Rightarrow \frac{1}{2} m v_x(t_1)^2 = (0.5 \cdot 0.05 \cdot 0.8^2) \text{ J} = 0.016 \text{ J}$
- $m g z(t_1) = (0.05 \cdot 9.8 \cdot 2) \text{ J} = 0.98 \text{ J}$

$$\Rightarrow E(t_1) = U(t_0) + 0.05 \text{ J} + 0.016 \text{ J} + \frac{1}{2} m v_z(t_1)^2 + 0.98 \text{ J} = U(t_0) + \frac{1}{2} m v_z(t_1)^2 + 1.046 \text{ J}$$

- The **constitutive relation for energy flux in matter** is $\Phi = Q + F \cdot v$ and the problem says that $F \cdot v$ is zero (no mechanical power) and gives us $\int Q(t) dt = -0.4 \text{ J} \Rightarrow \int \Phi(t) dt = \int Q(t) dt + \int F(t) \cdot v(t) dt = -0.4 \text{ J}$

Substituting all this in the energy balance:

$$\underbrace{U(t_0) + \frac{1}{2} m v_z(t_1)^2 + 1.046 \text{ J}}_{E(t_1)} = \underbrace{U(t_0) + 1.52 \text{ J}}_{E(t_0)} \underbrace{- 0.4 \text{ J}}_{\int \Phi(t) dt}$$

From which we finally find $v_z(t_1)^2 = [(1.52 - 0.4 - 1.046) \cdot 2 / 0.05] \text{ J} = 2.96 \text{ J} \Rightarrow v_z(t_1) = \pm 1.72 \text{ m/s}$ with "+" because the problem says the ball is going upward at t_1 .



A football with mass 0.4 kg is connected to a vertical 1.5-metre pole by a stick of length 1 m , as depicted in the side figure. The football can move horizontally and vertically, but its velocity v is always perpendicular to the stick. The stick exerts a force F on the football, parallel to the stick. There are no heat fluxes in this system, and the gravitational acceleration is 9.8 m/s^2 .

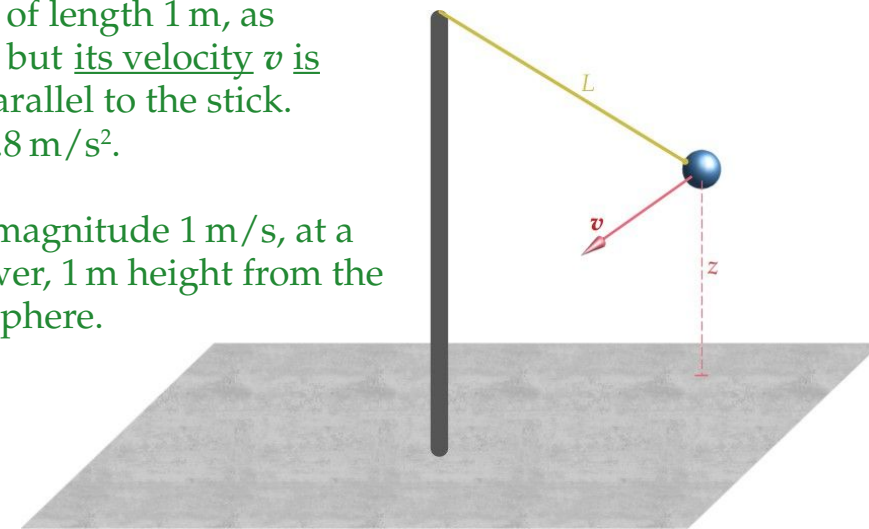
At time t_0 the football is moving with a velocity which is horizontal and has magnitude 1 m/s , at a 1.5 m height from the ground. At a later time t_1 the football is moving at a lower, 1 m height from the ground; its internal energy has remained constant. Suppose there is no atmosphere.

- (1) How much is the flux of total energy between stick and football?
- (2) What is the magnitude of the football's velocity at t_1 ?
- (3) How much would this magnitude be if the football weighed 2 kg ?

In a realistic situation we would expect that the football, unless pushed or kicked, will end up being at rest at about 0.5 m from the ground, besides the pole.

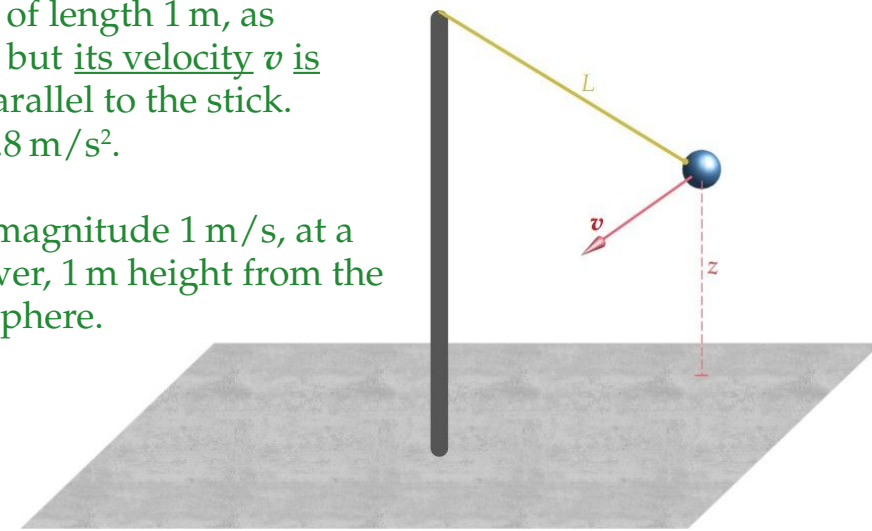
- (4) Is this final situation possible, according to the conditions of the present problem? If not, why not?
- (5) Can the more realistic final situation be reached, if the total-energy flux is non-zero? How much should the time-integrated energy influx to the football be?

(Explain which physical laws and principles you are using in your calculations.)



A football with mass 0.4 kg is connected to a vertical 1.5-metre pole by a stick of length 1 m , as depicted in the side figure. The football can move horizontally and vertically, but its velocity v is always perpendicular to the stick. The stick exerts a force F on the football, parallel to the stick. There are no heat fluxes in this system, and the gravitational acceleration is 9.8 m/s^2 .

At time t_0 the football is moving with a velocity which is horizontal and has magnitude 1 m/s , at a 1.5 m height from the ground. At a later time t_1 the football is moving at a lower, 1 m height from the ground; its internal energy has remained constant. Suppose there is no atmosphere.



- (1) How much is the flux of total energy between stick and football?
- (2) What is the magnitude of the football's velocity at t_1 ?
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In a realistic situation we would expect that the football, unless pushed or kicked, will end up being at rest at about 0.5 m from the ground, besides the pole.

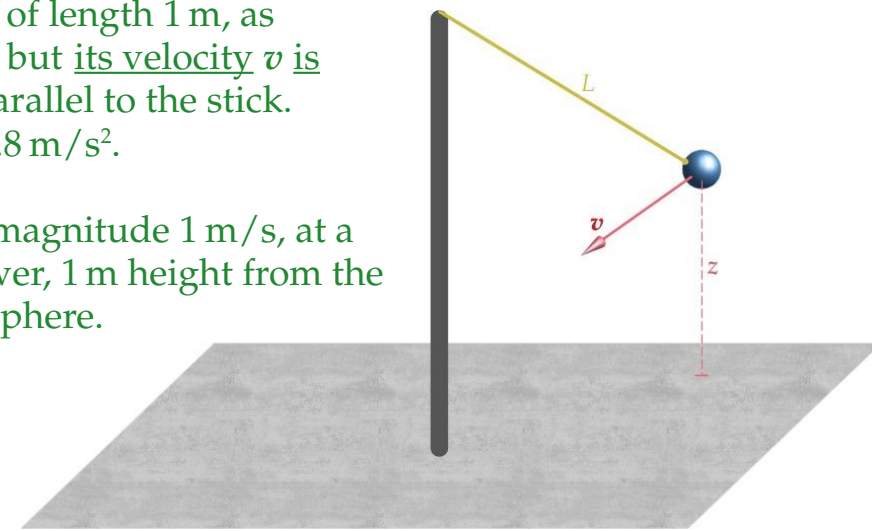
- (4) Is this final situation possible, according to the conditions of the present problem? If not, why not?
- (5) Can the more realistic final situation be reached, if the total-energy flux is non-zero? How much should the time-integrated energy influx to the football be?

(Explain which physical laws and principles you are using in your calculations.)

(1) The constitutive relation for total-energy flux is $\Phi = Q + F \cdot v$. This flux can only occur between stick and football, because there is no atmosphere. According to the problem, the heat flux Q is zero. The football's velocity v is always perpendicular to the force F exerted by the stick, and therefore $F \cdot v = 0$. We conclude that the total-energy flux Φ is zero.

A football with mass 0.4 kg is connected to a vertical 1.5-metre pole by a stick of length 1 m, as depicted in the side figure. The football can move horizontally and vertically, but its velocity v is always perpendicular to the stick. The stick exerts a force F on the football, parallel to the stick. There are no heat fluxes in this system, and the gravitational acceleration is 9.8 m/s^2 .

At time t_0 the football is moving with a velocity which is horizontal and has magnitude 1 m/s , at a 1.5 m height from the ground. At a later time t_1 the football is moving at a lower, 1 m height from the ground; its internal energy has remained constant. Suppose there is no atmosphere.



- (1) How much is the flux of total energy between stick and football?
- (2) What is the magnitude of the football's velocity at t_1 ?
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- (4) Is this final situation possible, according to the conditions of the present problem? If not, why not?
- (5) Can the more realistic final situation be reached, if the total-energy flux is non-zero? How much should the time-integrated energy influx to the football be?

(Explain which physical laws and principles you are using in your calculations.)

(2) We can apply the balance of energy in integral form: $E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt$

where in this case the flux Φ is zero, and the total-energy content is given by the constitutive relation $E = U + \frac{1}{2}mv^2 + mgz$ applied to the control volume consisting of the football itself. So we have

$$E(t_1) = E(t_0) \quad \Rightarrow \quad U(t_1) + \frac{1}{2}mv(t_1)^2 + mgz(t_1) = U(t_0) + \frac{1}{2}mv(t_0)^2 + mgz(t_0)$$

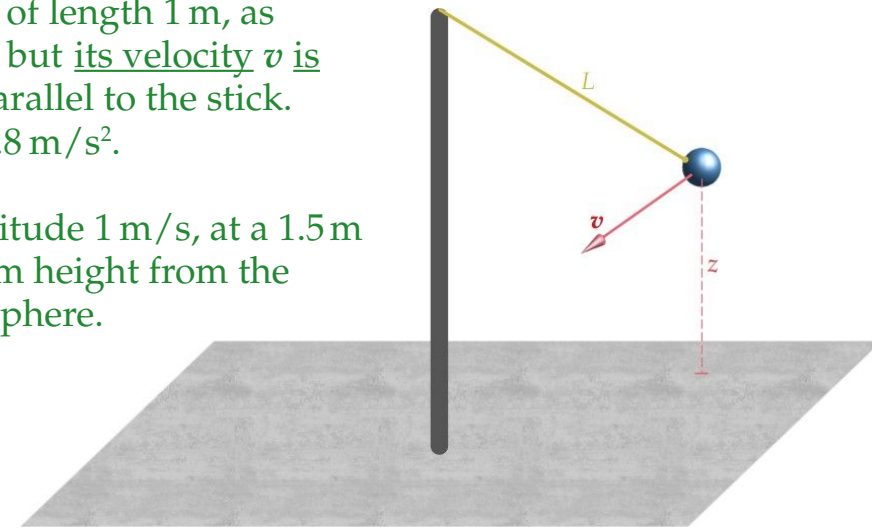
According to the problem, $m = 0.4 \text{ kg}$ $v(t_0) = 1 \text{ m/s}$ $z(t_0) = 1.5 \text{ m}$ $z(t_1) = 1 \text{ m}$ $U(t_1) = U(t_0)$ $g = 9.8 \text{ m/s}^2$

We substitute these values in the balance equation, we notice that the internal energies cancel out, and we have all quantities except $v(t_1)$:

$$v(t_1)^2 = \frac{\cancel{U(t_0)} - \cancel{U(t_1)} + \cancel{1/2 m} v(t_0)^2 + \cancel{m} g z(t_0) - \cancel{m} g z(t_1)}{\cancel{1/2 m}} \approx \frac{1/2 \cdot 1 + 9.8 \cdot 1.5 - 9.8 \cdot 1}{1/2} \text{ m}^2/\text{s}^2 \approx 11 \text{ m}^2/\text{s}^2 \quad \Rightarrow \quad v(t_1) \approx 3.3 \text{ m/s}$$

A football with mass 0.4 kg is connected to a vertical 1.5-metre pole by a stick of length 1 m, as depicted in the side figure. The football can move horizontally and vertically, but its velocity v is always perpendicular to the stick. The stick exerts a force F on the football, parallel to the stick. There are no heat fluxes in this system, and the gravitational acceleration is 9.8 m/s^2 .

At time t_0 the football is moving a velocity which is horizontal and has magnitude 1 m/s, at a 1.5 m height from the ground. At a later time t_1 the football is moving at a lower, 1 m height from the ground; its internal energy has remained constant. Suppose there is no atmosphere.



- (1) How much is the flux of total energy between stick and football?
- (2) What is the magnitude of the football's velocity at t_1 ?
- (3) How much would this magnitude be if the football weighed 2 kg?

In a realistic situation we would expect that the football, unless pushed or kicked, will end up being at rest at about 0.5 m from the ground, besides the pole.

- (4) Is this final situation possible, according to the conditions of the present problem? If not, why not?
- (5) Can the more realistic final situation be reached, if the total-energy flux is non-zero? How much should the time-integrated energy influx to the football be?

(Explain which physical laws and principles you are using in your calculations.)

(4) In our calculations for question (2) we arrived at the equation $U(t_1) + \frac{1}{2}mv(t_1)^2 + mgz(t_1) = U(t_0) + \frac{1}{2}mv(t_0)^2 + mgz(t_0)$, representing the balance of energy.

In a final situation with $v(t_1) = 0 \text{ m/s}$ $z(t_1) = 0.5 \text{ m}$ $U(t_1) = U(t_0)$ the balance would become

$$U(t_1) + \frac{1}{2}mv(t_1)^2 + mgz(t_1) = U(t_0) + \frac{1}{2}mv(t_0)^2 + mgz(t_0)$$

~~$$U(t_0) + \frac{1}{2}mv(t_1)^2 + mgz(t_1) = U(t_0) + \frac{1}{2}mv(t_0)^2 + mgz(t_0)$$~~

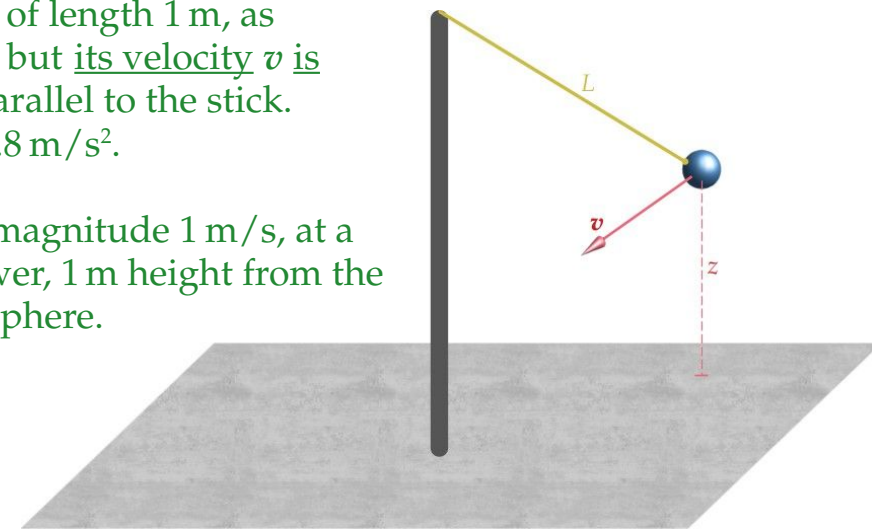
$$(\frac{1}{2} \cdot 0.4 \cdot 0^2 + 0.4 \cdot 9.8 \cdot 0.5) \text{ J} \approx (\frac{1}{2} \cdot 0.4 \cdot 1^2 + 0.4 \cdot 9.8 \cdot 1.5) \text{ J}$$

$$2.0 \text{ J} \approx 6.1 \text{ J} \text{ impossible!}$$

so this final situation would break the balance of energy.

A football with mass 0.4 kg is connected to a vertical 1.5-metre pole by a stick of length 1 m, as depicted in the side figure. The football can move horizontally and vertically, but its velocity v is always perpendicular to the stick. The stick exerts a force F on the football, parallel to the stick. There are no heat fluxes in this system, and the gravitational acceleration is 9.8 m/s^2 .

At time t_0 the football is moving with a velocity which is horizontal and has magnitude 1 m/s , at a 1.5 m height from the ground. At a later time t_1 the football is moving at a lower, 1 m height from the ground; its internal energy has remained constant. Suppose there is no atmosphere.



- (1) How much is the flux of total energy between stick and football?
- (2) What is the magnitude of the football's velocity at t_1 ?
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In a realistic situation we would expect that the football, unless pushed or kicked, will end up being at rest at about 0.5 m from the ground, besides the pole.

- (4) Is this final situation possible, according to the conditions of the present problem? If not, why not?
- (5) Can the more realistic final situation be reached, if the total-energy flux is non-zero? How much should the time-integrated energy influx to the football be?

(Explain which physical laws and principles you are using in your calculations.)

- (5) If we have a non-zero energy flux, then the balance of energy derived in question (4) would be

$$U(t_0) + 2.0 \text{ J} = U(t_0) + 6.1 \text{ J} + \int \Phi dt \quad \Rightarrow \quad \int \Phi dt = -4.1 \text{ J}$$

which means that there should be a flux of energy from the football.

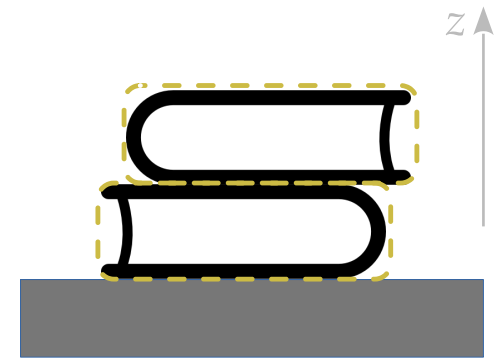
In a realistic scenario, the football eventually comes to rest because of the force of friction from the air. This force is opposite to the football's velocity, so $F \cdot v < 0$, a negative *influx* of mechanical power, that is, a positive *efflux* from the football to the air. (In a realistic scenario this friction, and possibly a heat flux, also changes slightly the internal energy of the football.)



Two books, each of mass $m = 0.2 \text{ kg}$, are stacked upon each other on a table. The whole system is at rest. Consider a control volume for each book, and a control surface between the lower book and table.

Calculate all influxes of momentum in these control volumes and the influx of momentum to the table. Assume atmospheric pressure to be negligible, and the gravitational acceleration to be $g = 9.8 \text{ N/kg}$. Use just one vertical coordinate.

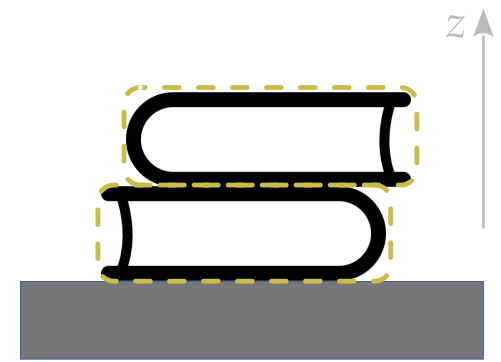
Explain which physical laws and principles you are using in your calculations.



Two books, each of mass $m = 0.2 \text{ kg}$, are stacked upon each other on a table. The whole system is at rest. Consider a control volume for each book, and a control surface between the lower book and table.

Calculate all influxes of momentum in these control volumes and the influx of momentum to the table. Assume atmospheric pressure to be negligible, and the gravitational acceleration to be $g = 9.8 \text{ N/kg}$. Use just one vertical coordinate.

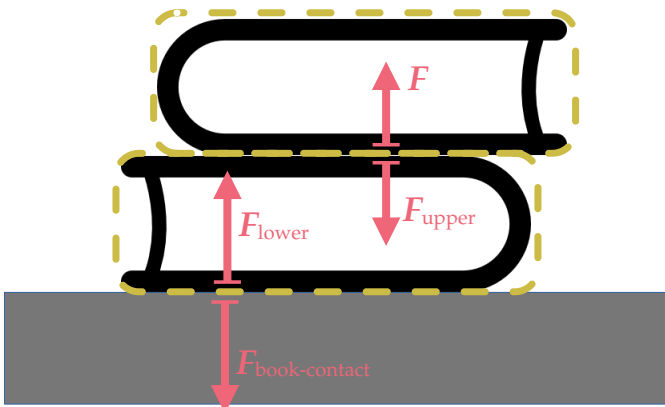
Explain which physical laws and principles you are using in your calculations.



**Example short answer
that could be given at the exam**

**If you have extra time at the
end you can add some extra
comments or explanations**

You don't need to use "explanation arrows" as shown here. It's just an example. Alternatively you could write "sum because of extensivity of flux" or something similar.



Upper book: balance of mom. $\frac{dP}{dt} = F(t) + G(t)$

$P = m v$ (Newton's formula for mom.), but $v = 0 \text{ m/s}$ constant $\Rightarrow dP/dt = 0 \text{ N}$

$F = ?$ mom. influx contact surface with lower book

$G = -mg = -1.96 \text{ N}$ (const. formula for gravity)

$$\Rightarrow 0 \text{ N} = F - 1.96 \text{ N} \Rightarrow F = +1.96 \text{ N}$$

Lower book: balance of mom. again

$dP/dt = 0 \text{ N}$: same reasoning as for upper book

$$F_{\text{tot}} = F_{\text{upper}} + F_{\text{lower}}$$

$\swarrow$ mom. influx from upper book $\swarrow$ mom. influx from table
 $\nwarrow$ extensivity of fluxes

$F_{\text{upper}} = -F = -1.96 \text{ N}$ (found before)

$\nwarrow$ symmetry of fluxes

$F_{\text{lower}} = ?$

$G = -1.96 \text{ N}$ like for upper book

$$\Rightarrow 0 \text{ N} = -1.96 \text{ N} + F_{\text{lower}} - 1.96 \text{ N} \Rightarrow F_{\text{lower}} = +3.92 \text{ N}$$

Table:

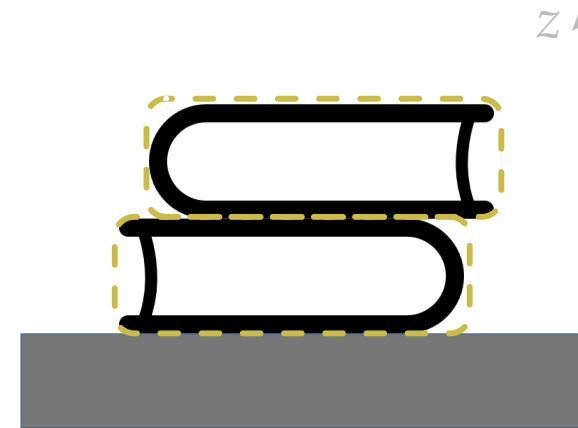
$$F_{\text{book-contact}} = -F_{\text{lower}} = +3.92 \text{ N}$$

$\nwarrow$ symmetry of fluxes

Two books, each of mass $m = 0.2 \text{ kg}$, are stacked upon each other on a table. The whole system is at rest. Consider a control volume for each book, and a control surface between the lower book and table.

Calculate all influxes of momentum in these control volumes and the influx of momentum to the table. Assume atmospheric pressure to be negligible, and the gravitational acceleration to be $g = 9.8 \text{ N/kg}$. Use just one vertical coordinate.

Explain which physical laws and principles you are using in your calculations.



Obviously we must use the **balance of momentum** $\frac{dP}{dt} = F(t) + G(t)$ these vectors are all one-dimensional, with z the upward vertical coordinate.

Upper book ("ub"):

- Time-rate of momentum: the matter in this control volume is at rest, that is, its velocity is constantly zero. By **Newton's formula for the momentum of matter** $P_{ub} = m v_{ub}$, the total momentum content in this volume is also constantly zero. So the time-derivative $dP_{ub}/dt = 0$.
- Total influx: atmosphere is negligible, so the only influx of momentum F_{ub} comes at the contact surface with the lower book. We don't know how much this influx is, yet.
- Supply: from gravity, the constitutive relation in this case is $G_{ub} = -mg$.

The balance of momentum for the upper book gives $0 \text{ N} = F_{ub} - mg \Rightarrow F_{ub} = mg = +1.96 \text{ N}$

Lower book ("lb"):

- Time-rate of momentum: same reasoning as for upper book: $dP_{lb}/dt = 0$.
- Total influx: we must use the **principle of extensivity** because we have an influx through the contact surface with the upper book, and an influx through the contact surface with the table. The influx through the upper surface is, by the **symmetry of fluxes**, equal to minus the outflux which we found above: $-F_{ub}$. The influx through the lower surface, F_{tab} , is yet unknown.
- Supply: same reasoning as before: $G_{lb} = -mg$.

The balance of momentum for the lower book gives $0 \text{ N} = -F_{ub} + F_{tab} - mg \Rightarrow F_{tab} = +3.92 \text{ N}$

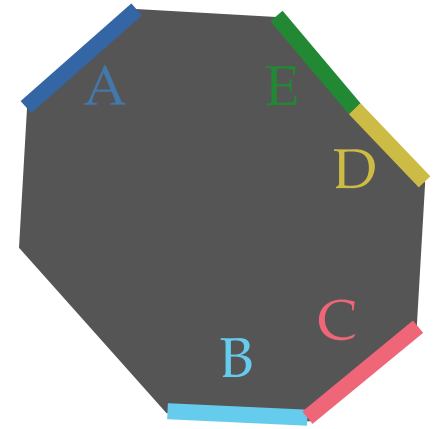
Table: by the **symmetry of fluxes**, the influx of momentum to the table through the contact surface with the lower book is $-F_{tab} = -3.92 \text{ N}$

?

Consider the control volume (simplified in 2D) in the picture. You know the following:

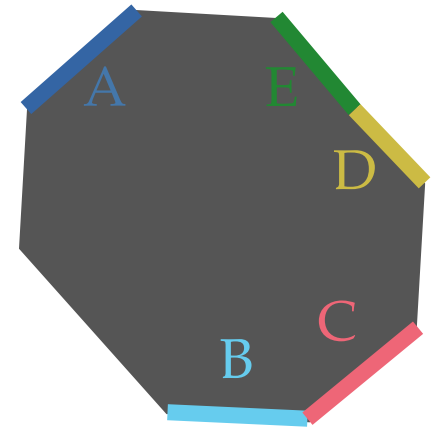
- The volume content of momentum is changing at a time rate $[4, -1, 3] \text{ N}$
- The momentum influx at surface A is $[3, 2, -7] \text{ N}$
- The momentum influx at surface B is $[-5, 0, 0] \text{ N}$
- The momentum **efflux** at surface C is $[4, 4, 4] \text{ N}$
- The momentum influx at surface D is $[0, 0, 1] \text{ N}$
- The momentum influx at the other surfaces is 0 N
- Gravity is negligible

Find the momentum influx at surface E



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- The momentum influx at the other surfaces is 0 N
- Gravity is negligible



Find the momentum influx at surface E

In order to find the unknown influx at E we use the **balance of momentum** in differential form:

$$\frac{dP}{dt} = F(t) + G(t)$$

(it seems appropriate because we're given fluxes and the time-rate of momentum). The problem gives us dP/dt , and says that gravity is negligible, which means $G=0 \text{ N}$. What's left is F .

In the balance, F is the *net* momentum influx through the whole surface, but we only know fluxes through pieces of the whole surface. We use the **principle of extensivity**: the net flux through a surface is equal to the sum of the fluxes through all pieces of the surface (no matter how we divide the surface). In our case this means

$$F = F_A + F_B + F_D + F_E + F_{\text{rest}}$$

we must be careful about the crossing direction: from outside to inside (they're all **influxes**). We find:

$$\frac{dP}{dt} = F_A(t) + F_B(t) + F_C(t) + F_D(t) + F_E(t) + F_{\text{rest}}(t) + G(t)$$

$$\begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix} \text{ N} = \begin{bmatrix} 3 \\ 2 \\ -7 \end{bmatrix} \text{ N} + \begin{bmatrix} -5 \\ 0 \\ 0 \end{bmatrix} \text{ N} - \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} \text{ N} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ N} + F_E(t) + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ N}$$

this is an **efflux**

Which means $F_E = [10, 1, 13] \text{ N}$

(c)

You and a friend of yours synchronize your clocks. Your friend goes to live for a while in an orbiting space station. After one year of your time, your friend comes back and you two meet again. You notice that your friend has aged three years instead of one.

How large is the mass M of the hypothetical planet in which you two live? Use the approximate formula

$$\Delta t_{\text{friend}} = \left[1 + \frac{G}{c^2} \frac{M}{R} \right] \Delta t_{\text{you}}$$

where Δt_{friend} is your friend's proper time while in orbit, Δt_{you} is your proper time, $G = 6.7 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$, $c = 3 \times 10^8 \text{ m/s}$, and assume your planet has radius $R = 7 \times 10^7 \text{ m}$.

Finding M is just a matter of algebra. To use consistent units, first let's express the times in seconds:

$$\Delta t_{\text{friend}} = 3 \cdot 365 \cdot 24 \cdot 60 \cdot 60 \text{ s} \approx 9.5 \times 10^7 \text{ s}$$

$$\Delta t_{\text{you}} = 1 \cdot 365 \cdot 24 \cdot 60 \cdot 60 \text{ s} \approx 3.2 \times 10^7 \text{ s}$$

Finally we find:

$$\frac{\Delta t_{\text{friend}}}{\Delta t_{\text{you}}} = 1 + \frac{G}{c^2} \frac{M}{R}$$

$$\Rightarrow M = R \frac{c^2}{G} \left[\frac{\Delta t_{\text{friend}}}{\Delta t_{\text{you}}} - 1 \right] \approx 7 \times 10^7 \text{ m} \frac{(3 \times 10^8 \text{ m/s})^2}{6.7 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)} \left[\frac{9.5 \times 10^7 \text{ s}}{3.2 \times 10^7 \text{ s}} - 1 \right] \approx 1.9 \times 10^{35} \text{ kg}$$

all units simplify correctly, leaving only “kg”.

(The mass of our sun is about $2 \cdot 10^{30} \text{ kg}$, so this planet has about 100 000 times the mass of our sun! On the other hand it has a radius of roughly only 10 times Earth's radius.)

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Assume a constant gravity acceleration $g = 9.8 \text{ m/s}^2$.

Explain which balance laws and which constitutive relations you use.

(i) At some instant in time, a tennis ball is at a height of 10 m above ground (no atmosphere), with a downward velocity of 1 m/s. The tennis ball's mass is 0.06 kg.

At a later instant of time, the ball is at a height of 8 m above ground. Its internal energy has not changed, and there have been no energy fluxes.

How much is the velocity of the ball at this later instant of time?

(ii) Another identical tennis ball is also at a height of 10 m above ground (no atmosphere), with an *upward* velocity of 1 m/s.

At a later instant of time (not the same as the first ball), the ball is at a height of 8 m above ground. Its internal energy has not changed, and there have been no energy fluxes.

How much is the velocity of this second ball at this later instant of time?



Assume a constant gravity acceleration $g = 9.8 \text{ m/s}^2$.

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How much is the velocity of this second ball at this later instant of time?

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt$$

$$E = U + \frac{1}{2}mv^2 + mgz \quad \text{given, zero}$$

$$U(t_1) + \frac{1}{2}mv(t_1)^2 + mgz(t_1) = U(t_0) + \frac{1}{2}mv(t_0)^2 + mgz(t_0)$$

$$\underbrace{U(t_1) - U(t_0)}_{0 \text{ J}} + \frac{1}{2}mv(t_1)^2 - \frac{1}{2}mv(t_0)^2 + \underbrace{mgz(t_1) - mgz(t_0)}_{0.06 \text{ kg} \cdot 9.8 \text{ m/s}^2 \cdot (8 \text{ m} - 10 \text{ m})} = 0 \text{ J}$$

$$= -1.176 \text{ J}$$

$$\frac{1}{2}mv(t_1)^2 = \underbrace{\frac{1}{2}mv(t_0)^2}_{\frac{1}{2} \cdot 0.06 \text{ kg} \cdot (1 \text{ m/s})^2} + 1.176 \text{ J}$$

$$= 0.03 \text{ J}$$

$$\frac{1}{2}mv(t_1)^2 = 1.206 \text{ J}$$

$$v(t_1) = \sqrt{2 \frac{1.206 \text{ J}}{0.06 \text{ kg}}} = 6.34 \text{ m/s}$$



A smooth slab of glass rests above a table made of glass. The table is anchored to the floor. The slab of glass has mass 10 kg and the contact surface with the table is 0.01 m^2 .

Because the glass is smooth, there is no air between the two glass surfaces, so the air pressure above the slab is **not** compensated by any air pressure between slab and table.

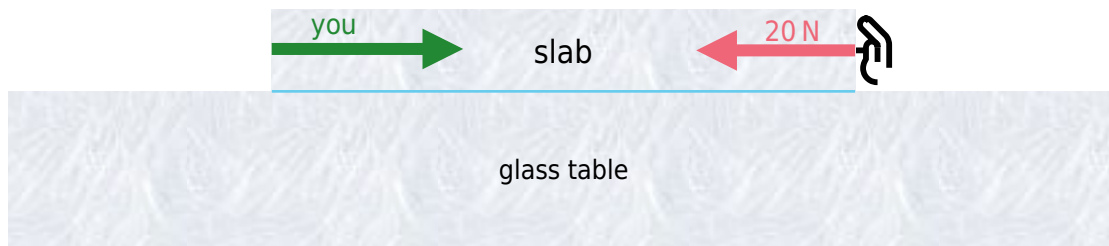
One person is exerting a constant leftward force of 20 N on the right side of the slab.

You would like to set the slab into motion to the right, by pushing horizontally on the left side of the slab. What is the minimal amount of horizontal force you need to exert, to put the slab into motion?

Assume: • gravity acceleration 9.8 m/s^2 , • atmospheric pressure 100000 N/m^2 ,
• coefficient of static friction glass-on-glass is 0.9.

Explain which balance laws and which constitutive relations you use.





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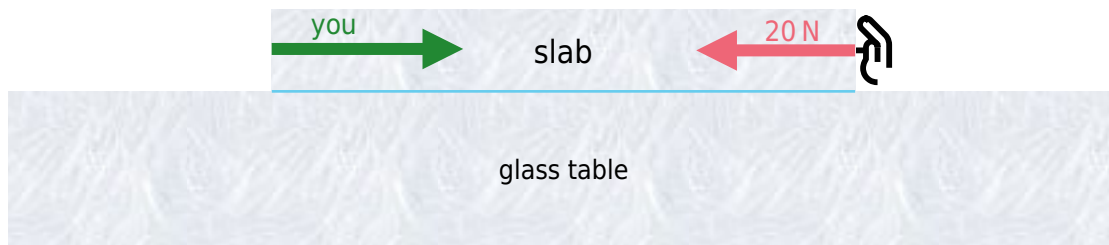
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The exercise asks about “minimal horizontal force to put the slab into motion”. This is because of **static friction**: that force (momentum flux) that cancels any other force that tries to move the slab – but only up to a particular amount. Let’s first find out how much is this minimal amount.

The magnitude of static friction is proportional to the normal force, by the **constitutive equation for static friction**: $F_s = \mu_s F_n$. The coefficient of static friction μ_s is given in the problem. We need to find the **normal force**. This is given by *minus* the sum of all the **vertical** components of all forces acting on the surface of the slab, plus gravity. We have three surface forces:

- force by you (left side): it has zero vertical component
- force by other person (right side): zero vertical component
- force by atmosphere (top side): it’s completely vertical. Its magnitude is given by pressure×area:

$$F_{\text{atm}} = p_{\text{atm}} \cdot \text{area} = 100\,000 \text{ N/m}^2 \cdot 0.01 \text{ m}^2 = 1000 \text{ N}$$

The vertical component of the total force on the slab surface is therefore –1000 N (minus because downward). The gravity supply is also completely downward:

$$G_z = -m_{\text{slab}} \cdot g = -10 \text{ kg} \cdot 9.8 \text{ N/kg} = -98 \text{ N}$$

So we find the normal force: $F_n = -(-1000 \text{ N} - 98 \text{ N}) = +1098 \text{ N}$

The minimal force necessary to move the slab horizontally is

$$F_s = \mu_s F_n = 0.9 \cdot 1098 \text{ N} \approx 988 \text{ N}$$

Remember that there’s a person exerting a force of 20 N on the right side. In order to move the slab we must have

$$\begin{aligned} F_{\text{you}} + F_{\text{person}} &\geq 988 \text{ N} \\ F_{\text{you}} - 20 \text{ N} &\geq 988 \text{ N} \\ F_{\text{you}} &\geq 1008 \text{ N} \end{aligned}$$

At some instant in time, a tennis ball is at a height of 1000 m above ground, with a downward velocity of 12 m/s. The tennis ball's mass is 0.057 kg.

Besides the force of gravity (momentum supply), air is exerting on the ball a total friction force (momentum flux), constant in time, equal to 0.5586 N and oriented upward.

How much is the velocity of the ball one second later?

Assume a constant gravity acceleration $g = 9.8 \text{ m/s}^2$.

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This question can be answered using the balance of momentum in its differential expression, or in its integral expression. Here are two example answers using either expression.

We can use just one coordinate, z

$$\frac{dP(t)}{dt} = \overset{\text{given}}{F(t)} + G(t)$$

$P = mv$
Newton's relation for momentum

$$G = -mg \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

gravitational momentum supply
near Earth's surface

$$\frac{dmv(t)}{dt} = 0.5586 \text{ N} - 0.057 \text{ kg} \cdot 9.8 \text{ m/s}^2$$

$$\frac{dmv(t)}{dt} = 0 \text{ N}$$

time-derivative is zero

→ velocity must be constant

→ same as before, 12 m/s

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$$\frac{dP(t)}{dt} = \overset{\text{given}}{F(t)} + G(t)$$

$$\underset{\text{Newton's relation for momentum}}{P = mv} \quad \underset{\text{gravitational momentum supply near Earth's surface}}{G = -mg \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}$$

$$\frac{dmv(t)}{dt} = 0.5586 \text{ N} - 0.057 \text{ kg} \cdot 9.8 \text{ m/s}^2$$

$$\frac{dmv(t)}{dt} = 0 \text{ N}$$

time-derivative is zero

→ velocity must be constant

→ same as before, 12 m/s

$$P(t_1) = P(t_0) + \overset{\text{given}}{\int_{t_0}^{t_1} F(t) dt} + \int_{t_0}^{t_1} \underset{\substack{\text{Newton's relation for momentum} \\ G = -mg \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ \text{gravitational momentum supply near Earth's surface}}}{G(t) dt}$$

$$P(1 \text{ s}) = P(0 \text{ s}) + \underset{\substack{0.057 \text{ kg} \cdot 12 \text{ m/s} \\ = -0.684 \text{ N s}}}{\int_{0 \text{ s}}^{1 \text{ s}} F(t) dt} + \underset{\substack{0.5586 \text{ N} \\ -0.057 \text{ kg} \cdot 9.8 \text{ m/s}^2 \\ = -0.5586 \text{ N}}}{\int_{0 \text{ s}}^{1 \text{ s}} G(t) dt}$$

$$= -0.684 \text{ N s} + 0.5586 \text{ N} \cdot 1 \text{ s} - 0.5586 \text{ N} \cdot 1 \text{ s}$$

$$= -0.684 \text{ N s}$$

final momentum is same as before

→ velocity must be the same, 12 m/s



At some instant in time, a tennis ball is at a height of 1000 m above ground, with a downward velocity of 12 m/s. The tennis ball's mass is 0.057 kg.

At a later instant of time, the ball is at a height of 880 m above ground, and its internal energy has increased by 10 J.

During this time there has been a total energy flow of 57.032 J from the ball to the surrounding air.

How much is the velocity of the ball at this later instant of time?

Assume a constant gravity acceleration $g = 9.8 \text{ m/s}^2$.

Explain which balance laws and which constitutive relations you use.



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Explain which balance laws and which constitutive relations you use.

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt$$

given

$$E = U + \frac{1}{2}mv^2 + mgz$$

Constitutive formula for energy content of matter

$$U(t_1) + \frac{1}{2}mv(t_1)^2 + mgz(t_1) = U(t_0) + \frac{1}{2}mv(t_0)^2 + mgz(t_0) - 57.032 \text{ J}$$

$$\underbrace{U(t_1) - U(t_0)}_{10 \text{ J}} + \frac{1}{2}mv(t_1)^2 - \frac{1}{2}mv(t_0)^2 + \underbrace{mgz(t_1) - mgz(t_0)}_{0.057 \text{ kg} \cdot 9.8 \text{ m/s}^2 \cdot (880 \text{ m} - 1000 \text{ m})} = -57.032 \text{ J}$$

$$= -67.032 \text{ J}$$

$$\frac{1}{2}mv(t_1)^2 - \frac{1}{2}mv(t_0)^2 = 0 \text{ J}$$

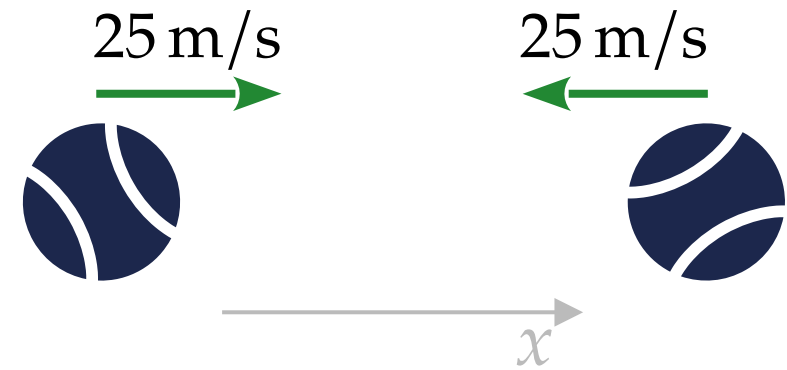
final kinetic energy is the same as before

→ velocity must be the same, 12 m/s

(This can also be solved numerically, finding $v(t_1) = \sqrt{\dots} = \pm 12 \text{ m/s}$)



In a 1D coordinate system x , at a particular time two tennis balls are moving towards each other in outer space with opposite velocities of magnitude 25 m/s , at some distance from each other. Each ball has mass 0.06 kg . There is no gravity and no other matter or electromagnetic fields outside of the balls.



(i) Consider the control volume, made of two separate parts, consisting of the two balls. What is the **total** momentum content of this volume?

(ii) Which physical principle and which constitutive relation did/would you use to calculate the total momentum content?

After some time, the tennis balls have collided and are moving away from each other, in the opposite directions they came from.

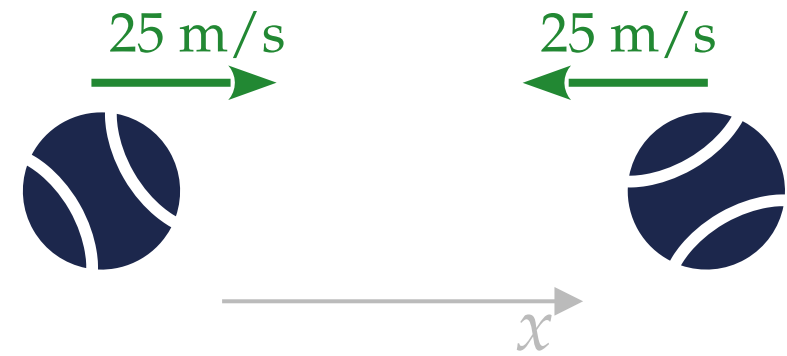
(iii) How much is the total momentum content after the collision?

(iv) Is it possible to find the magnitude of the velocity of each ball after the collision, using the balance of momentum?

(v) We are told that the **internal energy of each ball** has increased by 12 J after the collision. Find the velocities after the collision.

(vi) Which balance law, which constitutive relation, and which physical principle did/would you use to solve (v)?

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- (i) Consider the control volume, made of two separate parts, consisting of the two balls. What is the **total** momentum content of this volume?
- (ii) Which physical principle and which constitutive relation did/would you use to calculate the total momentum content?

(i, ii) To calculate the momentum content we can use **Newton's relation for momentum**:
$$P = m v$$

but it can only be used if the velocity is the same throughout the control volume. This is true for each ball separately:

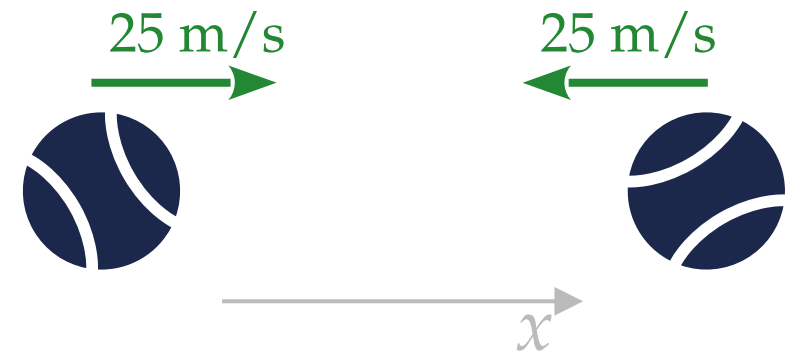
$$P_{\text{left}} = (0.06 \text{ kg}) \cdot (25 \text{ m/s}) = 1.5 \text{ N s}$$

$$P_{\text{right}} = (0.06 \text{ kg}) \cdot (-25 \text{ m/s}) = -1.5 \text{ N s}$$

To find the total momentum we use **the principle of extensivity**: the total is simply the sum of the separate parts:

$$P_{\text{tot,initial}} = P_{\text{left}} + P_{\text{right}} = 0 \text{ N s}$$

In a 1D coordinate system x , at a particular time two tennis balls are moving towards each other in outer space with opposite velocities of magnitude 25 m/s , at some distance from each other. Each ball has mass 0.06 kg . There is no gravity and no other matter or electromagnetic fields outside of the balls.



After some time, the tennis balls have collided and are moving away from each other, in the opposite directions they came from.

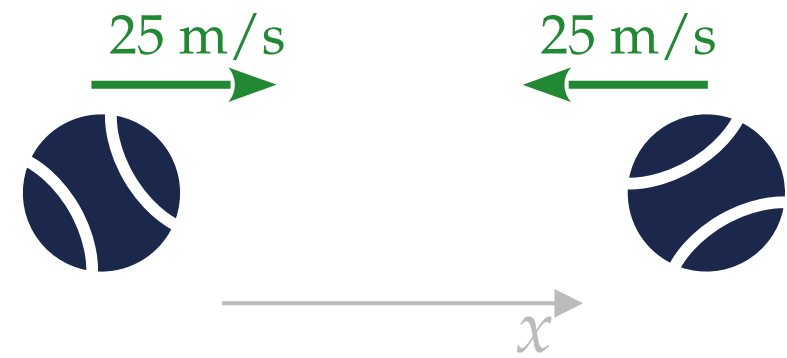
(iii) How much is the total momentum content after the collision?

(iii) We use the **balance of momentum** in integral form, knowing that the momentum flux into the **total** volume, and the momentum supply are zero:

$$P_{\text{tot}}(t_1) = P_{\text{tot}}(t_0) + \underbrace{\int_{t_0}^{t_1} F(t) dt}_0 + \underbrace{\int_{t_0}^{t_1} G(t) dt}_0 \quad \Rightarrow \quad P_{\text{tot}}(t_1) = P_{\text{tot}}(t_0) = 0 \text{ N s}$$

The momentum after the collision is still zero.

In a 1D coordinate system x , at a particular time two tennis balls are moving towards each other in outer space with opposite velocities of magnitude 25 m/s , at some distance from each other. Each ball has mass 0.06 kg . There is no gravity and no other matter or electromagnetic fields outside of the balls.



After some time, the tennis balls have collided and are moving away from each other, in the opposite directions they came from.

(iv) Is it possible to find the magnitude of the velocity of each ball after the collision, using the balance of momentum?

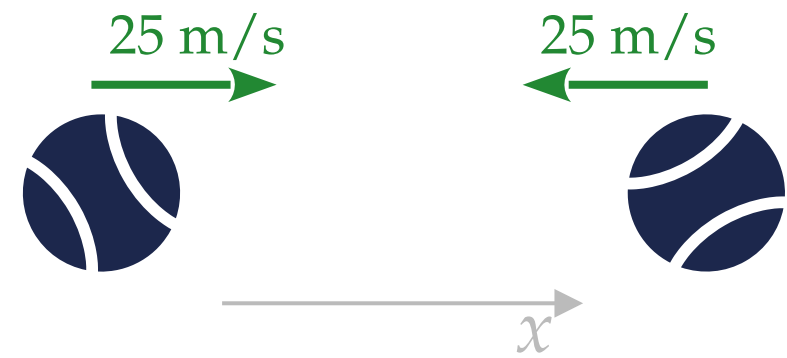
(iv) We can apply Newton's relation for momentum $P = mv$ and the principle of extensivity, as in question (i):

$$0 \text{ Ns} = P_{\text{tot,final}} = P_{\text{left,final}} + P_{\text{right,final}} = m v_{\text{left,final}} + m v_{\text{right,final}}$$

$$\Rightarrow 0 \text{ Ns} = m v_{\text{left,final}} + m v_{\text{right,final}} \quad \Rightarrow \quad v_{\text{left,final}} = -v_{\text{right,final}}$$

So we find that their final velocities must be equal in magnitude but opposite. But we can't say how much their magnitude is; the balance of momentum doesn't tell us.

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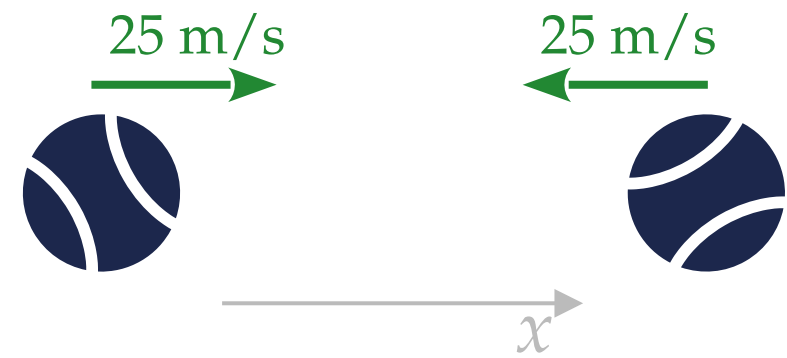
(v,vi) We can use the **balance of energy** in integral form, together with the **constitutive relation for the energy content of matter**, without gravitational-energy term: $E = U + \frac{1}{2}mv^2$. This can be used only for regions where the velocity is the same. So we use the **principle of extensivity** to find the total energy

We know that there are no fluxes of energy into the total control volume, and the energy supply is practically zero in our small region of universe:

$$E_{\text{tot}}(t_1) = E_{\text{tot}}(t_0) + \underbrace{\int_{t_0}^{t_1} \Phi(t) dt}_0 \quad \Rightarrow \quad E_{\text{tot}}(t_1) = E_{\text{tot}}(t_0) \quad \Rightarrow \quad E_{\text{left}}(t_1) + E_{\text{right}}(t_1) = E_{\text{left}}(t_0) + E_{\text{right}}(t_0)$$

$$U_{\text{left}}(t_1) + \frac{1}{2} m v_{\text{left}}(t_1)^2 + U_{\text{right}}(t_1) + \frac{1}{2} m v_{\text{right}}(t_1)^2 = U_{\text{left}}(t_0) + \frac{1}{2} m v_{\text{left}}(t_0)^2 + U_{\text{right}}(t_0) + \frac{1}{2} m v_{\text{right}}(t_0)^2$$

In a 1D coordinate system x , at a particular time two tennis balls are moving towards each other in outer space with opposite velocities of magnitude 25 m/s, at some distance from each other. Each ball has mass 0.06 kg. There is no gravity and no other matter or electromagnetic fields outside of the balls.



After some time, the tennis balls have collided and are moving away from each other, in the opposite directions they came from.

(v) We are told that the internal energy of each ball has increased by 12 J after the collision. Find the velocities after the collision.

(vi) Which balance law, which constitutive relation, and which physical principle did/would you use to solve (v)?

(v,vi) We can use the **balance of energy** in integral form, together with the **constitutive relation for the energy content of matter**, without gravitational-energy term: $E = U + \frac{1}{2}mv^2$. This can be used only for regions where the velocity is the same. So we use the **principle of extensivity** to find the total energy

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$$v_{\text{left}}(t_1) = -v_{\text{right}}(t_1)$$

$$U_{\text{left}}(t_1) = U_{\text{left}}(t_0) + 12 \text{ J}$$

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$$U_{\text{left}}(t_0) + 12 \text{ J} + \frac{1}{2} m v_{\text{right}}(t_1)^2 + U_{\text{right}}(t_0) + 12 \text{ J} + \frac{1}{2} m v_{\text{right}}(t_1)^2 =$$

$$U_{\text{left}}(t_0) + \frac{1}{2} m v_{\text{left}}(t_0)^2 + U_{\text{right}}(t_0) + \frac{1}{2} m v_{\text{right}}(t_0)^2$$

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$$\cancel{U_{\text{left}}(t_0)} + 12 \text{ J} + \frac{1}{2} m v_{\text{right}}^? (t_1)^2 + \cancel{U_{\text{right}}(t_0)} + 12 \text{ J} + \frac{1}{2} m v_{\text{right}}^? (t_1)^2 =$$

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$$m v_{\text{right}}(t_1)^2 = -24 \text{ J} + \frac{1}{2} m v_{\text{left}}(t_0)^2 + \frac{1}{2} m v_{\text{right}}(t_0)^2$$

$$v_{\text{right}}(t_1)^2 = -24 \text{ J}/m + \frac{1}{2} v_{\text{left}}(t_0)^2 + \frac{1}{2} v_{\text{right}}(t_0)^2$$

$$v_{\text{right}}(t_1)^2 = -24 \text{ J}/(0.06 \text{ kg}) + 0.5 \cdot (25 \text{ m/s})^2 + 0.5 \cdot (25 \text{ m/s})^2$$

$$v_{\text{right}}(t_1)^2 = 225 \text{ m}^2/\text{s}^2$$

$$v_{\text{right}}(t_1) = \pm 15 \text{ m/s} \quad v_{\text{left}}(t_1) = \mp 15 \text{ m/s}$$

The text says “the tennis balls have collided and are moving away from each other, in the opposite directions they came from”:

$$v_{\text{left}}(t_1) = -15 \text{ m/s} \quad v_{\text{right}}(t_1) = +15 \text{ m/s}$$

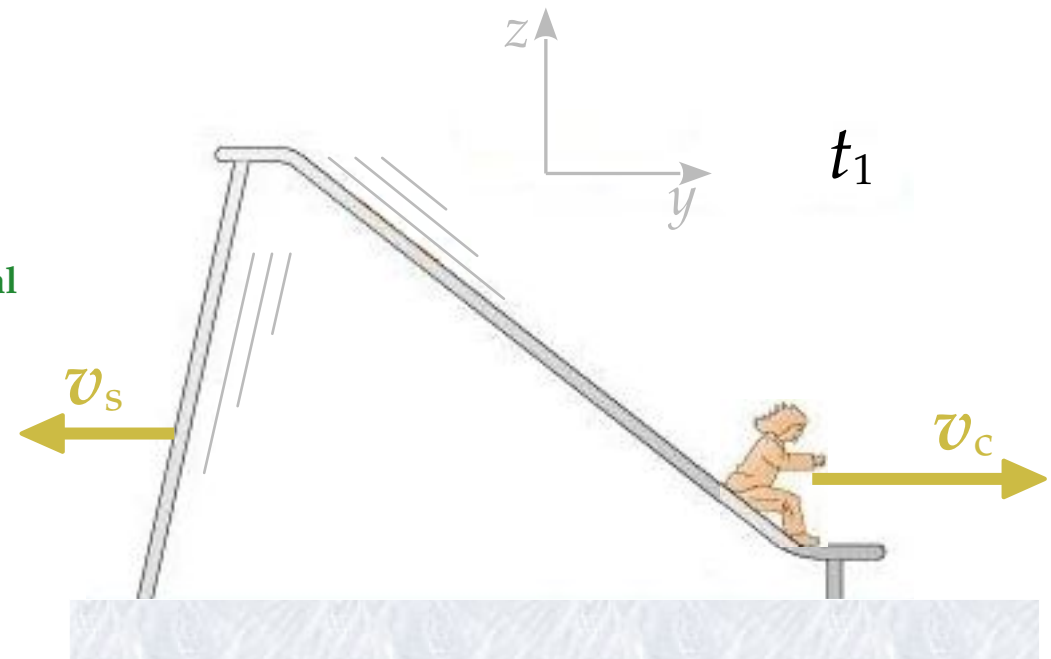
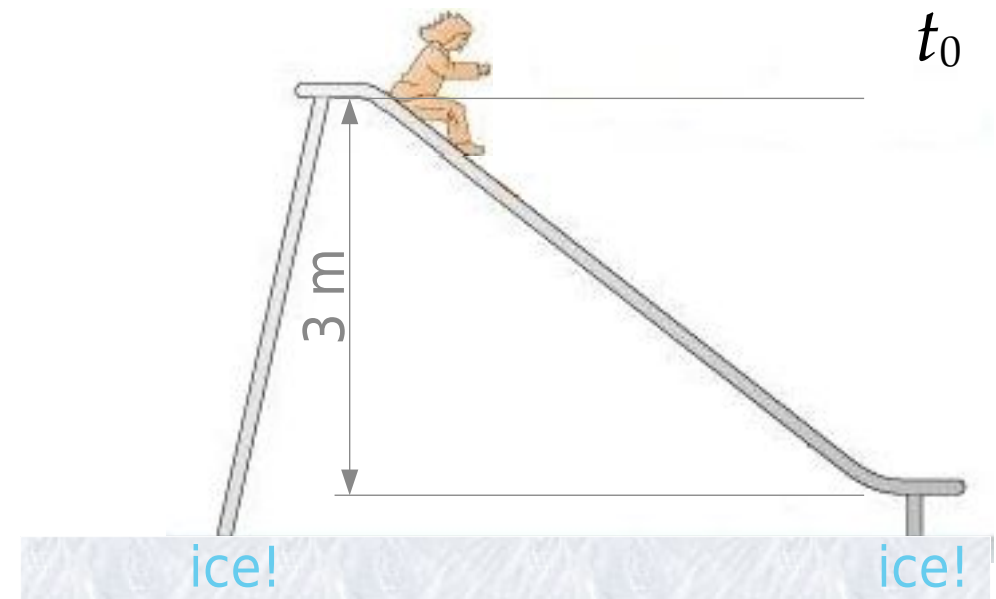
Someone had the brilliant idea of placing a slide on ice, without anchoring it to the ground. The slide ramp is 3 m high.

A child climbs up the slide, and at time t_0 sits at the top, at rest. It then slides down. At time t_1 the child is at the bottom, travelling with a horizontal velocity v_c ; at the same time the slide is gliding on the ice with velocity v_s .

The child has mass 30 kg, the slide has mass 50 kg. Consider air and friction between slide and ice to be negligible. There are no heat fluxes and any internal energy is constant. Consider the child as a point, so that it can be characterized by one vertical z and one horizontal y coordinate. The gravitational potential energy of the slide is constant.

(i) Consider the ramp & child together as a single, deformable control volume. Collect together and write down:

- the integral expressions of the balances of momentum and of energy for this control volume;
- all constitutive relations that seem appropriate to this physical situation;
- the expressions of the principles of extensivity and of flux symmetry, if they apply to this situation.

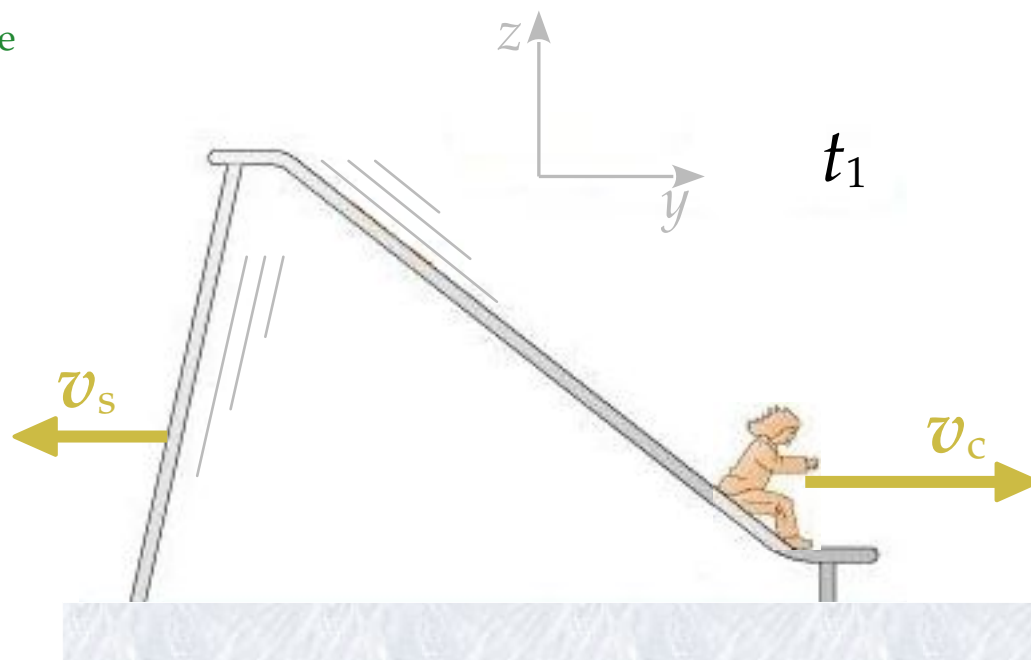
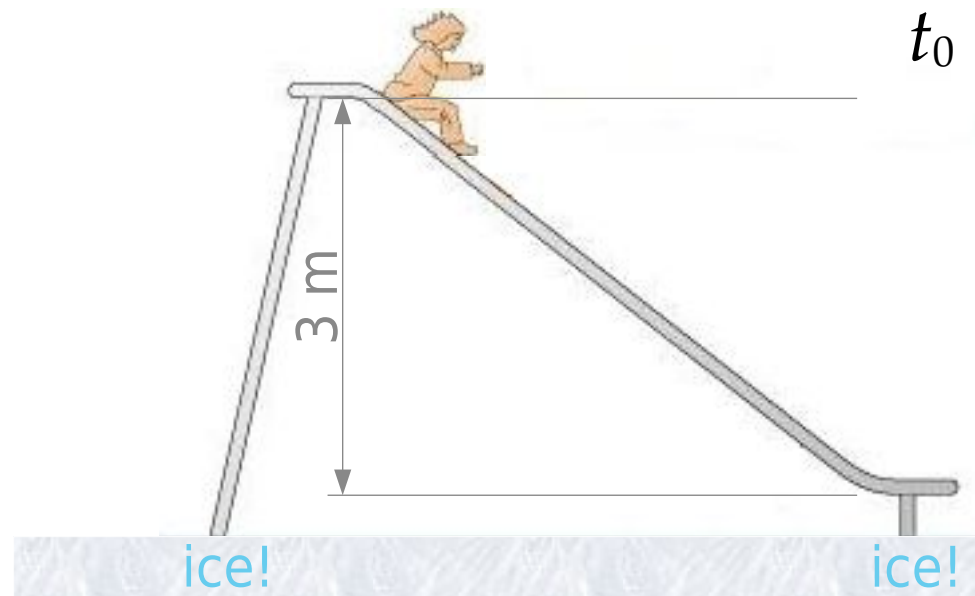


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(ii) Calculate v_s and v_c and report them in vector form (complete with units!) using coordinates (y, z) . **Explain all the physics laws and principles that you use in your calculations.**

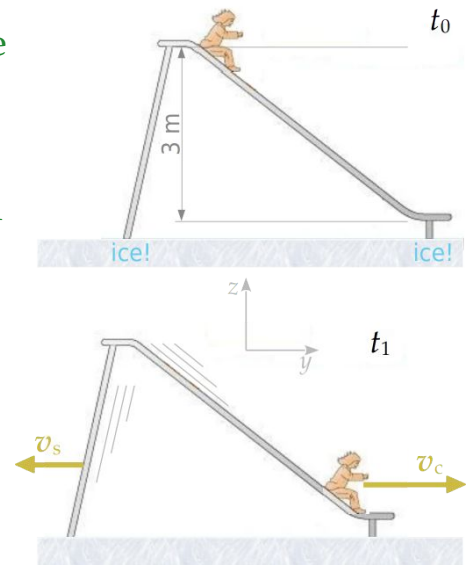


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Calculate v_s and v_c and report them in vector form (complete with units!) using coordinates (y, z) . Explain all the physics laws and principles that you use in your calculations.



Given the quantities involved, this problem seems to require the balance of momentum, or of energy, or both. The balance of momentum gives us two equations (one per coordinate), and the balance of energy gives us one. We know that the z -components of the velocities are zero, so the problem effectively asks for *two* numbers (horizontal components of two velocities), so probably we need the balances of energy and the balance of momentum for the y -component. Since data are given at two times, the integral expression of these balance is appropriate.

Which control volumes to choose? Note that there seems to be fluxes of momentum between the slide and the child, but they are *not* given in the problem. So the best idea seems to consider only one control volume, comprising child and slide. But we can still use the principle of extensivity with this volume if necessary. Let's use "c" for "child" and "s" for "slide".

Let's use the **balance of y -momentum** first. For simplicity I omit all " y " indices for the components, writing " P " instead of " P_y " etc.:

$$P(t_1) = P(t_0) + \int F(t) dt + \int G(t) dt \quad (\text{limits of integration between } t_0 \text{ and } t_1).$$

The total momentum of our control volume can be obtained by **extensivity**, considering the subvolume with the child and the subvolume with the slide. As constitutive relation, we use **Newton's formula for the momentum of matter**, $P = mv$.

Time t_0 : all velocities are zero, so $P(t_0) = P_c(t_0) + P_s(t_0) = 0 \text{ N}\cdot\text{s} + 0 \text{ N}\cdot\text{s}$ (the sum of momenta comes from **extensivity**)

Time t_1 : $P(t_1) = P_c(t_1) + P_s(t_1) = m_c v_c + m_s v_s$

Time-integrated flux: the forces from the air to the child and slide, and the horizontal forces on the slide are zero, so $\int F(t) dt = 0 \text{ N}\cdot\text{s}$

Time-integrated supply: gravity has no horizontal component, so $\int G(t) dt = 0 \text{ N}\cdot\text{s}$

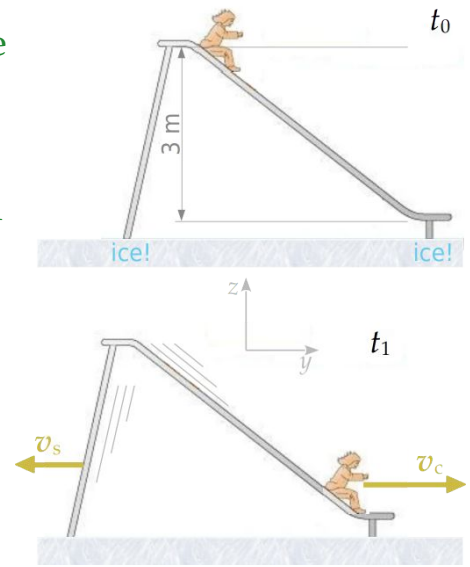
Putting all these back in the balance we have $m_c v_c + m_s v_s = 0 \text{ N}\cdot\text{s} + 0 \text{ N}\cdot\text{s} + 0 \text{ N}\cdot\text{s} \Rightarrow m_c v_c = -m_s v_s$ which we keep for later

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The child has mass 30 kg, the slide has mass 50 kg. Consider air and friction between slide and ice to be negligible; there are no heat fluxes and any internal energy is constant. Consider the child as a small volume so that it can be characterized by one vertical and one horizontal coordinate only. The gravitational potential energy of the slide is constant.

Calculate v_s and v_c and report them in vector form (complete with units!) using coordinates (y, z) . Explain all the physics laws and principles that you use in your calculations.



Now let's focus on the **balance of energy**: $U(t_1) = U(t_0) + \int \Phi(t) dt$. We use again the principle of **extensivity** for energy content, and use the **constitutive relation for total-energy content of matter**: $E = U + \frac{1}{2}mv^2 + mgz$, and for **total-energy flux**: $\Phi = Q + F \cdot v$. Note that the problem says that there are no heat fluxes, and all internal energies are constant. Also note that we don't know how to calculate the gravitational potential energy of the slide, but it is obviously constant because the slide doesn't move vertically.

We use again **extensivity** of energy content, summing the energy for the control volume with child and that with slide:

Time t_0 : all velocities are zero, so $E(t_0) = E_c(t_0) + E_s(t_0) = U_c + 0J + m_c g z_c + U_s + 0J + (\text{pot. energy slide})$ with $z_c = 3 \text{ m}$.

Time t_1 : $E(t_1) = E_c(t_1) + E_s(t_1) = U_c + \frac{1}{2}m_c v_c^2 + 0J + U_s + \frac{1}{2}m_s v_s^2 + (\text{pot. energy slide})$

Time-integrated flux: We know the heat flux is zero: $Q = 0J/s$. But what about the mechanical power? The force from ground to slide is vertical, but the velocity of the slide is horizontal: they are perpendicular. Therefore $F \cdot v = 0J/s$ for the slide. The time-integrated total-energy flux is therefore zero: $\int \Phi(t) dt = 0J$.

Putting all these in the balance of energy, we find

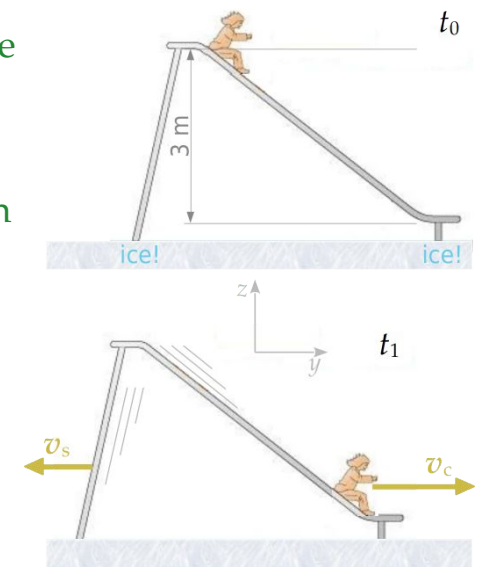
$$\begin{aligned}
 \cancel{U_c + \frac{1}{2}m_c v_c^2 + 0J} + \cancel{U_s + \frac{1}{2}m_s v_s^2 + (\text{pot. energy slide})} &= \cancel{U_c + 0J + m_c g z_c} + \cancel{U_s + 0J + (\text{pot. energy slide})} \\
 \Rightarrow \frac{1}{2}m_c v_c^2 + \frac{1}{2}m_s v_s^2 &= m_c g z_c \\
 \Rightarrow \frac{1}{2}m_c v_c^2 + \frac{1}{2}m_s v_s^2 &\approx 882J
 \end{aligned}$$

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Calculate v_s and v_c and report them in vector form (complete with units!) using coordinates (y, z) . Explain all the physics laws and principles that you use in your calculations.



We can finally put together the equation obtained from the momentum balance and the one obtained from the energy balance, and solve them:

$$\begin{cases} m_c v_c = -m_s v_s \\ \frac{1}{2} m_c v_c^2 + \frac{1}{2} m_s v_s^2 = 882 \text{ J} \end{cases} \quad \begin{cases} v_c = -\frac{m_s}{m_c} v_s \\ m_c v_c^2 + m_s v_s^2 = 1764 \text{ J} \end{cases} \quad \begin{cases} v_c = -\frac{m_s}{m_c} v_s \\ m_c \left(\frac{m_s}{m_c} v_s \right)^2 + m_s v_s^2 = 1764 \text{ J} \end{cases} \quad \begin{cases} v_c = -\frac{m_s}{m_c} v_s \\ \frac{m_s^2}{m_c} v_s^2 + m_s v_s^2 = 1764 \text{ J} \end{cases}$$

$$\begin{cases} v_c = -\frac{m_s}{m_c} v_s \\ \frac{(50 \text{ kg})^2}{30 \text{ kg}} v_s^2 + 50 \text{ kg } v_s^2 = 1764 \text{ J} \end{cases} \quad \begin{cases} v_c = -\frac{m_s}{m_c} v_s \\ 83 \text{ kg } v_s^2 + 50 \text{ kg } v_s^2 = 1764 \text{ J} \end{cases} \quad \begin{cases} v_c = -\frac{50 \text{ kg}}{30 \text{ kg}} v_s \\ 133 \text{ kg } v_s^2 = 1764 \text{ J} \end{cases} \quad \begin{cases} v_c = -1.6 v_s \\ v_s^2 = \frac{1764}{133} \text{ m}^2/\text{s}^2 \end{cases}$$

$$\begin{cases} v_c = -1.7 \times (\pm 3.6 \text{ m/s}) \\ v_s = \pm 3.6 \text{ m/s} \end{cases} \quad \begin{cases} v_c = +6.1 \text{ m/s} \\ v_s = -3.6 \text{ m/s} \end{cases}$$

slide moves left, negative y -direction

$$v_c = [+6.1, 0] \text{ m/s}$$

$$v_s = [-3.6, 0] \text{ m/s}$$

(Comment: the child's velocity is unrealistically high! In a real situation there's friction between child and slide, this causes a heat flux from child to slide, and an increase of internal energy of the slide. So the child has less kinetic energy.)



A skydiver jumps with zero initial velocity at an altitude of 3000 m. What would be the skydiver's velocity at 1000 m, if there were no changes in internal energy and no energy fluxes? Consider the skydiver as a small control volume that can be characterized by one vertical coordinate. The gravity acceleration is 9.8 m/s^2 .

Explain which laws you use.

(Compare the velocity you found and the typical velocity of 200 km/h that a skydiver may have at 1000 m, when the parachute is deployed.)





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We can use the **balance of energy** in integral form and the constitutive relation for the **energy content of matter** :

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same

$$\cancel{\frac{1}{2}mv(t_1)^2} + \cancel{mgz(t_1)} = \cancel{\frac{1}{2}mv(t_0)^2} + \cancel{mgz(t_0)}$$

$$\frac{1}{2}v(t_1)^2 + gz(t_1) = \frac{1}{2}v(t_0)^2 + gz(t_0)$$

$$v(t_1) = \sqrt{2 \left[\frac{1}{2}v(t_0)^2 + gz(t_0) - gz(t_1) \right]}$$

$$= \sqrt{2 \left[0 \text{ m}^2/\text{s}^2 + 9.8 \text{ m/s}^2 \cdot 3000 \text{ m} - 9.8 \text{ m/s}^2 \cdot 1000 \text{ m} \right]} \approx 198 \text{ m/s} \approx 713 \text{ km/h}$$



A skydiver, of mass 70 kg, jumps with zero initial velocity at an altitude of 3000 m. At 1000 m the skydiver has a velocity of 200 km/h, ready to deploy the parachute.

Assume that the *internal* energy of the skydiver is the same at 1000 m as at 3000 m.

(i) How much is the time-integrated energy *influx* to the skydiver from the surrounding air?

(ii) Can you tell how much are the heat flux and the flux of mechanical power, separately, given the data in this problem?





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$$\underbrace{\cancel{U(t_1)} + \frac{1}{2}mv(t_1)^2 + mgz(t_1)}_{\text{same}} = \underbrace{\cancel{U(t_0)} + \frac{1}{2}mv(t_0)^2 + mgz(t_0)}_{\text{same}} + \int_{t_0}^{t_1} \Phi(t) dt$$

$$\int_{t_0}^{t_1} \Phi(t) dt = \frac{1}{2}mv(t_1)^2 + mgz(t_1) - \frac{1}{2}mv(t_0)^2 - mgz(t_0)$$

$$= \frac{1}{2}70 \text{ kg} \cdot (55.6 \text{ m/s})^2 + 70 \text{ kg} \cdot 9.8 \text{ m/s}^2 \cdot 1000 \text{ m} - 0 \text{ J} - 70 \text{ kg} \cdot 9.8 \text{ m/s}^2 \cdot 3000 \text{ m}$$

$$\approx -1.26 \times 10^6 \text{ J} \quad \text{the skydiver has lost energy}$$

Impossible to say how much of this is heat and how much is mechanical power (from friction).

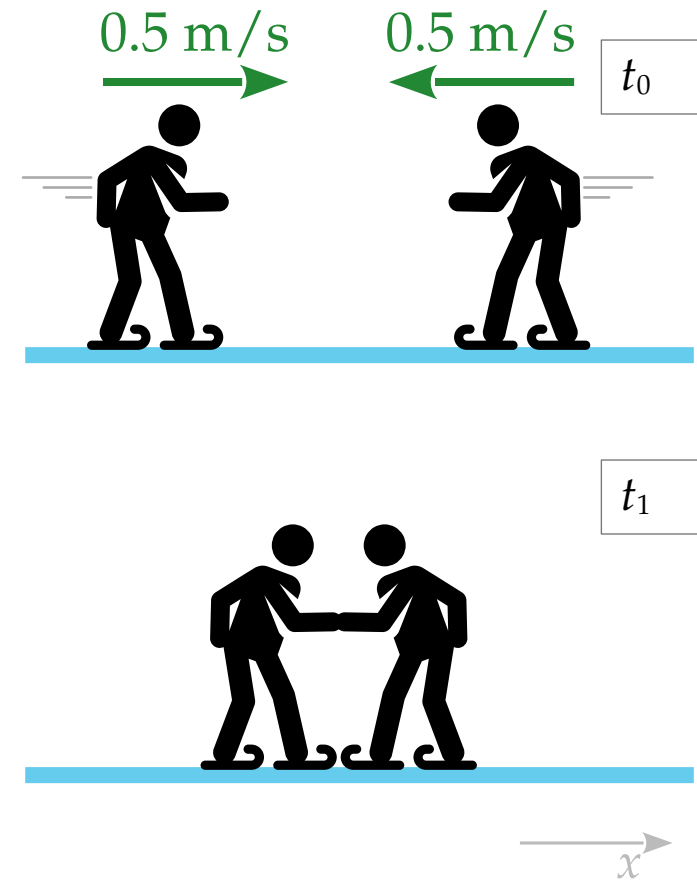


Two ice-skaters move towards each other with opposite speeds of magnitude 0.5 m/s . When they meet, they hold each other, slowing down and completely stopping their motion.

The ice exerts a normal force on each skater, but assume that **there's no friction**, and there's **no heat flux** between skaters and ice. The gravitational energy can be taken to be constant.

(i) Show that the *internal* energy of the skaters after they stop **cannot** be the same as when they were in motion toward each other. Explain which balance law and constitutive relations you have to use to show this.

(ii) Find out how much their total internal energy has increased.



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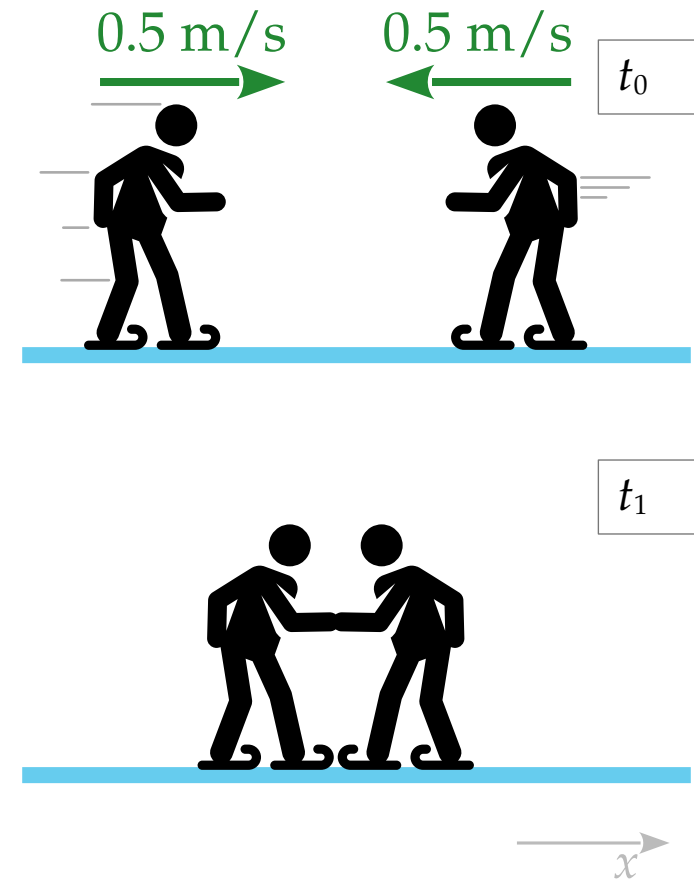
The ice exerts a normal force on each skater, but assume that **there's no friction**, and there's **no heat flux** between skaters and ice. The gravitational energy is negligible.

(i) Show that the *internal* energy of the skaters after they stop **cannot** be the same as when they were in motion toward each other. Explain which balance law and constitutive relations you have to use to show this.

(ii) Find out how much their total internal energy has increased, assuming their mass is 60 kg each.

We can use the **balance of energy** in integral form and the constitutive relations for the **energy content** and **energy flux of matter** :

$$E(t_1) = E(t_0) + \underbrace{\int_{t_0}^{t_1} \Phi(t) dt}_{\text{energy flux}} \quad E = U + \frac{1}{2}mv^2 + mgz \quad \Phi = Q + F \cdot v$$



These must be used with the **principle of extensivity** to find the totals

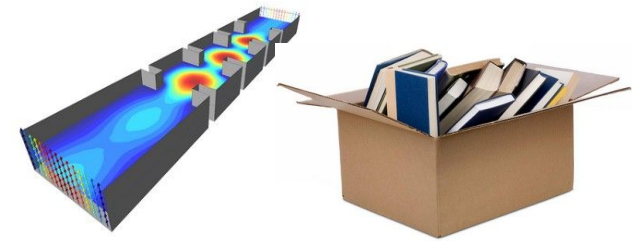
$$U_L(t_1) + \underbrace{\frac{1}{2}m_L v_L(t_1)^2}_0 + U_R(t_1) + \underbrace{\frac{1}{2}m_R v_R(t_1)^2}_0 = U_L(t_0) + \frac{1}{2}m_L v_L(t_0)^2 + U_R(t_0) + \frac{1}{2}m_R v_R(t_0)^2$$

$$U_L(t_1) + U_R(t_1) = U_L(t_0) + U_R(t_0) + \underbrace{\frac{1}{2}m_L v_L(t_0)^2 + \frac{1}{2}m_R v_R(t_0)^2}_{\text{non-zero}}$$

increase in total internal energy: $\frac{1}{2}m_L v_L(t_0)^2 + \frac{1}{2}m_R v_R(t_0)^2 \approx \frac{1}{2}60 \text{ kg} \cdot (0.5 \text{ m/s})^2 \times 2 \approx 15 \text{ J}$



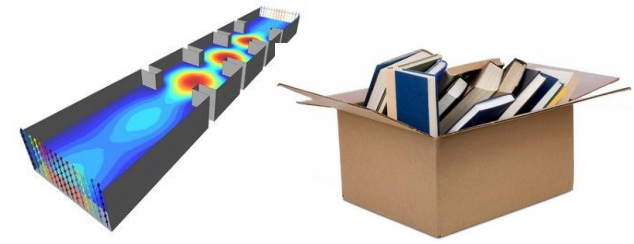
A box (of negligible mass) contains electromagnetic radiation having total momentum $[0, -1, 3]$ Ns. Another box (also of negligible mass) contains ten books. Each book has mass 0.2 kg and velocity $[0, 0.03, 0]$ m/s.



- (i) What is the total momentum content of the two boxes together?
- (ii) Which physical principles and constitutive relations did/would you use to find the result (i)?



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(i) What is the total momentum content of the two boxes together?

(ii) Which physical principles and constitutive relations did/would you use to find the result (i)?

We use the **principle of extensivity** for momentum content, for the two boxes together and for the ten books within the second box. For the ten books we use **Newton's constitutive relation for momentum** $\mathbf{P} = m \mathbf{v}$.

$$\mathbf{P}_{\text{tot}} = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix} \text{ N s} + 10 \times 0.2 \text{ kg} \cdot \begin{bmatrix} 0 \\ 0.03 \\ 0 \end{bmatrix} \text{ m/s} = \begin{bmatrix} 0 \\ -0.94 \\ 3 \end{bmatrix} \text{ N s}$$



A suction cup has been placed on a bathroom's wall. The total force exerted by the atmosphere on the cup corresponds to a pressure of 10^5 N/m^2 over an area of 0.004 m^2 . The cup has a mass of 0.04 kg . The coefficient of static friction between the cup and the wall is 0.8 .

A person would like to hang three towels on the cup, for a total mass of 2 kg . Will the cup manage to hold the towels? The gravitational acceleration is 9.8 N/kg



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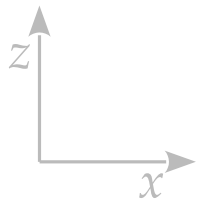
A person would like to hang three towels on the cup, for a total mass of 2 kg . Will the cup manage to hold the towels? The gravitational acceleration is 9.8 N/kg

The wall exerts a horizontal **normal force** F_n to the suction cup. This normal force must be opposite to the horizontal component of all other momentum influxes (because the cup isn't moving horizontally, so its horizontal momentum is constant, zero). Let's use coordinates (x, z) .

The only horizontal force on the cup is from the atmosphere:

$$F_{\text{atm}} = p_{\text{atm}} \cdot \text{area} = 10^5 \text{ N/m}^2 \cdot 0.004 \text{ m}^2 = 400 \text{ N}$$

$$\Rightarrow F_n = -400 \text{ N} \quad (x\text{-direction})$$



There is **static friction** between cup and wall. It's directed vertically. The minimum amount of vertical force necessary to move the cup is given by the formula for the maximal static friction:

$$F_s = 0.8 \cdot |F_n| = 320 \text{ N}$$

Attaching a mass of 2 kg , plus the mass 0.04 kg of the cup, would create a downward force, from the gravitational momentum supply, of magnitude

$$F_{\text{weight}} = (2+0.04) \text{ kg} \cdot 9.8 \text{ N/kg} \approx 20 \text{ N} \quad (\text{negative } z\text{-direction})$$

so the cup should be able to hold the towels without any problem.



A block of material is in a constant state, that is, its energy content, entropy content, matter content, and so on are **all constant in time**.

There are no fluxes of any kind, except a constant **heat influx of 100 J/s** at the top and a constant **heat efflux of 100 J/s** at the bottom.

According to experimental temperature report (A), the constant temperatures at the top and bottom are

$$T_{\text{top}} = 300 \text{ K}$$

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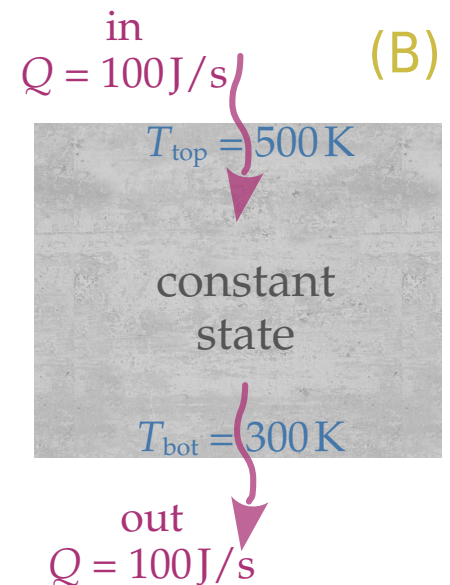
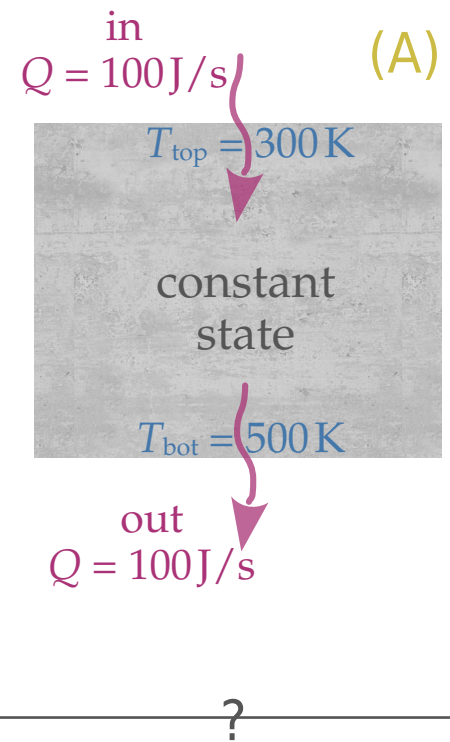
$$T_{\text{top}} = 500 \text{ K}$$

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One of the two reports is wrong.

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Relevant here are the **balances of energy and of entropy**

$$\frac{dE(t)}{dt} = \Phi(t) \qquad \frac{dS(t)}{dt} \geq \Pi(t)$$

together with the constitutive relation for **energy and entropy fluxes**

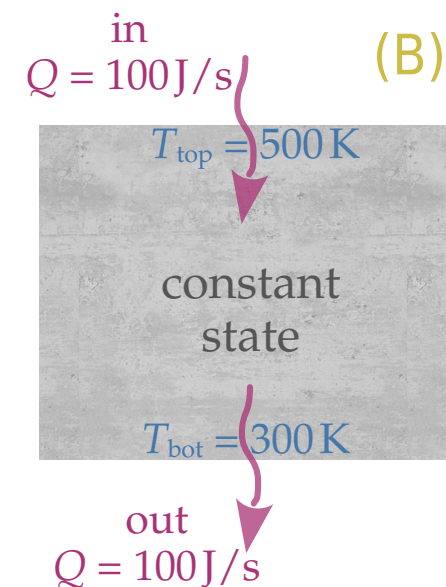
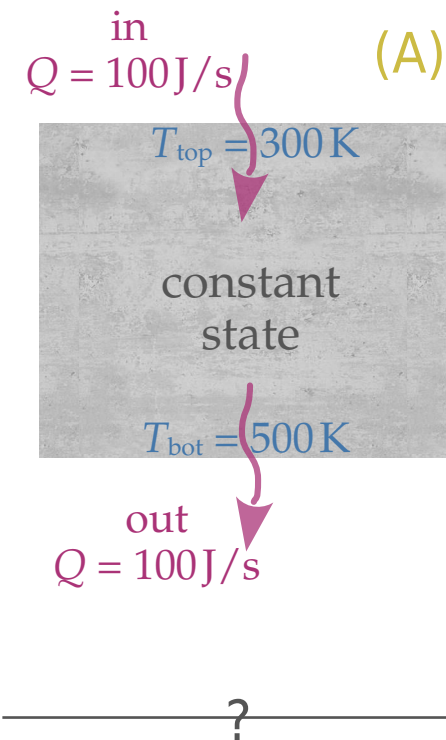
$$\Phi = Q + F \cdot v \qquad \Pi = \frac{Q}{T}$$

and the **principle of extensivity** to calculate the **total** fluxes. We have $F \cdot v = 0$.

The **time derivatives** of E and S are **zero** according to the assumptions.

(A):

(B):





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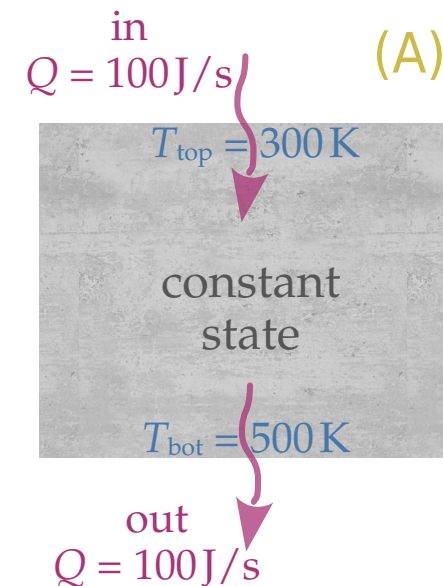
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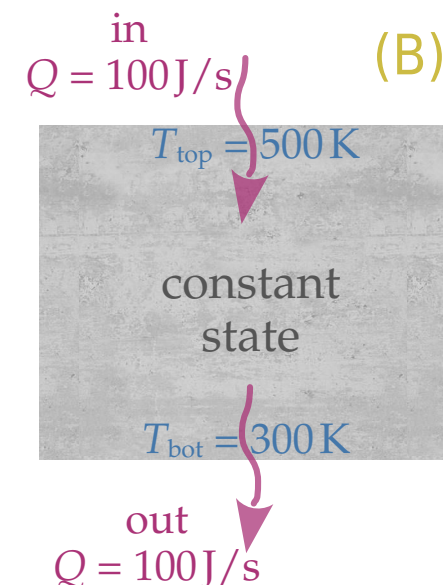
$$(A): \Phi_{\text{tot}} = +100 \text{ J/s} - 100 \text{ J/s} = 0 \text{ J/s} \quad \frac{dE(t)}{dt} = \Phi_{\text{tot}}(t) \Rightarrow 0 \text{ J/s} = 0 \text{ J/s}$$

top bottom

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_____ ? _____





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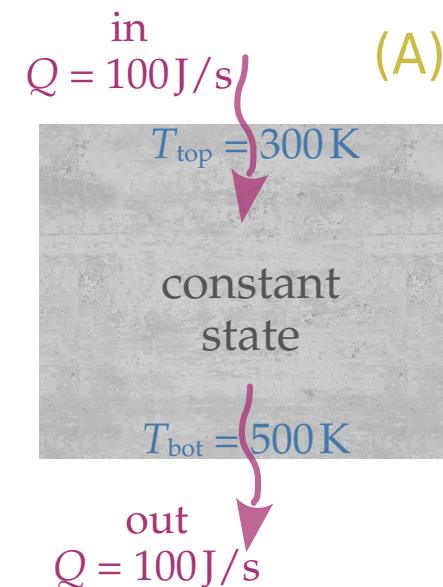
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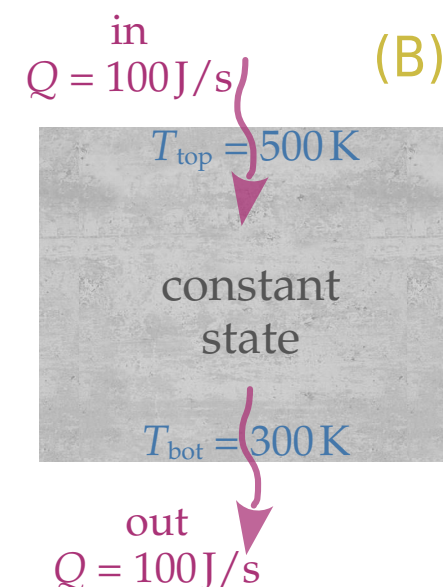
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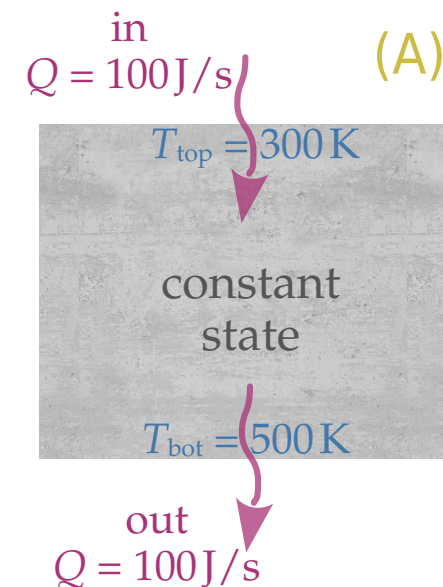
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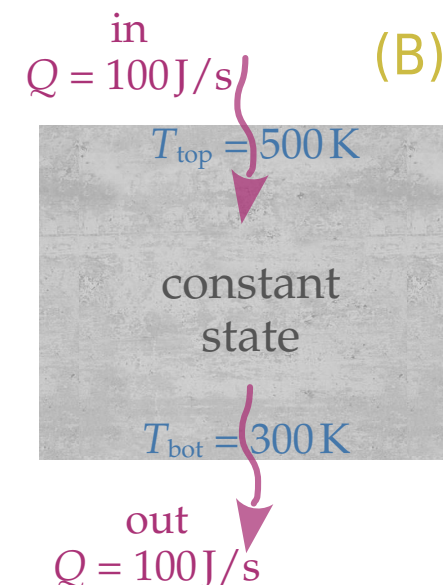
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! balance of entropy is broken!

(B):



_____ ? _____





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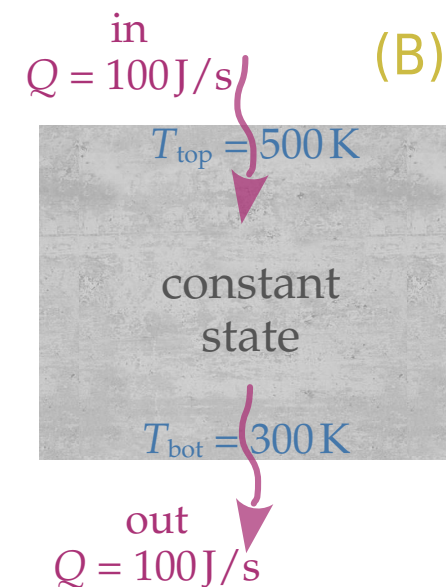
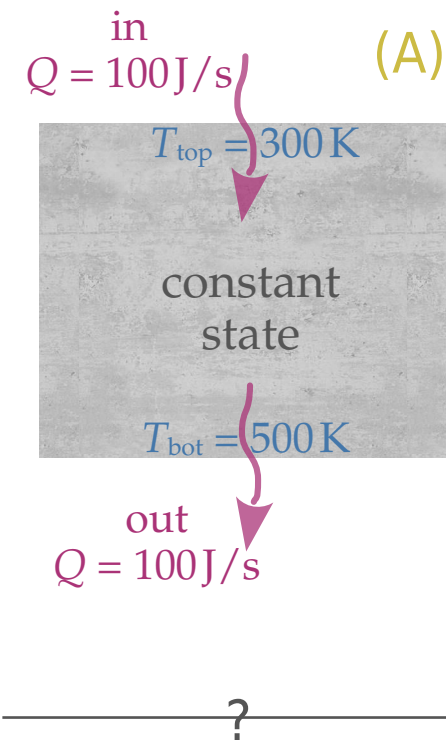
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! balance of entropy is broken!

$$(B): \Pi_{\text{tot}} = +\frac{100 \text{ J/s}}{500 \text{ K}} - \frac{100 \text{ J/s}}{300 \text{ K}} \approx -0.13 \text{ J/(K s)} \quad \frac{dS(t)}{dt} \geq \Pi_{\text{tot}}(t) \Rightarrow 0 \text{ J/(K s)} \geq -0.13 \text{ J/(K s)}$$

✓

(B) is the correct report!

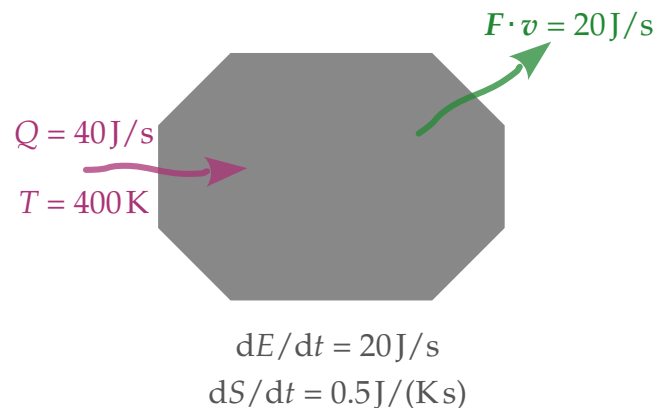




The following picture illustrate a control volume with one or more fluxes of heat Q and momentum F , temperatures T , velocities of matter v at different portions of its bounding surface, as well as the time-rate of change of the total-energy and entropy contents, dE/dt and dS/dt .

Assume that there's matter both inside and outside, but *no* matter fluxes, and no electromagnetic phenomena. Be careful about the directions of all fluxes!

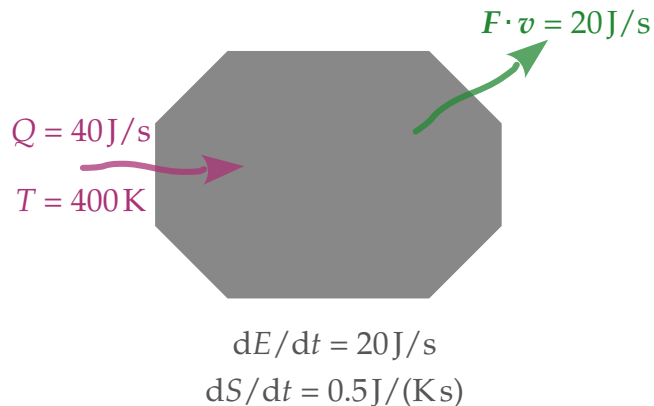
Establish whether the balances of energy and entropy are satisfied for this control volume.



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Teaching note for this and the following solutions of this kind of physics problem:

☞ In order to check the balance of energy, we need the net influx of energy through the whole control surface.

☞ The influx of energy through a surface is calculated (**in this problem**) using the **constitutive formula for the energy flux of matter**: $\Phi = Q + F \cdot v$. We can use this because there are **no matter fluxes** and no electromagnetic phenomena.

☞ The net influx through the whole surface usually is calculated by using the **principle of extensivity**: calculate the influxes through different parts of the surface, then sum up to find the total influx.

☞ In order to check the balance of entropy, we need the net influx of entropy through the whole control surface.

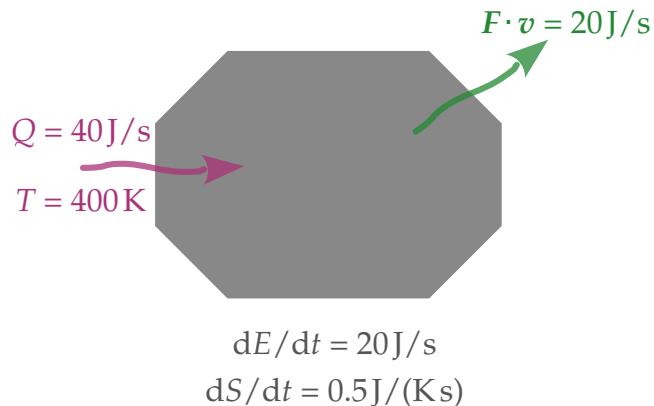
☞ The influx of energy through a surface is calculated (**in this problem**) using the **constitutive formula for the entropy flux of matter**: $\Pi = Q/T$. We can use this because there are **no matter fluxes** and no electromagnetic phenomena.

☞ The net influx through the whole surface usually is calculated by using the **principle of extensivity**: calculate the influxes through different parts of the surface, then sum up to find the total influx.

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Establish whether the balances of energy and entropy are satisfied for this control volume.



Balance of energy:

$$dE/dt = Q_{\text{tot}} + (F \cdot v)_{\text{tot}}$$

$$dE/dt = 20 \text{ J/s}$$

$$Q_{\text{tot}} = +40 \text{ J/s}$$

$$(F \cdot v)_{\text{tot}} = -20 \text{ J/s}$$

$$20 \text{ J/s} = +40 \text{ J/s} - 20 \text{ J/s} \quad \checkmark \text{ YES}$$

Balance of entropy:

$$dS/dt \geq \Pi_{\text{tot}}$$

$$dS/dt = 0.5 \text{ J/(Ks)}$$

$$\Pi_{\text{tot}} = +40/400 \text{ J/(Ks)} = 0.1 \text{ J/(Ks)}$$

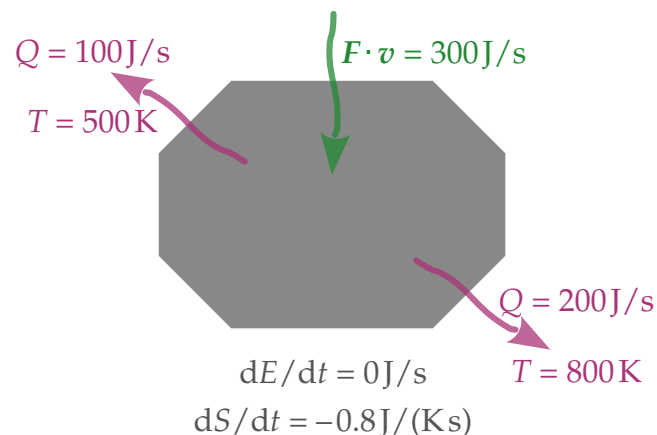
$$0.5 \text{ J/(Ks)} \geq 0.1 \text{ J/(Ks)} \quad \checkmark \text{ YES}$$

?

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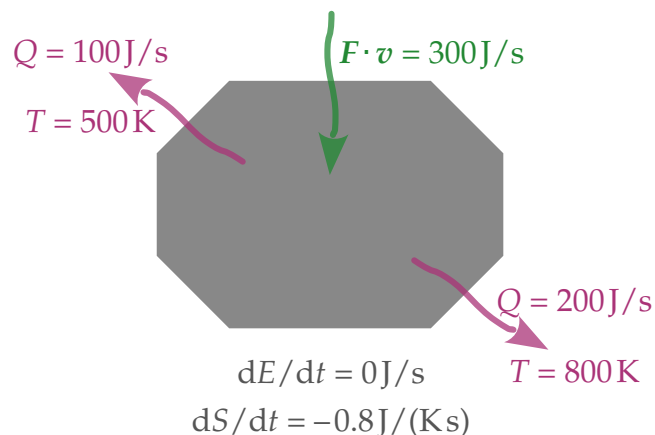


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Establish whether the balances of energy and entropy are satisfied for this control volume.



Balance of energy:

$$dE/dt = Q_{\text{tot}} + (F \cdot v)_{\text{tot}}$$

$$dE/dt = 0 \text{ J/s}$$

$$Q_{\text{tot}} = (-100 - 200) \text{ J/s} = -300 \text{ J/s}$$

$$(F \cdot v)_{\text{tot}} = +300 \text{ J/s}$$

$$0 \text{ J/s} = -300 \text{ J/s} + 300 \text{ J/s} \quad \checkmark \text{ YES}$$

Balance of entropy:

$$dS/dt \geq \Pi_{\text{tot}}$$

$$dS/dt = -0.8 \text{ J/(Ks)}$$

$$\Pi_{\text{tot}} = (-100/500 - 200/800) \text{ J/(Ks)} = -0.45 \text{ J/(Ks)}$$

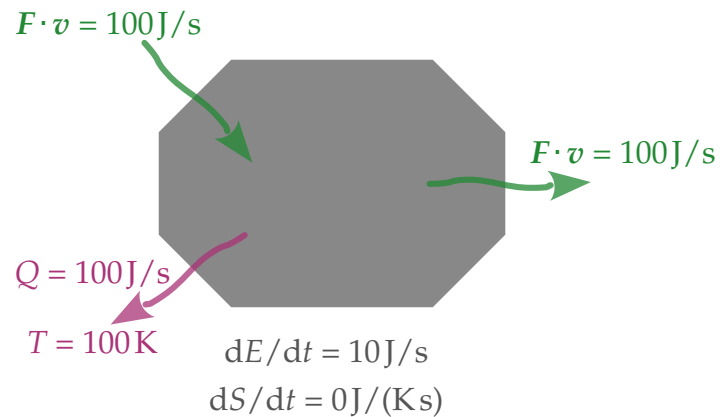
$$-0.8 \text{ J/(Ks)} \geq -0.45 \text{ J/(Ks)} \quad \times \text{ NO!}$$



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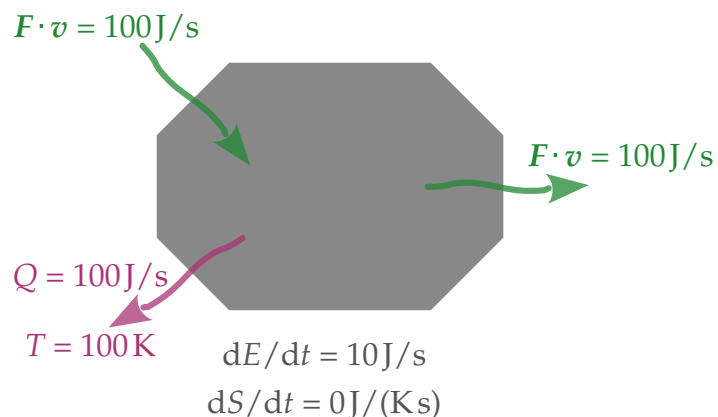


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Balance of energy:

$$dE/dt = Q_{\text{tot}} + (F \cdot v)_{\text{tot}}$$

$$dE/dt = 10 \text{ J/s}$$

$$Q_{\text{tot}} = -100 \text{ J/s}$$

$$(F \cdot v)_{\text{tot}} = (+100 - 100) \text{ J/s} = 0 \text{ J/s}$$

$$10 \text{ J/s} = -100 \text{ J/s} + 0 \text{ J/s} \quad \times \text{ NO!}$$

Balance of entropy:

$$dS/dt \geq \Pi_{\text{tot}}$$

$$dS/dt = 0 \text{ J/(Ks)}$$

$$\Pi_{\text{tot}} = -100/100 \text{ J/(Ks)} = -1 \text{ J/(Ks)}$$

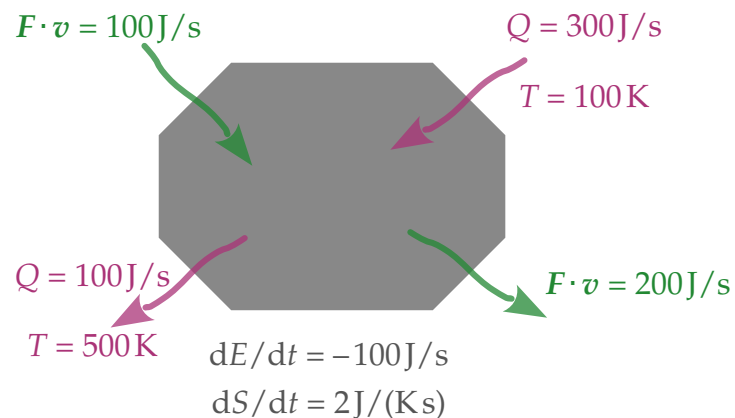
$$0 \text{ J/(Ks)} \geq -1 \text{ J/(Ks)} \quad \checkmark \text{ YES}$$

?

The following picture illustrate a control volume with one or more fluxes of heat Q and momentum F , temperatures T , velocities of matter v at different portions of its bounding surface, as well as the time-rate of change of the total-energy and entropy contents, dE/dt and dS/dt .

Assume that there's matter both inside and outside, but *no* matter fluxes, and no electromagnetic phenomena. Be careful about the directions of all fluxes!

Establish whether the balances of energy and entropy are satisfied for this control volume.

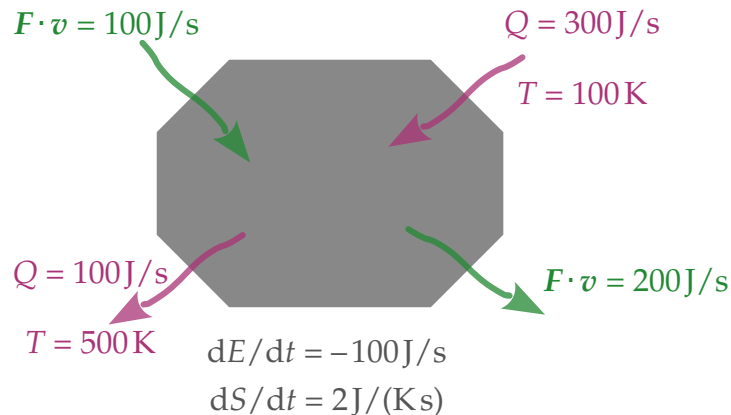


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Balance of energy:

$$dE/dt = Q_{\text{tot}} + (F \cdot v)_{\text{tot}}$$

$$dE/dt = -100 \text{ J/s}$$

$$Q_{\text{tot}} = (-100 + 300) \text{ J/s} = +200 \text{ J/s}$$

$$(F \cdot v)_{\text{tot}} = (+100 - 200) \text{ J/s} = -100 \text{ J/s}$$

$$-100 \text{ J/s} = +200 \text{ J/s} - 100 \text{ J/s} \quad \times \text{ NO!}$$

Balance of entropy:

$$dS/dt \geq \Pi_{\text{tot}}$$

$$dS/dt = 2 \text{ J/(Ks)}$$

$$\Pi_{\text{tot}} = (-100/500 + 300/100) \text{ J/(Ks)} = +2.8 \text{ J/(Ks)}$$

$$2 \text{ J/(Ks)} \geq +2.8 \text{ J/(Ks)} \quad \times \text{ NO!}$$

Oppgave 1

A robot arm must lift, vertically, a block of concrete having mass $m = 100\text{ kg}$ (see figure). Take the block itself as a control volume, and use coordinates (x, z) as illustrated; $z = 0\text{ m}$ corresponds to the floor. Assume that there is no atmosphere. The gravitational acceleration is $g = 9.8\text{ N/kg}$.

(a)

- i. Calculate the **total** force (momentum flux) that the gripper (the robot's "hand") must exert on the block in order to lift it with a constant upward speed.

Express this total force as a vector with x - and z -components.

Which law do you use to calculate the total force? which constitutive relations?



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Which law do you use to calculate the total force? which constitutive relations?



The question involves motion and forces, so use of the **balance of momentum** seems to be required.

(i) If the block must have *constant* upward speed, then it must have *constant* upward momentum content because of **Newton's formula for momentum** $P = m v$. Given that momentum is constant and the question requires the total force (momentum flux), it's best to use the balance of momentum in differential form:

$$\frac{dP_x}{dt} = F_x + G_x$$

$$\frac{dP_z}{dt} = F_z + G_z$$

The question asks for $[F_x, F_z]$. We do have dP/dt : it's zero because the velocity and therefore the momentum is constant (the derivative of a constant is zero). We also have G : it's given by the constitutive relation for **gravitational force** $G = -mg[0,1]$. Therefore

$$\frac{dP_x}{dt} = F_x + G_x$$

$$0 = F_x + 0$$

$$\frac{dP_z}{dt} = F_z + G_z$$

$$0 = F_z - mg$$

and we find $F = [0\text{ N}, mg]$, with $mg \approx 100 \cdot 9.8\text{ N} = 980\text{ N}$.

- ii. The left and right fingers of the gripper exert forces having these unknown components:

$$\begin{array}{cc} \text{left} & \text{right} \\ \begin{bmatrix} A \\ \mu A \end{bmatrix} & \begin{bmatrix} -B \\ \mu B \end{bmatrix} \end{array}$$

with $A > 0$ and $B > 0$, where $\mu = 0.21$ is called sliding-friction coefficient.

Find A and B ; don't forget the physical units.

Which principle about fluxes and volume contents are you using, in order to find these two forces from the total one?

- iii. What is the **magnitude** of the force exerted by the **left** finger of the gripper?

- ii. The left and right fingers of the gripper exert forces having these unknown components:

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(ii) The total force $F = [0, mg]$ found in (i) is, by the principle of **extensivity of fluxes**, the sum of all forces (that is, momentum fluxes) acting on the block on all its surfaces. There's no air, so the only surfaces with a momentum flux are the sides, where the gripper's fingers grip the block. Let's call them $F_{\text{left}} = [A, \mu A]$ and $F_{\text{right}} = [B, \mu B]$. So we must have:

$$F = F_{\text{left}} + F_{\text{right}}$$

or explicitly in components:

$$\begin{bmatrix} 0 \\ mg \end{bmatrix} = \begin{bmatrix} A \\ \mu A \end{bmatrix} + \begin{bmatrix} -B \\ \mu B \end{bmatrix} \Rightarrow \begin{cases} 0 = A - B \\ mg = \mu A + \mu B \end{cases} \Rightarrow \begin{cases} B = A \\ mg = 2\mu A \end{cases} \Rightarrow A = B = \frac{mg}{2\mu}$$

with $mg/(2\mu) \approx 980 \text{ N}/(2 \cdot 0.21) \approx 2333 \text{ N}$.

(iii) The magnitude of the force by the left gripper is

$$\sqrt{A^2 + (\mu A)^2} = \sqrt{(1 + \mu^2) A^2} = \sqrt{1 + \mu^2} A = \sqrt{1 + 0.21^2} \cdot 2333 \text{ N} \approx 2384 \text{ N}$$

(b)

The block is being lifted at a constant upward speed of 0.01 m/s. When the bottom of the block of concrete is at $z = 1$ m from the ground, the gripper suddenly breaks and does not exert any force anymore.

Find the magnitude of the velocity $\boldsymbol{v}$ that the block possesses right before it hits the ground:

- i. Find the magnitude of $\boldsymbol{v}$ using the balance of energy. Assume that the internal energy of the block is constant in time. (*Warning: remember that the block had an upward velocity while it was lifted!*)

Explain how you use the energy balance: what is the energy content? fluxes? supply?

(i) The question asks for quantities at a later time, so it seems best to use the **balance of energy** in integral form:

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt + \int_{t_0}^{t_1} R(t) dt$$

applied to the block itself as control volume. The time t_0 is when the gripper breaks, and t_1 when the block touches the ground.

We need constitutive relations for energy content E , energy flux Φ , energy supply R :

- E : The velocity is the same for the whole block. Different parts of the block are at different heights z , but maybe we can consider the block as enough small to just use the same height z throughout. We therefore use the constitutive relation for the **energy content of matter**:

$$E = U + \frac{1}{2}mv^2 + mgz$$

The problem gives us $m, g, z(t_0)=1\text{ m}, z(t_1)=0\text{ m}, v(t_0)=0.01\text{ m/s}$, and says that $U(t_1)=U(t_0)$. We want $v(t_1)$.

- Φ : The block is not in contact with anything, so the **energy flux** is zero. Its time-integral is then also zero.
- R : In our small region of the universe, the **energy supply** is approximately zero.

$$E(t_1) = E(t_0) + \int_{t_0}^{t_1} \Phi(t) dt + \int_{t_0}^{t_1} R(t) dt$$

$$\cancel{U(t_1)} + \frac{1}{2}mv(t_1)^2 + mgz(t_1) = \cancel{U(t_0)} + \frac{1}{2}mv(t_0)^2 + mgz(t_0) + 0 + 0$$

?

$$\Rightarrow \frac{1}{2}mv(t_1)^2 = \frac{1}{2}mv(t_0)^2 + mgz(t_0) - mgz(t_1)$$

$$\Rightarrow v(t_1) = \sqrt{\frac{2}{m} \left[\frac{1}{2}mv(t_0)^2 + mgz(t_0) - mgz(t_1) \right]} \approx \sqrt{2 \left[\frac{1}{2} (0.01)^2 + 9.8 \cdot 1 - 9.8 \cdot 0 \right]} \text{ m/s} \approx 4.4 \text{ m/s}$$

(b)

The block is being lifted at a constant upward speed of 0.01 m/s. When the bottom of the block of concrete is at $z = 1\text{ m}$ from the ground, the gripper suddenly breaks and does not exert any force anymore.

Find the magnitude of the velocity v that the block possesses right before it hits the ground:

- Find the magnitude of v using the balance of energy. Assume that the internal energy of the block is constant in time. (Warning: remember that the block had an upward velocity while it was lifted!)

Explain how you use the energy balance: what is the energy content? fluxes? supply?

- ?
- ii. Using *only* the balance of energy (and no intuitive reasoning), can you find not only the magnitude of $\mathbf{v}$, but also its x - and z -components? Explain your reasoning about why this is or is not possible. *[This question does not have a right or wrong answer; what's important is to show your reasoning.]*
 - iii. Use the balance of momentum to explain why the x -component of $\mathbf{v}$ must be zero. *(Hint: had the initial momentum any x -component? Did the total flux or supply of momentum have any x -component during the fall?)*

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(ii) The balance of energy gives us **only the magnitude** (squared) of the velocity, not its individual components or its direction. In fact, the velocity at time t_0 (when the gripper breaks) could have been completely horizontal. Then at time t_1 it would probably be partly horizontal, partly vertical.

In the present problem we can say that the velocity $\mathbf{v}$ is purely vertical and downward by a *hidden application of the balance of momentum*, but not with the application of only the balance of energy:

- there are no horizontal momentum fluxes, and the initial velocity had no x -component initially. So it can never acquire any x -component later. Therefore $v_x = 0$ m/s always. **This answers question (iii).**
- the vertical momentum can only change because of the gravitational force, which points downward. So this force adds negative values to the vertical component of velocity. The vertical component can therefore never exceed $v_z = +0.01$ m/s. The final magnitude we find is 4.4 m/s, so we must have $v_z = -4.4$ m/s.

(c)

The floor is made of a special material that deforms (see picture) and gently stops the fall of the block.

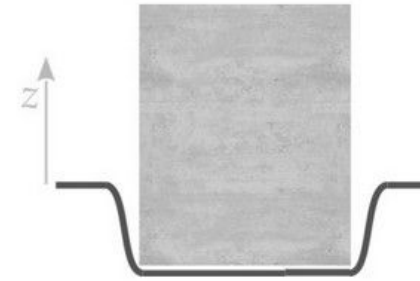
Disregard the horizontal dimension, and consider a z -coordinate directed upward.

As soon as the block is in contact with the floor, the floor starts exerting a vertical force (momentum flux) F on the block. This force has constitutive equation

$$F(t) = -k z(t) - \mu v(t)$$

where $z(t) \leq 0$ is the position of the block's bottom and $v(t)$ its velocity, at time t . The velocity is positive if it points upward. The constant k is an elastic constant, and μ is a dampening constant.

- i. Find the units of k and those of μ .



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i. Find the units of k and those of μ .

(i) The terms in a sum, and the two sides of an equality, *must all have the same units*. F is a force (momentum flux), with units *newtons* (N); z is a position, with units *metres* (m); v is a velocity, with units *metres/second* (m/s). So we must have

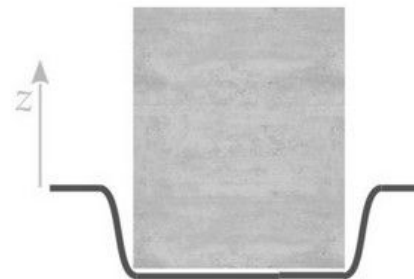
$$\text{N} = [\text{units of } k] \cdot \text{m} = [\text{units of } \mu] \cdot \text{m/s}$$

from this we find

$$[\text{units of } k] = \text{N/m} \quad \text{or equivalently} \quad = \text{kg/s}^2$$

$$[\text{units of } \mu] = \text{N}/(\text{m/s}) \equiv \text{N} \cdot \text{s}/\text{m} \quad \text{or equivalently} \quad = \text{kg/s}$$

The last equalities come from $\text{N} \equiv \text{kg} \cdot \text{m}/\text{s}^2$



?

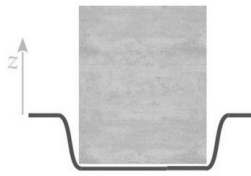
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- ii. Assume that $k = 1500$ and $\mu = 1000$, measured in the units you found. How much is the force $F(t_0)$ (including its sign) at the very moment t_0 the block hits the floor? (*Hint: you already calculated the velocity at impact in a previous question, and the floor correspond to the zero of the z coordinate.*)

?

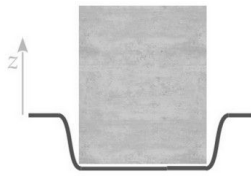
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(ii) (This question uses t_0 for the time the block hits the floor). We found that the velocity of the block right when it hits the floor is $v(t_0) = -4.4 \text{ m/s}$ (minus because it's downwards). The coordinate is $z(t_0) = 0 \text{ m}$ then. So we find

$$F(t_0) = -(1500 \text{ N/m}) \cdot (0 \text{ m}) - (1000 \text{ N}\cdot\text{s/m}) \cdot (-4.4 \text{ m/s}) = +4400 \text{ N}$$

This force, as the plus-sign shows, points upwards, which makes intuitive sense.

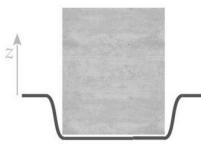
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- iii. Between the moment t_0 when the block hits the floor, and the moment t_1 when it stops, the integrated flux of momentum given to the block by the floor is

$$\int_{t_0}^{t_1} F(t) dt = +3392.42 \text{ N s}$$

Use the balance of momentum in integral form to find the integrated supply of momentum G between t_0 and t_1 . (Hint: How much is the final momentum, when the block has stopped? how much was the initial momentum? Note that you do not need to solve any integrals.)

- iv. The integrated supply of momentum that you just found is, by analytical integration, also equal to

$$\int_{t_0}^{t_1} G(t) dt = -m g (t_1 - t_0)$$

From this expression and your previous result, find the time $(t_1 - t_0)$ that it took for the floor to completely stop the fall of the block of concrete.

(iii) The exercise suggests to use

$$P(t_1) = P(t_0) + \int_{t_0}^{t_1} F(t) dt + \int_{t_0}^{t_1} G(t) dt$$

Which quantities to we know in this equation?

- At t_1 the block has stopped, which means $P(t_1) = 0 \text{ N s}$.
- At t_0 (the moment of hitting the floor) the block has velocity $v = -4.4 \text{ m/s}$ calculated in a previous question. Using Newton's relation for momentum we have

$$P(t_0) = m v = 100 \text{ kg} \cdot (-4.4) \text{ m/s} = -440 \text{ N s}$$

- The question tells us that

$$\int_{t_0}^{t_1} F(t) dt = +3392.42 \text{ N s}$$

We can therefore easily find

$$\begin{aligned} \int_{t_0}^{t_1} G(t) dt &= P(t_1) - P(t_0) - \int_{t_0}^{t_1} F(t) dt \\ &= 0 \text{ N s} - (-440 \text{ N s}) - 3392.42 \text{ N s} \approx -2952 \text{ N s} \end{aligned}$$

(iv) We find

$$-2952 \text{ N s} = -m g (t_1 - t_0) \quad \Rightarrow \quad (t_1 - t_0) = \frac{2952 \text{ N s}}{m g} = \frac{2952 \text{ N s}}{100 \text{ kg} \cdot 9.8 \text{ N/kg}} \approx 3 \text{ s}$$

That is, the deforming floor stops the fall of the block in 3 seconds.

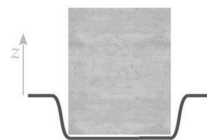
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Examples of simulation/code-based questions

The script on the side simulates a piston-gas system, using the constitutive relations shown under the picture and the balances of momentum (piston) and of energy (gas). The symbols are explained below.

(1) Which lines implement balance laws? Which balance laws?

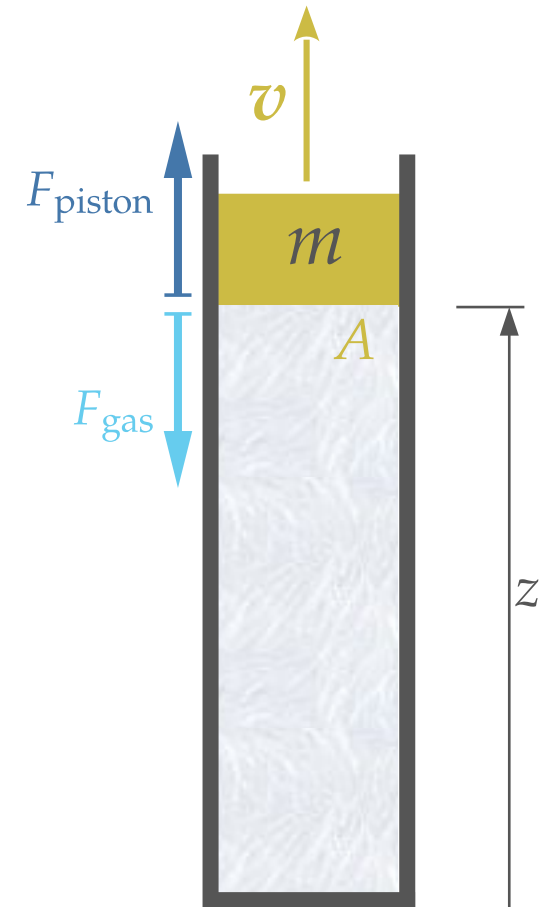
(2) Which lines implement constitutive relations? Which constitutive relations?

(3) Which lines implement the principle of symmetry of fluxes?

```

...
10 %% constant quantities
11 G = -m*g; % gravity force on piston
12 Q = 0.1; % heat flux to gas
13 %% initialize state & time
14 t = 0;
15 z = ...;
16 v = ...;
17 T = ...;
18 %% time loop
19 while t < t_final
20     %% transform from variables z,v,T
21     %% to P,F_piston,z,v,E,F_gas
22     P = m*v;
23     pr = N*R*T/(z*A);
24     F_piston = pr*A;
25     F_gas = -F_piston;
26     E = C*N*T;
27     %% update
28     P = P + (F_piston + G)*dt;
29     z = z + v*dt;
30     E = E + (Q + F_gas*v)*dt;
31     t = t + dt;
32     %% transform from P,z,E
33     %% back to variables z,v,T
34     v = P/m;
35     T = E/(C*N);
36 end

```



P : piston momentum
 z : piston position
 v : piston velocity
 A : piston area
 m : piston mass
 pr : gas pressure on piston
 F_piston : force on piston
 F_gas : force on gas
 E : gas energy
 T : gas temperature
 Q : heat flux (zero)
 G : gravitational force (fixed)

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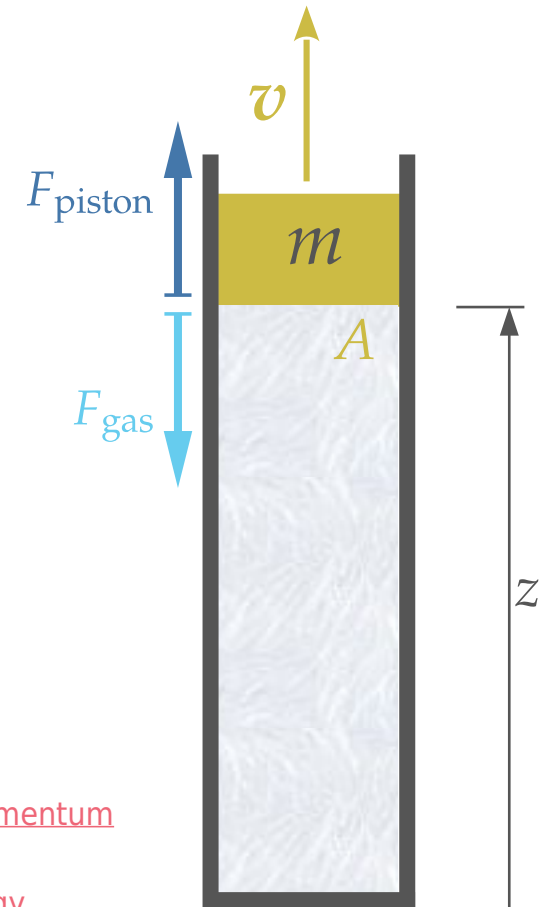
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26     E = C*N*T;
27     %% update
28     P = P + (F_piston + G)*dt; balance of momentum
29     z = z + v*dt; (balance of matter)
30     E = E + (Q + F_gas*v)*dt; balance of energy
31     t = t + dt;
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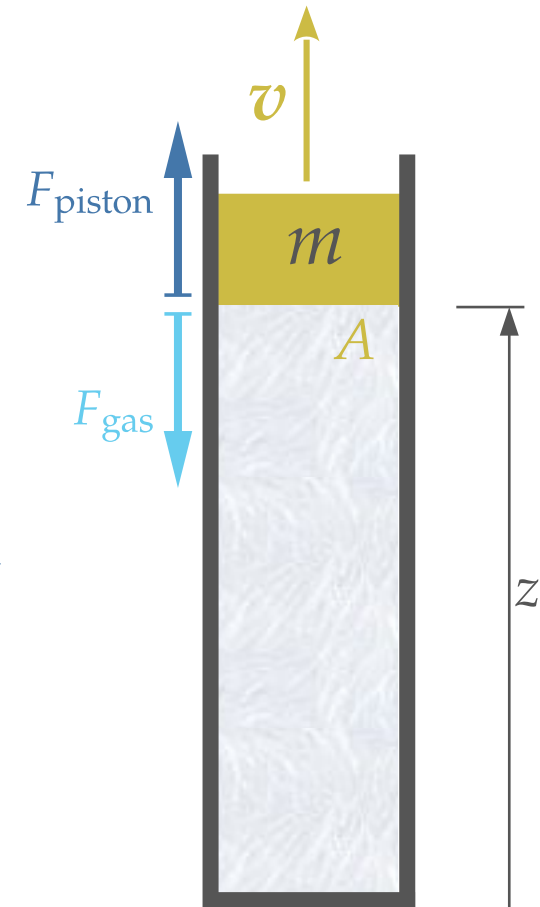
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21     %% to P,F_piston,z,v,E,F_gas
22     P = m*v; Newton's const. relation for momentum
23     pr = N*R*T/(z*A); ideal-gas law
24     F_piston = pr*A;
25     F_gas = -F_piston;
26     E = C*N*T; const. relation for energy of ideal gas
27     %% update
28     P = P + (F_piston + G)*dt;
29     z = z + v*dt;
30     E = E + (Q + F_gas*v)*dt;
31     t = t + dt;
32     %% transform from P,z,E
33     %% back to variables z,v,T
34     v = P/m; Newton's const. relation for momentum
35     T = E/(C*N); const. relation for energy of ideal gas
36 end

```



P : piston momentum
 z : piston position
 v : piston velocity
 A : piston area
 m : piston mass
 pr : gas pressure on piston
 F_{piston} : force on piston
 F_{gas} : force on gas
 E : gas energy
 T : gas temperature
 Q : heat flux (zero)
 G : gravitational force (fixed)

The script on the side simulates a piston-gas system, using the constitutive relations shown under the picture and the balances of momentum (piston) and of energy (gas). The symbols are explained below.

(1) Which lines implement balance laws? Which balance laws?

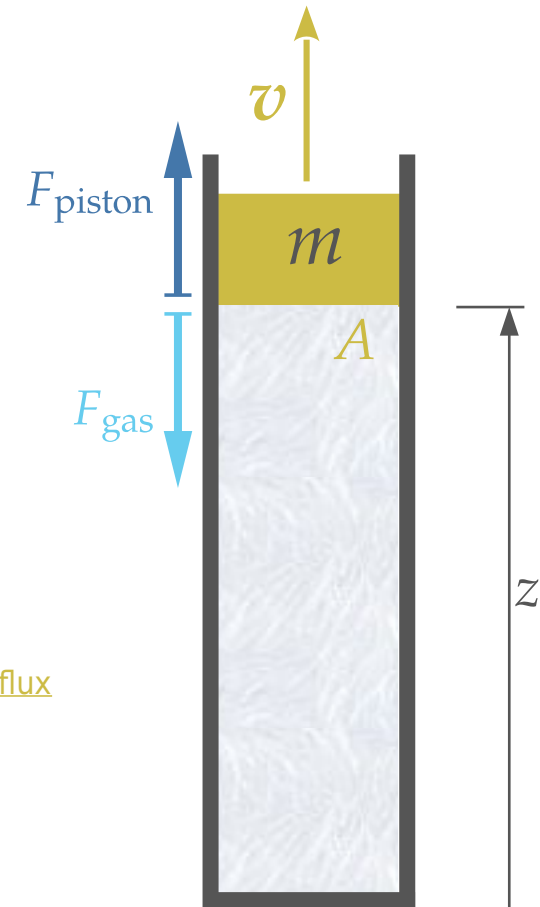
(2) Which lines implement constitutive relations? Which constitutive relations?

(3) Which lines implement the principle of symmetry of fluxes?

```

...
10 %% constant quantities
11 G = -m*g; % gravity force on piston
12 Q = 0.1; % heat flux to gas
13 %% initialize state & time
14 t = 0;
15 z = ...;
16 v = ...;
17 T = ...;
18 %% time loop
19 while t < t_final
20     %% transform from variables z,v,T
21     %% to P,F_piston,z,v,E,F_gas
22     P = m*v;
23     pr = N*R*T/(z*A);
24     F_piston = pr*A;
25     F_gas = -F_piston; symmetry of momentum flux
26     E = C*N*T;
27     %% update
28     P = P + (F_piston + G)*dt;
29     z = z + v*dt;
30     E = E + (Q + F_gas*v)*dt;
31     t = t + dt;
32     %% transform from P,z,E
33     %% back to variables z,v,T
34     v = P/m;
35     T = E/(C*N);
36 end

```



P : piston momentum
 z : piston position
 v : piston velocity
 A : piston area
 m : piston mass
 pr : gas pressure on piston
 F_piston : force on piston
 F_gas : force on gas
 E : gas energy
 T : gas temperature
 Q : heat flux (zero)
 G : gravitational force (fixed)

?

We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

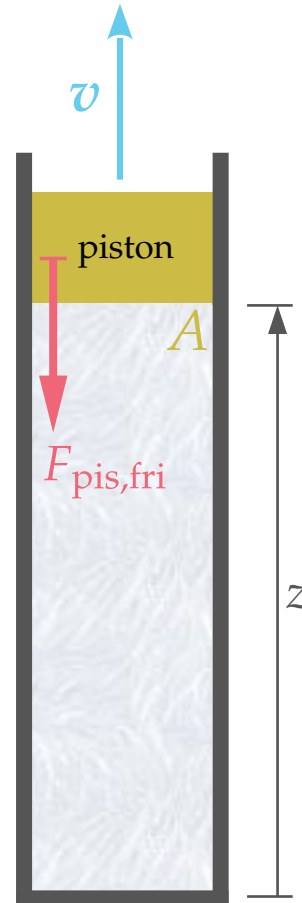
- The piston has mass m , area A , vertical coord. z
- Gravity is *non-negligible* for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(i) Which lines in the while-loop use constitutive relations? Report line number and name of the equation

```

1  %%% Numerical time integration of piston-gas with friction
2  %% SI units used throughout
3  %% Vertical coordinate z, positive upward
4
5  m = 50; % mass of piston
6  N = 0.04; % amount of ideal gas
7  A = 0.1^2; % area of piston
8  R = 8.31446261815; % molar gas constant
9  C = 20; % molar heat capacity of gas
10 mu = 0.00004; % gas viscosity
11 T_out = 300; % temperature of environment
12 mu = 0.8; % kinetic friction coefficient
13 g = 9.8; % gravitational acceleration
14
15 t_end = 11; % final time
16 dt = 0.0001; % time step
17
18 G = -m*g; % gravitational supply of momentum
19 Fn = 5; % normal force on piston
20 Q = 0; % heat flux for ideal gas (adiabatic)
21
22 t = 0; % initial time
23 z = 0.15; % initial position of piston
24 v = 0; % initial velocity of piston
25 T = 300; % initial temperature of gas
26
27 while t < t_end
28     P = m*v;
29     F_pis_fri = -sign(v) * mu * Fn;
30     F_pis_gas = A * ( N*R*T/(A*z) - mu*A*v/(A*z) );
31     F_pis_tot = F_pis_gas + F_pis_fri;
32     E = C*N*T;
33     F_gas_pis = -F_pis_gas;
34     phi = Q + F_gas_pis * v;
35     P = P + (F_pis_tot + G) * dt;
36     E = E + phi * dt;
37     z = z + v * dt;
38     t = t + dt;
39     v = P/m;
40     T = E/(C*N);
41 end

```



We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

- The piston has mass m , area A , vertical coord. z
- Gravity is *non*-negligible for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(i) Which lines in the while-loop use constitutive relations? Report line number and name of the equation

28: Newton's relation for momentum

29: Formula for kinetic friction

30: area of piston $\times$ ideal-gas law with viscosity

32: Total energy of matter with zero kinetic energy & grav. energy, & internal energy formula for ideal gases

34: energy flux for matter

39: Newton's relation for momentum

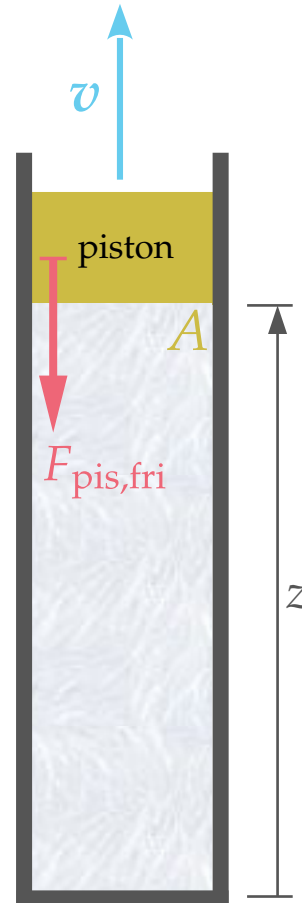
40: Total energy of matter with zero kinetic energy & grav. energy, & internal energy formula for ideal gases

34: energy flux for matter

```

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22 t = 0; % initial time
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24 v = 0; % initial velocity of piston
25 T = 300; % initial temperature of gas
26
27 while t < t_end
28     P = m*v;
29     F_pis_fri = -sign(v) * mu * Fn;
30     F_pis_gas = A * ( N*R*T/(A*z) - mu*A*v/(A*z) );
31     F_pis_tot = F_pis_gas + F_pis_fri;
32     E = C*N*T;
33     F_gas_pis = -F_pis_gas;
34     phi = Q + F_gas_pis * v;
35     P = P + (F_pis_tot + G) * dt;
36     E = E + phi * dt;
37     z = z + v * dt;
38     t = t + dt;
39     v = P/m;
40     T = E/(C*N);
41 end

```



?

We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

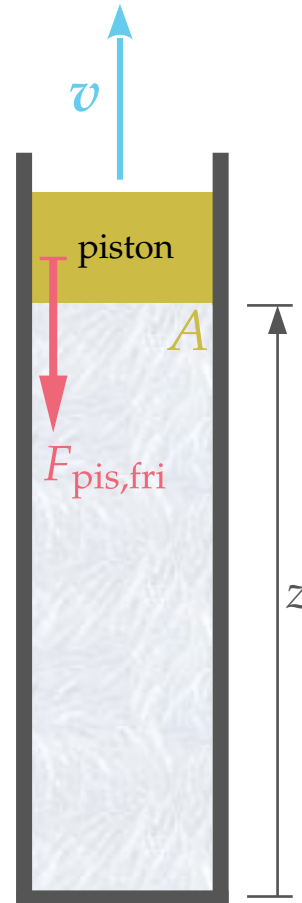
- The piston has mass m , area A , vertical coord. z
- Gravity is *non-negligible* for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(ii) Which lines in the while-loop apply extensivity and symmetry of fluxes?

```

1  %%% Numerical time integration of piston-gas with friction
2  %% SI units used throughout
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4
5  m = 50; % mass of piston
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27 while t < t_end
28     P = m*v;
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30     F_pis_gas = A * ( N*R*T/(A*z) - mu*A*v/(A*z) );
31     F_pis_tot = F_pis_gas + F_pis_fri;
32     E = C*N*T;
33     F_gas_pis = -F_pis_gas;
34     phi = Q + F_gas_pis * v;
35     P = P + (F_pis_tot + G) * dt;
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37     z = z + v * dt;
38     t = t + dt;
39     v = P/m;
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```



We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

- The piston has mass m , area A , vertical coord. z
- Gravity is *non*-negligible for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
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(ii) Which lines in the while-loop apply extensivity and symmetry of fluxes?

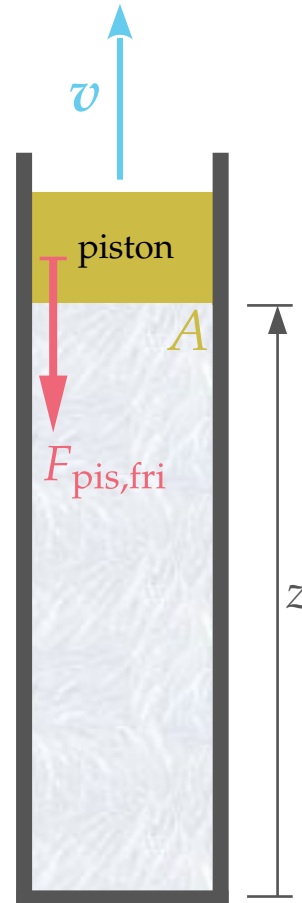
31: Extensivity: The total influx of momentum to the piston comes from the gas (pressure) and from the side (kinetic friction)

33: Flux symmetry: The influx of momentum from gas to piston is equal to minus the influx of momentum from piston to gas

```

1  %%% Numerical time integration of piston-gas with friction
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27 while t < t_end
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29     F_pis_fri = -sign(v) * mu * Fn;
30     F_pis_gas = A * ( N*R*T/(A*z) - mu*A*v/(A*z) );
31     F_pis_tot = F_pis_gas + F_pis_fri;
32     E = C*N*T;
33     F_gas_pis = -F_pis_gas;
34     phi = Q + F_gas_pis * v;
35     P = P + (F_pis_tot + G) * dt;
36     E = E + phi * dt;
37     z = z + v * dt;
38     t = t + dt;
39     v = P/m;
40     T = E/(C*N);
41 end

```



?

We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

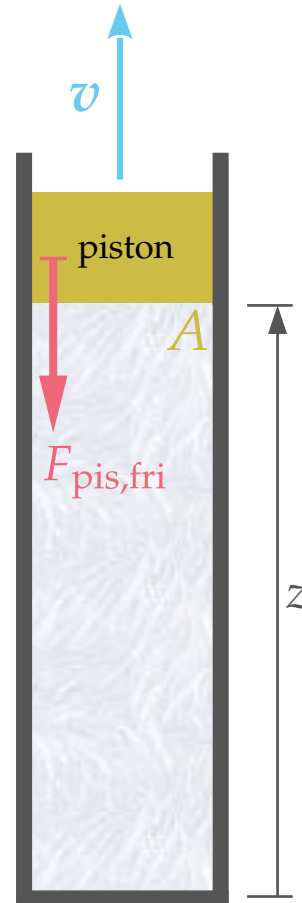
- The piston has mass m , area A , vertical coord. z
- Gravity is *non-negligible* for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(iii) Which lines in the while loop apply balance laws? Report line number and balance

```

1  %%% Numerical time integration of piston-gas with friction
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```





We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

- The piston has mass m , area A , vertical coord. z
- Gravity is *non*-negligible for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(iii) Which lines in the while loop apply balance laws? Report line number and balance

35: Balance of momentum for piston

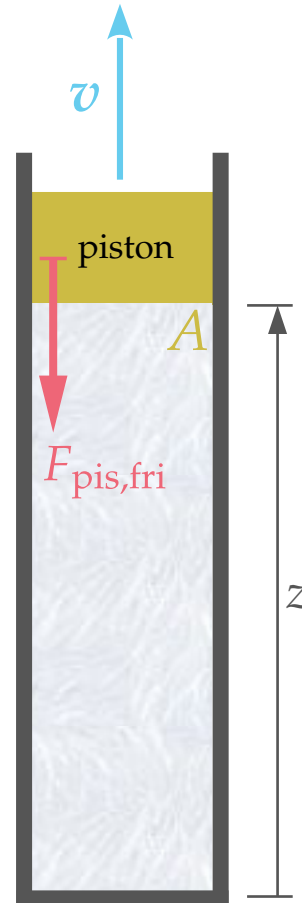
36: Balance of energy for gas

(37: This is balance of matter for gas and piston, but disguised – you don't need to know this)

```

1  %%% Numerical time integration of piston-gas with friction
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34     phi = Q + F_gas_pis * v;
35     P = P + (F_pis_tot + G) * dt;
36     E = E + phi * dt;
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38     t = t + dt;
39     v = P/m;
40     T = E/(C*N);
41 end

```



?

We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

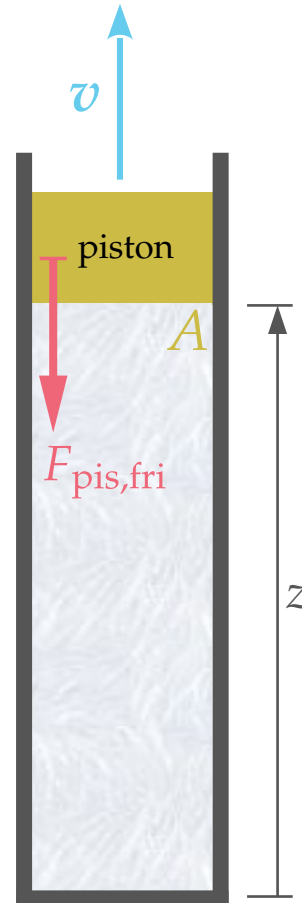
- The piston has mass m , area A , vertical coord. z
- Gravity is *non-negligible* for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(iv) Which quantities are used as state variables?

```

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27 while t < t_end
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```



?

We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

- The piston has mass m , area A , vertical coord. z
- Gravity is *non*-negligible for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(iv) Which quantities are used as state variables?

State variables are:

z: vertical position of piston
v: vertical velocity of piston
T: temperature of ideal gas

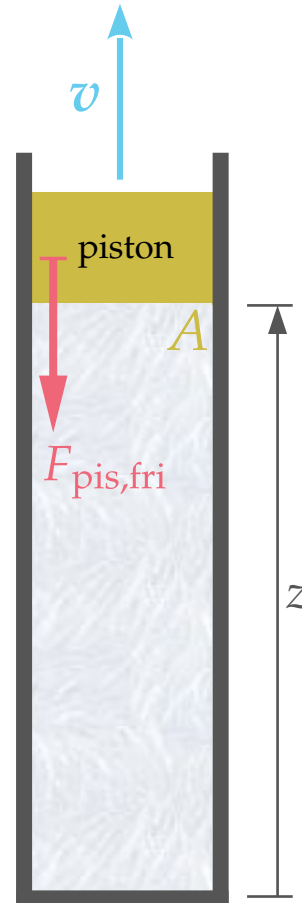
We can see this in three different ways:

- These quantities are variable, and their values are initialized on lines 23–25
- In the while-loop, before the timestep update with the balance laws, lines 28,29,30,32,34 require the values of (z,v,T) to be already known
- In the while-loop, after the timestep update with the balance laws, lines 39–40 gives us back the values of v,T from the updated values of P,E; z is already updated directly

```

1  %%% Numerical time integration of piston-gas with friction
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32     E = C*N*T;
33     F_gas_pis = -F_pis_gas;
34     phi = Q + F_gas_pis * v;
35     P = P + (F_pis_tot + G) * dt;
36     E = E + phi * dt;
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38     t = t + dt;
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40     T = E/(C*N);
41 end

```



?

We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

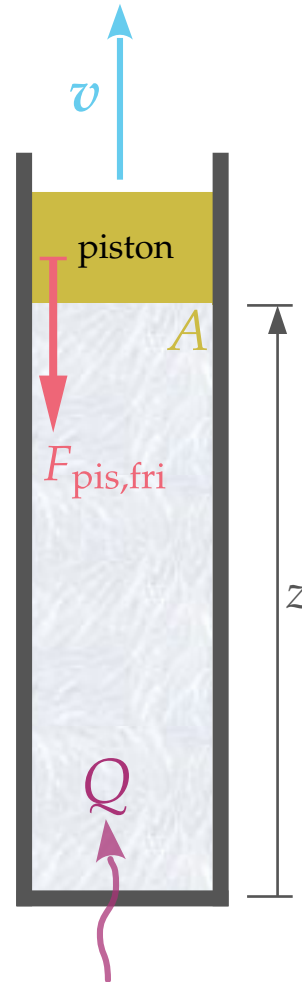
- The piston has mass m , area A , vertical coord. z
- Gravity is *non-negligible* for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n = 5\text{ N}$ is constant; the coefficient of kinetic friction is $\mu = 0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}} = pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(v) We want to add a heat flux Q to the gas at the bottom of the cylinder. Which constitutive relation could you use? Tell which lines you add or modify to implement this, and briefly explain their physical meaning. (If you need to assign constants, don't worry about their values)

```

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17
18 G = -m*g; % gravitational supply of momentum
19 Fn = 5; % normal force on piston
20 Q = 0; % heat flux for ideal gas (adiabatic)
21
22 t = 0; % initial time
23 z = 0.15; % initial position of piston
24 v = 0; % initial velocity of piston
25 T = 300; % initial temperature of gas
26
27 while t < t_end
28     P = m*v;
29     F_pis_fri = -sign(v) * mu * Fn;
30     F_pis_gas = A * ( N*R*T/(A*z) - mu*A*v/(A*z) );
31     F_pis_tot = F_pis_gas + F_pis_fri;
32     E = C*N*T;
33     F_gas_pis = -F_pis_gas;
34     phi = Q + F_gas_pis * v;
35     P = P + (F_pis_tot + G) * dt;
36     E = E + phi * dt;
37     z = z + v * dt;
38     t = t + dt;
39     v = P/m;
40     T = E/(C*N);
41 end

```



We have a system consisting in an ideal gas in a vertical cylinder with a movable piston. We know the following:

- The piston has mass m , area A , vertical coord. z
- Gravity is *non*-negligible for the piston
- There is kinetic friction between piston and the cylinder's wall; the normal force $F_n=5\text{N}$ is constant; the coefficient of kinetic friction is $\mu=0.8$
- The gas has pressure p given by the ideal-gas law with viscosity, and exerts an upward force on the piston $F_{\text{pis,gas}}=pA$
- The mass of the gas is negligible
- All energy fluxes are adiabatic

(v) We want to add a heat flux Q to the gas at the bottom of the cylinder. Which constitutive relation could you use? Tell which lines you add or modify to implement this, and briefly explain their physical meaning. (If you need to assign constants, don't worry about their values)

One constutive relation for heat flux that could apply to this situation is Newton's law of cooling:

$$Q = A h (T_{\text{out}} - T)$$

where h is the heat-transfer coefficient, T_{out} is the temperature outside the cylinder, and T is the temperature of the gas.

One way to implement it:

- after line 13: give the constant value for heat-transfer coefficient and for external temperature

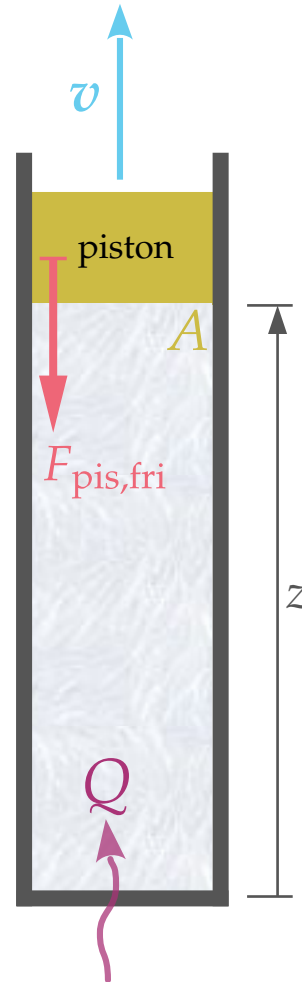
```
h = ...;
T_out = ...;
```

- line 20: remove, because now Q can now change with time.
- before line 34: Calculate heat flux Q from Newton's law of cooling:

```
Q = A * h * (T_out - T);
```

This must be done in the while loop, because T , Q can change in time.

```
1  %%% Numerical time integration of piston-gas with friction
2  %% SI units used throughout
3  %% Vertical coordinate z, positive upward
4
5  m = 50; % mass of piston
6  N = 0.04; % amount of ideal gas
7  A = 0.1^2; % area of piston
8  R = 8.31446261815; % molar gas constant
9  C = 20; % molar heat capacity of gas
10 mu = 0.00004; % gas viscosity
11 T_out = 300; % temperature of environment
12 mu = 0.8; % kinetic friction coefficient
13 g = 9.8; % gravitational acceleration
14 h = ...; % heat-transfer coeff.
15 T_out = ...; % outside temp.
16
17
18 G = -m*g; % gravitational supply of momentum
19 Fn = 5; % normal force on piston
20 Q = 0; % heat flux for ideal gas (adiabatic)
21
22 t = 0; % initial time
23 z = 0.15; % initial position of piston
24 v = 0; % initial velocity of piston
25 T = 300; % initial temperature of gas
26
27 while t < t_end
28     P = m*v;
29     F_pis_fri = -sign(v) * mu * Fn;
30     F_pis_gas = A * ( N*R*T/(A*z) - mu*A*v/(A*z) );
31     F_pis_tot = F_pis_gas + F_pis_fri;
32     E = C*N*T;
33     F_gas_pis = -F_pis_gas;
34     Q = A * h * (T_out - T); % Newton's law of cooling
35     phi = Q + F_gas_pis * v;
36     P = P + (F_pis_tot + G) * dt;
37     E = E + phi * dt;
38     z = z + v * dt;
39     t = t + dt;
40     v = P/m;
41     T = E/(C*N);
42 end
```



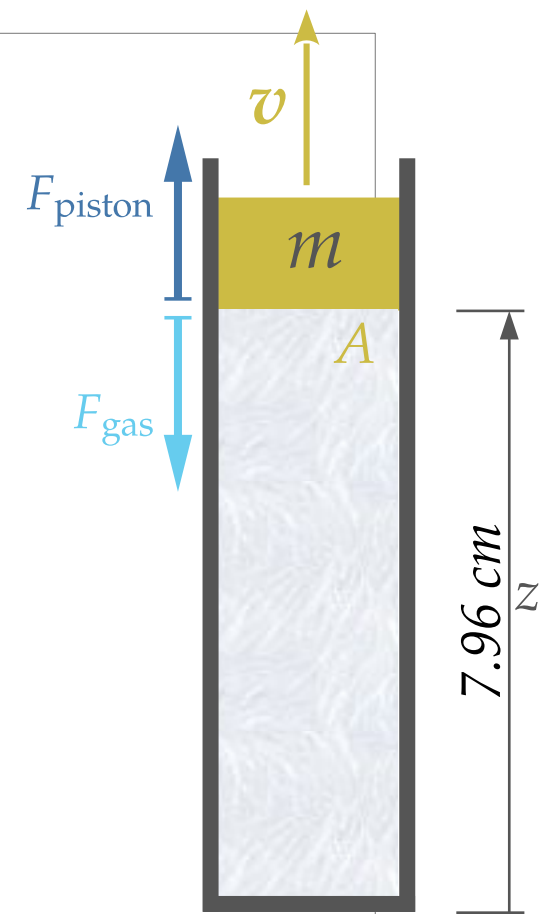
The script on the side simulates a piston-gas system, using the constitutive relations shown under the picture and the balances of momentum (piston) and of energy (gas).

The symbols are explained below.

The script uses the state z, v, T to iterate the simulation.

Modify it so that instead it uses the state z, v, pr

```
...
%% constant quantities
G = -m*g; % gravity force on piston
Q = 0; % heat flux to gas
%% initialize state & time
t = 0;
z = ...;
v = ...;
T = ...;
%% time loop
while t < t_final
    %% transform from state z,v,T
    %% to P,F_piston,z,v,E,F_gas
    P = m*v;
    pr = N*R*T/(z*A);
    F_piston = pr*A;
    F_gas = -F_piston;
    E = C*N*T;
    %% update
    P = P + (F_piston + G)*dt;
    z = z + v*dt;
    E = E + (Q + F_gas*v)*dt;
    t = t + dt;
    %% transform from P,z,E
    %% back to state z,v,T
    v = P/m;
    T = E/(C*N);
end
```



$$P = mv$$

$$E = C N T$$

$$pr = \frac{R N T}{z A}$$

$$F_{\text{piston}} = pr \cdot A$$

$$F_{\text{gas}} = -F_{\text{piston}}$$

$$Q = 0 \text{ J/s}$$

$$\Phi = Q + F_{\text{gas}} \cdot v$$

(Φ already substituted in script)

P : piston momentum
 z : piston position
 v : piston velocity
 A : piston area
 m : piston mass
 pr : gas pressure on piston
 F_piston : force on piston
 F_gas : force on gas
 E : gas energy
 T : gas temperature
 Q : heat flux (zero)
 G : gravitational force (fixed)

- We need the temperature T to calculate the energy E .

→ We get the temperature T from the pressure pr by inverting the ideal-gas law.

- We use the same law at the end, to get the pressure pr back from the temperature T .

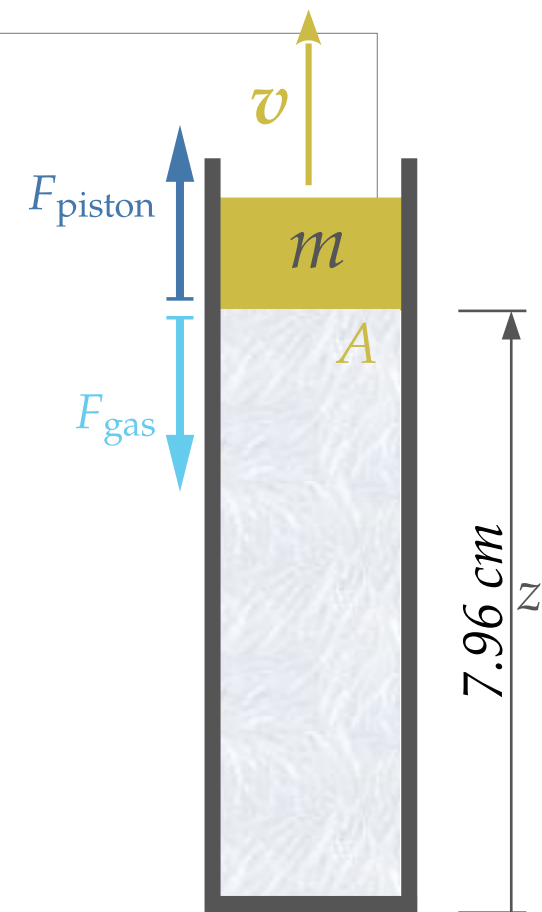
- Don't forget to initialize the pressure pr !

$$pr = \frac{R N T}{z A} \quad T = \frac{pr z a}{R N}$$

P : piston momentum
 z : piston position
 v : piston velocity
 A : piston area
 m : piston mass
 pr : gas pressure on piston
 F_{piston} : force on piston
 F_{gas} : force on gas
 E : gas energy
 T : gas temperature
 Q : heat flux (zero)
 G : gravitational force (fixed)

```

...
%% constant quantities
G = -m*g; % gravity force on piston
Q = 0; % heat flux to gas
%% initialize state & time
t = 0;
z = ...;
v = ...;
pr = ...;
%% time loop
while t < t_final
  %% transform from state z,v,pr
  %% to P,F_piston,z,v,E,F_gas
  P = m*v;
  pr = N*R*T/(z*A);
  F_piston = pr*A;
  F_gas = -F_piston;
  T = pr*z*A/(N*R);
  E = C*N*T;
  %% update
  P = P + (F_piston + G)*dt;
  z = z + v*dt;
  E = E + (Q + F_gas*v)*dt;
  t = t + dt;
  %% transform from P,z,E
  %% back to state z,v,pr
  v = P/m;
  T = E/(C*N);
  pr = R*N*T/(z*A);
end
  
```



$$P = mv$$

$$E = C N T$$

$$pr = \frac{R N T}{z A}$$

$$F_{\text{piston}} = pr \cdot A$$

$$F_{\text{gas}} = -F_{\text{piston}}$$

$$Q = 0 \text{ J/s}$$

$$\Phi = Q + F_{\text{gas}} \cdot v$$

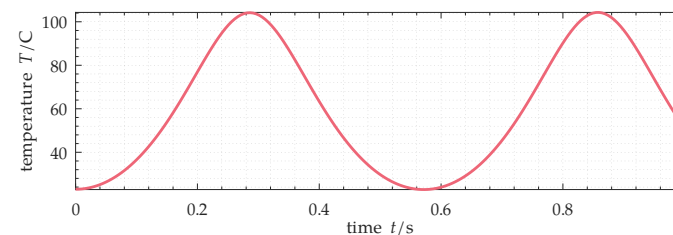
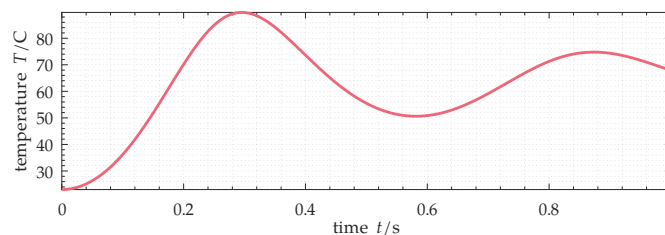
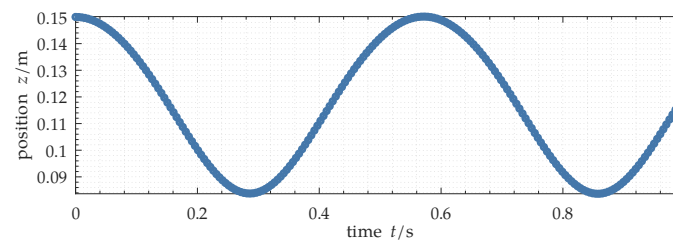
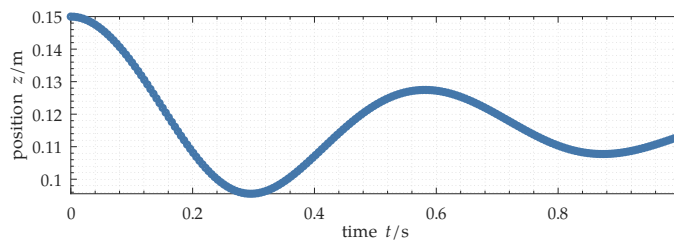
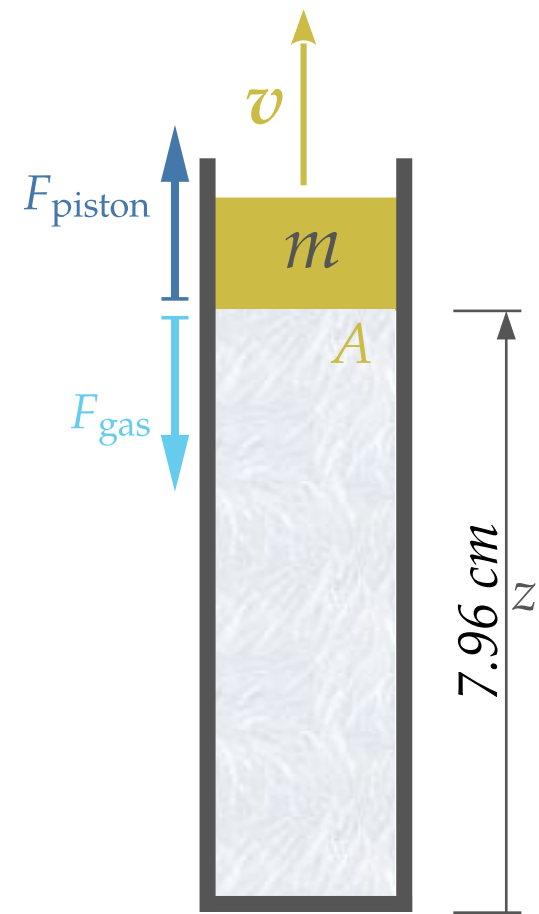
(Φ already substituted in script)



After the script is run, the position z of the piston and the temperature T of the gas are plotted against time.

The pair of plots obtained is one of these two. Which one? and why?

```
...
%% constant quantities
G = -m*g; % gravity force on piston
Q = 0; % heat flux to gas
%% initialize state & time
t = 0;
z = ...;
v = ...;
pr = ...;
%% time loop
while t < t_final
    %% transform from state z,v,pr to P,F_piston,z,v,E,F_gas
    P = m*v;
    pr = N*R*T/(z*A);
    F_piston = pr*A;
    F_gas = -F_piston;
    T = pr*z*A/(N*R);
    E = C*N*T;
    %% update
    P = P + (F_piston + G)*dt;
    z = z + v*dt;
    E = E + (Q + F_gas*v)*dt;
    t = t + dt;
    %% transform from P,z,E
    %% back to state z,v,pr
    v = P/m;
    T = E/(C*N);
    pr = R*N*T/(z*A);
end
```



$$P = mv$$

$$E = CNT$$

$$pr = \frac{RNT}{zA}$$

$$F_{\text{piston}} = pr \cdot A$$

$$F_{\text{gas}} = -F_{\text{piston}}$$

$$Q = 0 \text{ J/s}$$

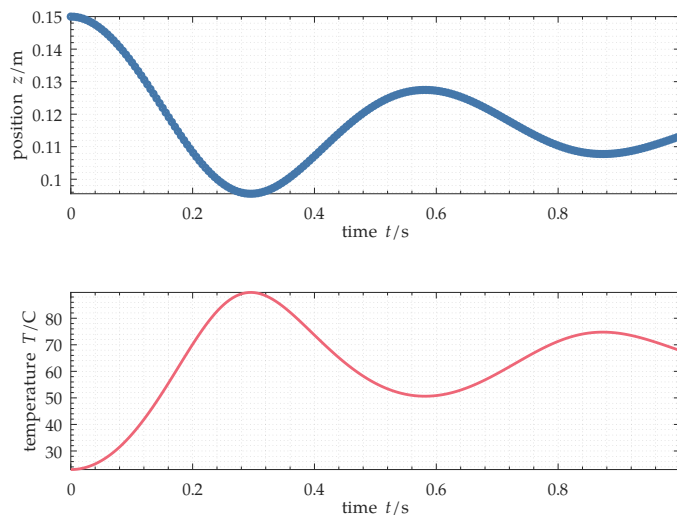
$$\Phi = Q + F_{\text{gas}} \cdot v$$

(Φ already substituted in script)



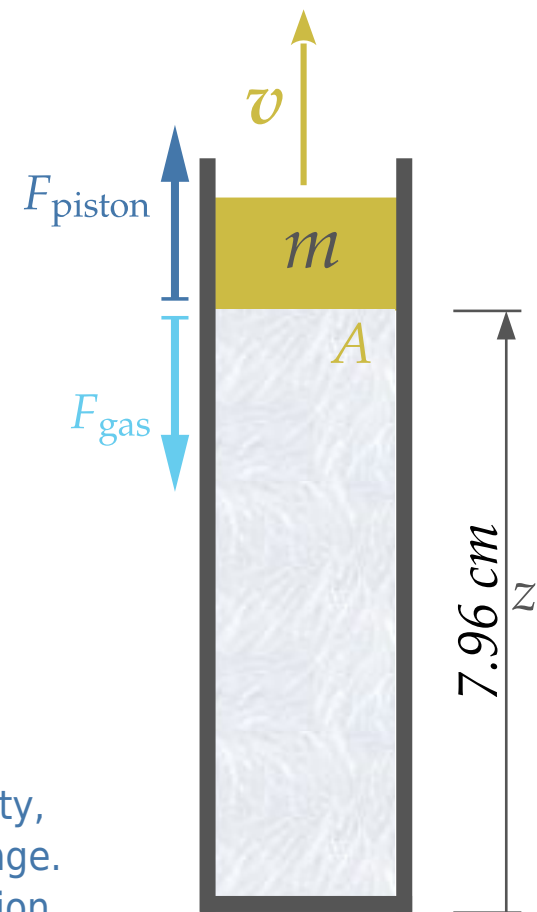
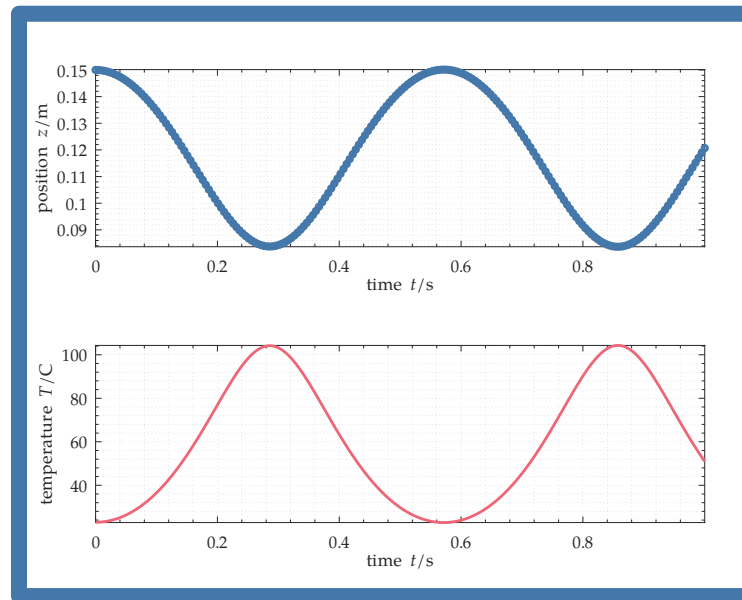
After the script is run, the position z of the piston and the temperature T of the gas are plotted against time.

The pair of plots obtained is one of these two. Which one? and why?



```
...
%% constant quantities
G = -m*g; % gravity force on piston
Q = 0; % heat flux to gas
%% initialize state & time
t = 0;
z = ...;
v = ...;
pr = ...;
%% time loop
while t < t_final
    %% transform from state z,v,pr to P,F_piston,z,v,E,F_gas
    P = m*v;
    pr = N*R*T/(z*A);
    F_piston = pr*A;
    F_gas = -F_piston;
    T = pr*z*A/(N*R);
    E = C*N*T;
    %% update
    P = P + (F_piston + G)*dt;
    z = z + v*dt;
    E = E + (Q + F_gas*v)*dt;
    t = t + dt;
    %% transform from P,z,E
    %% back to state z,v,pr
    v = P/m;
    T = E/(C*N);
    pr = R*N*T/(z*A);
end
```

This is correct.
The pressure has no viscosity,
and there is no heat exchange.
So there cannot be dissipation



$$P = mv$$

$$E = CNT$$

$$pr = \frac{RNT}{zA}$$

$$F_{\text{piston}} = pr \cdot A$$

$$F_{\text{gas}} = -F_{\text{piston}}$$

$$Q = 0 \text{ J/s}$$

$$\Phi = Q + F_{\text{gas}} \cdot v$$

(Φ already substituted in script)

Two small masses $m_a=2\text{ kg}$ and $m_b=2\text{ kg}$ are connected by a Hookean spring with elastic constant $k=5\text{ N/m}$. The constitutive relation for the spring is given below.

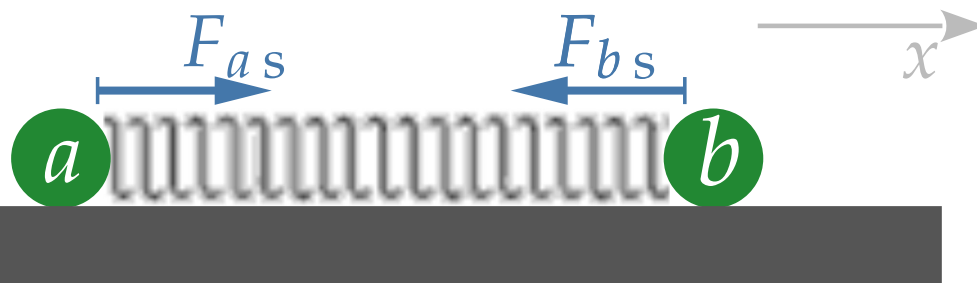
The two masses move horizontally on a table, and experience kinetic friction (there's no static friction). The common kinetic-friction coefficient is $\mu=0.1$ (μ).

Complete the script on the side with the appropriate expressions for the spring force, normal force, kinetic friction, and total force on each mass. Note that only the horizontal positions need to be updated.

*Hint: use the function $\text{sign}(v)$ to get the sign of a velocity v
Call the normal force on a F_{a_n} , the friction on a F_{a_f} , and similarly for b*

$$G_a = -m_a g \quad G_b = -m_b g$$

$$F_{a_s} = -k(x_a - x_b) \quad F_{b_s} = -F_{a_s}$$



```
ma = 2; % mass a
mb = 2; % mass b
g = 9.8; % gravit. acceleration
k = 5; % elastic const.
mu = 0.1; % kinetic-friction coeff.
t_final = 10; % simulation length
%% constant quantities
Ga = -ma*g; % gravity force on a
Gb = -mb*g; % gravity force on b
%% initialize state & time
t = 0;
xa = -3;
va = 0;
xb = 3;
vb = 0;
%% time loop
while t < t_final
    %% transform from state
    Pa = ma*va;
    Pb = mb*vb;
    Fa_s = ?????????????? % Hooke law
    Fb_s = ??????????????
    ?????????????? % normal force a
    ?????????????? % normal force b
    ?????????????? % friction a
    ?????????????? % friction b
    Fa_tot = ?????????????? % total force a
    Fb_tot = ?????????????? % total force b
    %% update in time
    Pa = Pa + Fa_tot*dt;
    Pb = Pb + Fb_tot*dt;
    xa = xa + va*dt;
    xb = xb + vb*dt;
    t = t + dt;
    %% transform back to state
    va = Pa/ma;
    vb = Pb/mb;
end
```

$$F_{a\,n} = -G_a$$

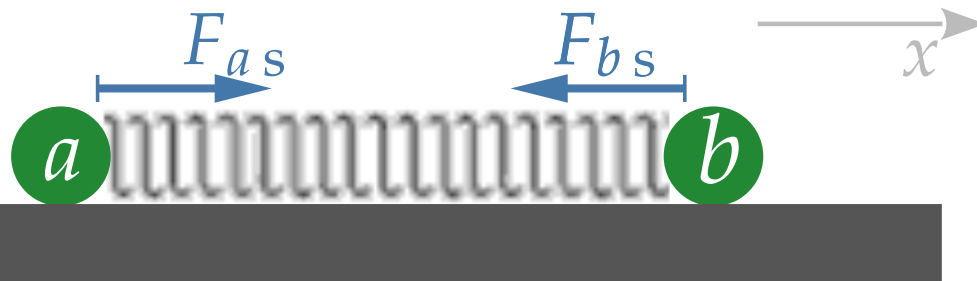
$$F_{b\,n} = -G_b$$

$$F_{a\,f} = -\mu \operatorname{sign}(v_a) F_{a\,n}$$

$$F_{b\,f} = -\mu \operatorname{sign}(v_b) F_{b\,n}$$

$$G_a = -m_a g \quad G_b = -m_b g$$

$$F_{a\,s} = -k(x_a - x_b) \quad F_{b\,s} = -F_{a\,s}$$



```

ma = 2; % mass a
mb = 2; % mass b
g = 9.8; % gravit. acceleration
k = 5; % elastic const.
mu = 0.1; % kinetic-friction coeff.
t_final = 10; % simulation length
%% constant quantities
Ga = -ma*g; % gravity force on a
Gb = -mb*g; % gravity force on b
%% initialize state & time
t = 0;
xa = -3;
va = 0;
xb = 3;
vb = 0;
%% time loop
while t < t_final
    %% transform from state
    Pa = ma*va;
    Pb = mb*vb;
    Fa_s = -k*(xa - xb); % Hooke law
    Fb_s = -Fa_s;
    Fa_n = -Ga; % normal force a
    Fb_n = -Gb; % normal force b
    Fa_f = -sign(va)*mu*Fa_n; % friction a
    Fb_f = -sign(vb)*mu*Fb_n; % friction b
    Fa_tot = Fa_s + Fa_f;
    Fb_tot = Fb_s + Fb_f;
    %% update in time
    Pa = Pa + Fa_tot*dt;
    Pb = Pb + Fb_tot*dt;
    xa = xa + va*dt;
    xb = xb + vb*dt;
    t = t + dt;
    %% transform back to state
    va = Pa/ma;
    vb = Pb/mb;
end

```


(a)

- i. Explain which balance law is implemented in the script.
- ii. Locate the line, in the while-loop, where the balance law is implemented. Explain the meaning of the various terms on this specific line.

Script for numerical simulation of floor deformation

```
1  %%% Numerical time integration of dampened fall
2  %% vertical z coordinate, positive upward
3  %% SI base units used throughout
4  m = 100; % mass
5  z0 = 0; % initial position
6  v0 = -4.4272; % initial velocity
7  k = 1500; % elastic constant
8  h = 1000; % viscosity coefficient
9  g = 9.8; % gravitational acceleration
10 t0 = 0; % initial time
11 dt = 0.01; % time step
12 %%% Numerical time integration
13 %%% Initialize
14 t = t0;
15 z = z0;
16 v = v0;
17 P = m*v0; % initial momentum
18 G = -m*g; % constant momentum supply
19 %% stop when |v| < 1 mm/s
20 while (abs(v) > 0.001)
21     t = t + dt;
22     F = -k*z - h*v;
23     P = P + (F + G)*dt;
24     z = z + v*dt;
25     v = P/m;
26 end
27 disp(z)
28 disp(t)
```

(a)

- i. Explain which balance law is implemented in the script.
- ii. Locate the line, in the while-loop, where the balance law is implemented. Explain the meaning of the various terms on this specific line.

On line 23 we have the balance of momentum in the approximate form

$$P(t + \Delta t) \approx P(t) + [F(t) + G(t)] \Delta t$$

The right side of the “=” sign has quantities of the present time t in the time-loop:

- F : momentum flux
- G : momentum supply
- P : present momentum
- Δt : time increment Δt

The left side of the “=” sign has the updated momentum P of time $t + \Delta t$

Script for numerical simulation of floor deformation

```
1  %%% Numerical time integration of dampened fall
2  %% vertical z coordinate, positive upward
3  %% SI base units used throughout
4  m = 100; % mass
5  z0 = 0; % initial position
6  v0 = -4.4272; % initial velocity
7  k = 1500; % elastic constant
8  h = 1000; % viscosity coefficient
9  g = 9.8; % gravitational acceleration
10 t0 = 0; % initial time
11 dt = 0.01; % time step
12 %%% Numerical time integration
13 %%% Initialize
14 t = t0;
15 z = z0;
16 v = v0;
17 P = m*v0; % initial momentum
18 G = -m*g; % constant momentum supply
19 %% stop when |v| < 1 mm/s
20 while (abs(v) > 0.001)
21     t = t + dt;
22     F = -k*z - h*v;
23     P = P + (F + G)*dt;
24     z = z + v*dt;
25     v = P/m;
26 end
27 disp(z)
28 disp(t)
```

(a)

- Explain which balance law is implemented in the script.
- Locate the line, in the while-loop, where the balance law is implemented. Explain the meaning of the various terms on this specific line.

On line 23 we have the balance of momentum in the approximate form

$$P(t + \Delta t) \approx P(t) + [F(t) + G(t)] \Delta t$$

The right side of the “=” sign has quantities of the present time t in the time-loop:

- F : momentum flux
- G : momentum supply
- P : present momentum
- dt : time increment Δt

The left side of the “=” sign has the updated momentum P of time $t+\Delta t$

(b)

- Which constitutive relations are used in the while-loop?
- Add comments to each of the lines 21–25, shortly explaining what each does.

Script for numerical simulation of floor deformation

```
1  %%% Numerical time integration of dampened fall
2  %% vertical z coordinate, positive upward
3  %% SI base units used throughout
4  m = 100; % mass
5  z0 = 0; % initial position
6  v0 = -4.4272; % initial velocity
7  k = 1500; % elastic constant
8  h = 1000; % viscosity coefficient
9  g = 9.8; % gravitational acceleration
10 t0 = 0; % initial time
11 dt = 0.01; % time step
12 %%% Numerical time integration
13 %%% Initialize
14 t = t0;
15 z = z0;
16 v = v0;
17 P = m*v0; % initial momentum
18 G = -m*g; % constant momentum supply
19 %% stop when |v| < 1 mm/s
20 while (abs(v) > 0.001)
21     t = t + dt;
22     F = -k*z - h*v;
23     P = P + (F + G)*dt;
24     z = z + v*dt;
25     v = P/m;
26 end
27 disp(z)
28 disp(t)
```


(a)

- Explain which balance law is implemented in the script.
- Locate the line, in the while-loop, where the balance law is implemented. Explain the meaning of the various terms on this specific line.

On line 23 we have the **balance of momentum** in the approximate form

$$P(t + \Delta t) \approx P(t) + [F(t) + G(t)] \Delta t$$

The right side of the “=” sign has quantities of the present time t in the time-loop:

- F : momentum flux
- G : momentum supply
- P : present momentum
- dt : time increment Δt

The left side of the “=” sign has the updated momentum P of time $t+\Delta t$

(b)

- Which constitutive relations are used in the while-loop?
- Add comments to each of the lines 21–25, shortly explaining what each does.

- The constitutive equation for the flux F given in the previous question is used on line 22
- **Newton's relation for momentum** $P = m v$ is used on line 25

Script for numerical simulation of floor deformation

```
1  %%% Numerical time integration of dampened fall
2  %% vertical z coordinate, positive upward
3  %% SI base units used throughout
4  m = 100; % mass
5  z0 = 0; % initial position
6  v0 = -4.4272; % initial velocity
7  k = 1500; % elastic constant
8  h = 1000; % viscosity coefficient
9  g = 9.8; % gravitational acceleration
10 t0 = 0; % initial time
11 dt = 0.01; % time step
12 %%% Numerical time integration
13 %%% Initialize
14 t = t0;
15 z = z0;
16 v = v0;
17 P = m*v0; % initial momentum
18 G = -m*g; % constant momentum supply
19 %% stop when |v| < 1 mm/s
20 while (abs(v) > 0.001)
21     t = t + dt; % update time
22     F = -k*z - h*v; % const. relation for momentum flux (previous oppg.)
23     P = P + (F + G)*dt; % balance of momentum: update momentum
24     z = z + v*dt; % update position
25     v = P/m; % find velocity from momentum with Newton's relation
26 end
27 disp(z)
28 disp(t)
```

?(c)

Assume that the total energy E of the floor consists only of internal energy U (no kinetic or potential energies). Also assume that this energy increases only from the mechanical power $F \cdot v$ coming from the force exerted by the block on the floor; that is the energy flux is adiabatic.

Take the initial energy of the floor to be $U(t_0) = 0$ J. Add two lines to the script, in appropriate places:

- one that initializes the value of the internal energy
- one that updates the internal energy, using the balance of energy.

that is, tell between which lines of the script above you would insert your lines, and what the code of your new lines would be.

Script for numerical simulation of floor deformation

```
1  %% Numerical time integration of dampened fall
2  %% vertical z coordinate, positive upward
3  %% SI base units used throughout
4  m = 100; % mass
5  z0 = 0; % initial position
6  v0 = -4.4272; % initial velocity
7  k = 1500; % elastic constant
8  h = 1000; % viscosity coefficient
9  g = 9.8; % gravitational acceleration
10 t0 = 0; % initial time
11 dt = 0.01; % time step
12 %% Numerical time integration
13 %% Initialize
14 t = t0;
15 z = z0;
16 v = v0;
17 P = m*v0; % initial momentum
18 G = -m*g; % constant momentum supply
19 %% stop when |v| < 1 mm/s
20 while (abs(v) > 0.001)
21     t = t + dt;
22     F = -k*z - h*v;
23     P = P + (F + G)*dt;
24     z = z + v*dt;
25     v = P/m;
26 end
27 disp(z)
28 disp(t)
```

?(c)

Assume that the total energy E of the floor consists only of internal energy U (no kinetic or potential energies). Also assume that this energy increases only from the mechanical power $F \cdot v$ coming from the force exerted by the block on the floor; that is the energy flux is adiabatic.

Take the initial energy of the floor to be $U(t_0) = 0$ J. Add two lines to the script, in appropriate places:

- one that initializes the value of the internal energy
- one that updates the internal energy, using the balance of energy.

that is, tell between which lines of the script above you would insert your lines, and what the code of your new lines would be.

The exercise says that there's no kinetic or gravitational energy, so

$$E = U + \frac{1}{2}mv^2 + mgz$$

simplifies to $E = U$. This means we can use E or U in the script. Let's say E .

The energy needs to be initialized before the while-loop, for example between lines 16 & 17, with

$E = 0;$

To update it we use $E(t + \Delta t) \approx E(t) + \Phi(t) \cdot \Delta t$ but we need a formula for the flux Φ . We can use the **constitutive relation for the flux of energy in matter**

$$\Phi = Q + F \cdot v$$

and consider that the flux is adiabatic, so $Q=0$. We can insert

$E = E + (0 - F \cdot v) * dt;$

for example **between lines 23 & 24** . On this line:

- “0” is the heat flux Q , zero
- “ $-F$ ” is the force on the floor. It must be opposite to the force on the block because of the **symmetry of fluxes**.
- “ v ” is the velocity of the floor, same as the block

Script for numerical simulation of floor deformation

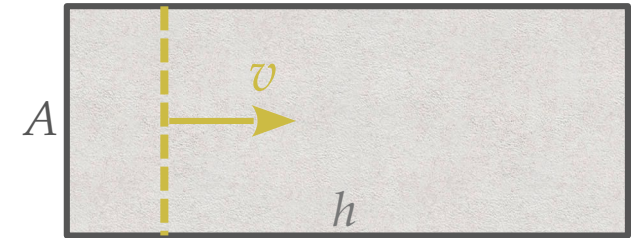
```
1  %% Numerical time integration of dampened fall
2  %% vertical z coordinate, positive upward
3  %% SI base units used throughout
4  m = 100; % mass
5  z0 = 0; % initial position
6  v0 = -4.4272; % initial velocity
7  k = 1500; % elastic constant
8  h = 1000; % viscosity coefficient
9  g = 9.8; % gravitational acceleration
10 t0 = 0; % initial time
11 dt = 0.01; % time step
12 %% Numerical time integration
13 %% Initialize
14 t = t0;
15 z = z0;
16 v = v0; E = 0;
17 P = m*v0; % initial momentum
18 G = -m*g; % constant momentum supply
19 %% stop when |v| < 1 mm/s
20 while (abs(v) > 0.001)
21     t = t + dt;
22     F = -k*z - h*v;
23     P = P + (F + G)*dt;
24     z = z + v*dt; E = E + (0 - F*v) * dt;
25     v = P/m;
26 end
27 disp(z)
28 disp(t)
```




A rectangular box of lateral area A and horizontal length h contains an amount of gas.

- The gas is in complete equilibrium, at rest.
- Total amount of gas is N , total energy content E , internal pressure p ; all constant in time.
- There is no heat flux through any control surface within or around the box.

Consider an imaginary control surface in the box,
moving from left to right at constant velocity v (see picture)
To repeat: the surface is *purely imaginary*,
so it is not displacing any gas.



I asked ChatGPT 4 how much is the flux of energy Φ through this imaginary control surface.
ChatGPT used the formula

$$\Phi = Q + F \cdot v \quad \text{with} \quad Q = 0 \quad \text{and} \quad F = pA$$

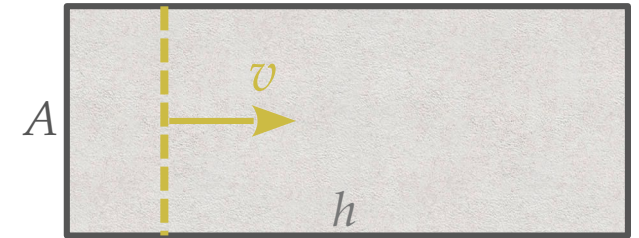
and said that the energy flux through the surface is $pA \cdot v$ (in fact it even used the ideal-gas law).

ChatGPT's answer is wrong. Explain why. (You don't need to find the correct answer.)

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The formula $\Phi = Q + F \cdot v$ for the energy flux through a surface **cannot** be used if there's matter flow!

(The correct answer is $E \cdot v / h$ but we haven't studied this formula in the course)